Boehringer Ingelheim

Clinical Overview

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LIST OF ABBREVIATIONS

AE	Adverse event
ANCOVA	Analysis of covariance
ALT	Alanine transaminase; SGPT
AST	Aspartate transaminase; SGOT
AUC	Area under the curve
BI	Boehringer Ingelheim
CI	Confidence Interval
Cmax	Maximum concentration
CYP3A4	Cytochrome P450 3A4
DBP	Diastolic blood pressure
DPP-4	Dipeptidyl-peptidase 4
EFF	Efficacy analysis group
eGFR	Estimated glomerular filtration rate
FAS	Full analysis set
FPG	Fasting plasma glucose
GGT	Gamma glutamyltransferase
Glim	Glimepiride
GLP-1	Glucagon-like peptide 1
HbA_{1c}	Glycosylated haemoglobin
ICH	International Conference on Harmonization
Lina	Linagliptin
L+M	Combination of linagliptin plus metformin (given together as individual components)
L/M	Fixed-dose (single-tablet) combination of linagliptin and metformin
LOCF	Last observation carried forward
Met	Metformin
MDRD	Modification of diet in renal disease
MTT	Meal tolerance test
OAD	Oral antidiabetic drug
OC	Observed cases
P-gp	Permeability glycoprotein
PK	Pharmacokinetics
PPS	Per protocol set
SAE	Serious adverse event
SAF	Safety analysis group
SBP	Systolic blood pressure
SCE	Summary of Clinical Efficacy
SCS	Summary of Clinical Safety
SD	Standard Deviation
SE	Standard error (of the mean)
SEA-2	Side Effect Analysis
SMQ	Standard MedDRA query
SOC	MedDRA System organ class
SU	Sulfonylurea
Tmax	Time of maximum concentration
ULN	Upper limit of normal

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1. PRODUCT DEVELOPMENT RATIONALE

1.1 INTRODUCTION AND INTENDED INDICATION

Boehringer Ingelheim has submitted an application for marketing authorisation for linagliptin monotherapy to the European Medicines Agency on 1 July 2010; this application is currently being reviewed. The company is submitting this application for a fixed-dose combination (FDC) of linagliptin/metformin with 3 different dose strengths (2.5 mg/500 mg, 2.5 mg/850 mg, 2.5 mg/1000mg) for the treatment of adults with type 2 diabetes mellitus (T2DM) in patients who are inadequately controlled on their maximum tolerated dose of metformin alone or in patients who are already treated with linagliptin and metformin. In addition, the linagliptin/metformin FDC is intended for the treatment of patients inadequately controlled on their maximum tolerated dose of metformin and a sulphonylurea (SU) or in patients already being treated with linagliptin, metformin, and an SU.

1.2 MEDICAL BACKGROUND AND DRUG PROFILES

Type 2 diabetes mellitus is a disease with a prevalence of more than 180 million people worldwide, a number expected to rise to 300 million people by the year 2025 [R07-0040, Module 2.7.5]. It is characterised by multiple metabolic dysfunctions involving insulin resistance, impaired insulin secretion, and increased glucose production. T2DM is associated with macrovascular complications (e.g. cardiovascular disease) and microvascular complications (e.g. retinopathy, neuropathy, and nephropathy) [R04-2186, Module 2.7.5]. Linagliptin (BI 1356) is a selective, orally administered, xanthine-based inhibitor of dipeptidylpeptidase-4 (DPP-4) with a 50% Inhibitor Concentration (IC50) of 1 nM. DPP-4 inhibitors are antidiabetic agents that lower blood glucose by extending the short half life of glucagon-like peptide 1 (GLP-1), which is secreted by intestinal L-cells in response to a meal [P08-07949, Module 2.7.5], and glucose-dependent insulinotropic peptide (GIP), both of which exert glucose-dependent insulinotropic effects and thereby contribute to the maintenance of post-meal glycaemic control. GLP-1 lowers blood glucose by augmenting the glucose-stimulated insulin release and limiting glucagon secretion to slow gastric emptying and to induce satiety. DPP-4 inhibitors maintain long-term β-cell function, which has been demonstrated in animal models. Inhibition of DPP-4 bears little risk of hypoglycaemia in patients with diabetes mellitus because GLP-1 activity ceases when plasma glucose concentration falls below 55 mg/dL. Linagliptin is predominantly excreted unchanged via the faeces. Renal excretion is a minor pathway of elimination of linagliptin at therapeutic doses.

Metformin has been used in Europe for over 50 years and, together with lifestyle modification, is recommended by the European Association for the Study of Diabetes (EASD) as the first-line treatment for T2DM [R10-4894, Module 2.7.5]. Although its mechanism of action is not yet fully understood, metformin lowers blood glucose levels primarily by suppressing hepatic gluconeogenesis; it is believed that this is achieved through metformin-induced activation of adenosine monophosphate-activated protein kinase, an energy-regulating enzyme in the liver. Furthermore, metformin improves the insulin sensitivity of peripheral tissues, decreases gastrointestinal tract glucose absorption, and acts as an insulin sensitiser without exerting any direct effect on pancreatic β-cell insulin secretion. Through these mechanisms, metformin therapy typically leads to substantial

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reductions in glycosylated HbA_{1c} but it does not promote weight gain or increase the risk of hypoglycaemia [R10-4882, R10-4896, Module 2.7.5].

The combination of linagliptin with metformin provides clinically meaningful treatment benefits by lowering glucose and reducing glycosylated haemoglobin (HbA_{1c}) further than monotherapy with either component at corresponding doses. The proposed doses of linagliptin/metformin hydrochloride are 2.5 mg/500 mg, 2.5 mg/850 mg, and 2.5 mg/ 1000 mg, all taken twice daily (bid). Combining linagliptin with metformin hydrochloride simplifies the antidiabetic therapy by decreasing the number of tablets to be taken and is expected to improve patients' compliance with medication; fixed-dose combination therapy has resulted in improved compliance in patients previously treated with oral antidiabetics [R10-1386, R10-1387, Module 2.7.5].

1.3 CLINICAL DEVELOPMENT PROGRAMME

Clinical development of linagliptin monotherapy and combination therapy with metformin started in September 2004. The clinical programme that formed the basis for the application for marketing authorisation of linagliptin as monotherapy comprised 24 phase I-trials (among them 9 drug-drug interaction trials), 4 phase II-trials, and 9 large phase III-trials including an open-label extension trial of the 4 pivotal placebo-controlled trials. These studies established the pharmacokinetics, safety, and efficacy profiles of linagliptin monotherapy.

Pharmacokinetics, safety, and efficacy profiles of metformin monotherapy are well established and thoroughly described in the current Summary of Product Characteristics [R10-5341, Module 2.7.5]. For the development of the FDC of linagliptin/metformin, no dedicated studies with metformin monotherapy were conducted. However, information on metformin as monotherapy is included in some of the analysed safety and efficacy groupings in the respective treatment arms.

Evaluation of efficacy and safety

The evaluation of efficacy and safety of combination therapy with linagliptin and metformin comprised 14 clinical trials, among them 6 phase I-trials (1288.1, 1288.2, 1288.3, 1288.4, 1218.4, 1218.47), 2 phase II-trials (1218.6, 1218.62), 4 phase III-trials (1218.17, 1218.18, 1218.20, 1218.46), and 2 long-term follow-up trials of the phase III-studies (1218.40, 1218.52). At the time of the database lock for this submission (15 Oct 2010 for all studies but trial 1218.20 for which the database lock was 26 Jan 2011), all studies except the extension studies 1218.40 and 1218.52 had been completed. Data from a small phase I bioequivalence trial in 40 healthy subjects of 2 FDC batches (1288.6, [U11-1379], Module 5.3.1.1) were not included because they were not available at the time of database lock; however the trial report is available. Study 1218.57, which showed bioequivalence between the European (used in study 1218.46) and US Glucophage® reference products, was also conducted as part of the development of the FDC for linagliptin/metformin. This study is not included in the analysis but is discussed in Module 2.7.1 and the trial report [U09-2155, Module 5.3.1.2] is available.

Details on studies and their groupings used to establish efficacy are provided in Section 4.1, details on studies and groupings that form the basis of the safety analysis are provided in Section 5.

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In total 3529 patients with type 2 diabetes (SAF-C8) and 354 healthy volunteers received treatment with linagliptin and metformin (SAF-C9). Overall 2694 patients were treated for at least 24 weeks, 2081 patients for at least 52 weeks, and 1756 patients for 78 weeks or longer. In accordance with current regulatory requirements, this considerable exposure forms the basis for the application of marketing authorisation for the combination of linagliptin and metformin. Overall, 43.9% of patients were recruited in Europe, 33.2% in Asia, 11.2% in North America, 10.2% in South America, and 1.5% in Africa.

All trials employed up-to-date methodologies and all trials were conducted according to ICH GCP guidelines subsequent to the review and approval of the relevant ethics committees, institutional review boards, and regulatory authorities. The principal trials were randomised, double-blind, placebo-controlled, parallel-group studies and were conducted in a population of male or female patients with type 2 diabetes whose blood concentrations of glycosylated haemoglobin (HbA_{1c}) indicated insufficient glycaemic control. All principal trials used the same primary endpoint to assess antidiabetic efficacy: the change in HbA_{1c} from baseline to end of trial. To ensure a homogeneous distribution of patients with poor glycaemic control, randomisation was stratified by the HbA_{1c} level at the beginning of the placebo run-in period (<8.5% versus ≥8.5%) in most of the studies. Randomisation was also stratified by the number of oral antidiabetic drugs (OADs) at the time of enrolment in most of the trials. Also the main secondary endpoints were the same in the entire phase III program, i.e. the change in fasting plasma glucose (FPG) from baseline to end of trial, treat-to-target efficacy response (i.e. HbA_{1c} on treatment <7.0% or <6.5%), and relative efficacy response (i.e. HbA_{1c}-lowering ≥0.5%).

Evaluation of pharmacokinetics

For the development of the linagliptin/metformin FDC, 6 phase I studies (1288.1, 1288.2, 1288.3, 1288.4, 1288.6, and 1218.57) were performed that established the bioequivalence of the 3 linagliptin/metformin FDC dose strengths and their respective individual components. In addition, trial 1288.4 [U10-2236, Module 5.3.1.1] investigated the effect of a high fat, high caloric meal on the bioavailability of the Lina 2.5 mg/Metformin 1000 mg FDC. The bioequivalence of 2.5 mg linagliptin twice daily (bid) with 5 mg linagliptin once daily (qd) was established in study 1218.45, and the absence of an interaction between linagliptin and metformin was shown in study 1218.4. Further details are provided in Section 2 and Section 3 below.

1.4 INTERACTION WITH REGULATORY AUTHORITIES

During the clinical development of linagliptin and the linagliptin/metformin FDC there was close interaction with key regulatory agencies. Scientific advice for the development of linagliptin monotherapy was sought from the CHMP on 16 May 2007 (EMEA/CHMP/ SAWP/385760/2007) and on 23rd April 2008 (EMEA/CHMP/ SAWP/311988/2008). For the development of the linagliptin/metformin FDC, scientific advice was sought on 27th May 2008 (EMEA/CHMP/SAWP/ 472394/2008) and final responses were received on 25th September 2008 (EMEA/CHMP/SAWP/472394/2008). The CHMP requested a clinical study to show equivalence of twice daily dosing of linagliptin 2.5 mg with once daily dosing of linagliptin 5 mg. This study was subsequently conducted (1218.62), and its results are presented in the current dossier. On 1st July 2010 an application for marketing authorisation

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of linagliptin monotherapy in patients with type 2 diabetes mellitus was submitted to the EMA that is currently being reviewed. This application has received the Day 180 List of Outstanding Issues on 14th April 2011.

On 10th November 2010 a pre-submission meeting for the FDC linagliptin/metformin for the treatment of patients with type 2 diabetes mellitus was held with the EMA. It was confirmed that the application is a full-mixed application according to Article 8(3) of Directive 2001/83/EC which requires that the non-clinical and clinical overviews and the summaries of clinical efficacy (SCE) and safety (SCS) cover linagliptin, metformin, and the combination separately, and that each section of the documents must address each of the components. On 9th December 2010 a presubmission meeting with the co-rapporteur the Medicines Evaluation Board of The Netherlands was held. In addition to the endorsement to cover all 3 components (linagliptin, metformin, combination) in the documents separately, it was pointed out that the concerns raised in the list of questions to the linagliptin monotherapy application should be appropriately addressed in the FDC submission. A product-specific paediatric waiver was granted on 11th December 2009 (EMA/802455/2009). On 19th January 2011 an NDA for linagliptin/metformin FDC was filed with the US FDA and is currently under review. On 2 May 2011, Boehringer Ingelheim received approval of the FDA for the use of linagliptin as an adjunct to diet and exercise to improve glycaemic control in adults with T2DM.

2. OVERVIEW OF BIOPHARMACEUTICS

For the establishment of efficacy and safety, a film-coated tablet containing 5 mg of linagliptin was developed and characterised in regard to its pharmacokinetics and pharmacodynamics. Evidence of its clinical efficacy and safety was provided by the development program for linagliptin monotherapy.

For the fixed-dose combination, the therapeutic dose of linagliptin of 5 mg qd was split into two doses of 2.5 mg each in order to conform to the twice daily dosing required for metformin. Both linagliptin regimens were shown to be equivalent with regard to DPP-4 inhibition and total steady-state plasma exposure (1218.45 [U08-2132, Module 5.3.1.2]). This was confirmed in regard to clinical efficacy by study 1218.62 [U11-3093, Module 5.3.5.1].

The mono-layer tablet of the intended commercial formulation of linagliptin/metformin was used in the phase I clinical studies 1218.47, 1288.1, 1288.2, 1288.3, and 1288.4. The pivotal study 1218.17, as well as trials 1218.18, 1218.20, and 1218.40 used the free combination of linagliptin 5 mg qd and metformin background therapy in various posologies (total daily dose ≥1500 mg or maximum tolerated dose). Study 1218.62 used the free combination of linagliptin 2.5 mg bid or linagliptin 5 mg qd with metformin background therapy in a bid regimen (total daily dose ≥1500 mg or maximum tolerated dose). Study 1218.46 and its 54-week double-blind extension 1218.52 used the free combination of linagliptin 2.5 mg bid and metformin 500 mg bid or 1000 mg bid.

Bioequivalence between the fixed-dose formulations and the free combinations was established in studies 1288.1, 1288.2, and 1288.3; details are available in [Modules 2.7.1 and 2.7.2]. In the bioequivalence trials and in studies 1218.46 and 1218.52 the European metformin reference product was supplied as part of the study medication (Glucophage® Merck Serono). In the pivotal trial 1218.17 as well as in studies 1218.18, 1218.20, 1218.40,

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and 1218.62, patients continued with their local prescribed metformin tablets, i.e. metformin was not supplied by Boehringer Ingelheim as part of the study medication.

The relative bioavailability of linagliptin was not relevantly affected when the FDC tablet was administered under fed conditions (study 1288.4). Metformin extent of absorption was not relevantly altered, whereas maximum concentrations were reduced by about 18% when given with food. The decrease in the rate and extent of absorption of metformin under fed conditions, seen in this study, was less pronounced than previously reported for Glucophage®, but is well within the range of food-effects on other FDC tablets containing metformin; e.g., Actoplus Met®, Avandamet®, Eucreas®, and Metaglip® [R10-4454, R10-4455, R10-4456, and R10-4457, Module 2.7.5]. Because metformin is recommended to be given with meals to reduce gastrointestinal side effects, the FDC tablet is also proposed to be given with food. Single oral doses of 2.5 mg linagliptin/1000 mg metformin were safe and well tolerated when administered with and without food (1288.4).

3. OVERVIEW OF CLINICAL PHARMACOLOGY

Linagliptin

The pharmacokinetics and pharmacodynamics of linagliptin were evaluated in healthy subjects and patients with T2DM. In these studies, linagliptin was administered as a single oral dose up to 600 mg or multiple doses every 24 hours up to 10 mg/day. The studies were designed to address issues related to linagliptin single-dose and multiple-dose pharmacokinetics, dose proportionality, excretion/mass balance, metabolite identification, effects of food on bioavailability, pharmacodynamics, drug interactions, pharmacokinetics in special populations (gender, race, BMI/weight, elderly, and renal or hepatic impairment) and absolute/relative bioavailability. Details are available in Module 2.7.2.

There is no need for dose adjustments of linagliptin on the basis of age, gender, BMI, race, or renal or hepatic impairment. Linagliptin had no clinically relevant effect on the pharmacokinetics of metformin, glibenclamide, simvastatin, pioglitazone, warfarin, digoxin, or oral contraceptives. These findings constitute *in vivo* evidence of a low propensity for drug interactions with substrates of CYP3A4, CYP2C9, CYP2C8, P-glycoprotein, and organic cationic transporters (OCT).

Co-administration of a single 5 mg oral dose of linagliptin and multiple 200 mg oral doses of ritonavir, a potent inhibitor of P-gp and CYP3A4, increased the AUC and C_{max} of linagliptin approximately 2-fold and 3-fold. A likely explanation for this observation is an increase in the bioavailability of linagliptin due to an inhibition of P-gp. As ritonavir is one of the most potent P-gp/CYP3A4 inhibitors, clinically relevant interactions with weaker P-gp/CYP3A4 inhibitors are not expected. Furthermore, since linagliptin has a broad safety window, as demonstrated by tolerability of single doses of up to 600 mg, and a safety margin of at least 40-fold based on data from toxicology studies, the increase in AUC and C_{max} is not of clinical relevance.

Conversely, multiple co-administrations of linagliptin with rifampicin, a potent inducer of P-gp and CYP3A4, resulted in a decrease of about 40% in linagliptin steady-state exposure and a decrease of about 30% in DPP-4 inhibition. This observation is most likely due to an

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inductive effect on P-gp. Thus, linagliptin in combination with strong P-gp inducers might not reach full clinical efficacy.

Metformin

No studies investigating the possible influence of intrinsic or extrinsic factors on metformin pharmacokinetics and pharmacodynamics were performed because all relevant information is available in the scientific literature and in the SPC [R10-5537, R11-0761, Module 2.7.5].

Patients taking metformin should be closely observed for loss of blood glucose control when the following medications are taken concomitantly: thiazides and other diuretics, corticosteroids, phenothiazines, thyroid products, estrogens, oral contraceptives, phenytoin, nicotinic acid, sympathomimetics, calcium-channel-blocking drugs, and isoniazid.
For metformin no dosage adjustment is recommended based on age, gender, BMI, or race. However, metformin treatment should not be initiated in patients 80 years of age or older unless measurement of creatinine clearance demonstrates that renal function is not reduced. Metformin is contraindicated in patients with renal failure and renal dysfunction even if moderate (creatinine clearance <60 mL/min).

Linagliptin/metformin combination

A possible interaction of linagliptin and metformin was investigated in study 1218.4. It showed that co-administration of linagliptin and metformin did not have a clinically meaningful effect on linagliptin pharmacokinetics and pharmacodynamics. Also, linagliptin did not meaningfully affect metformin exposure.

No dedicated studies were performed that investigated the effect of intrinsic or extrinsic factors on the pharmacokinetics of the linagliptin/metformin FDC formulation. However, it can be assumed that the same conditions, limitations, and dose adjustments as reported for the single entities will apply to the fixed-dose single-tablet combination. Hence, for the linagliptin/metformin FDC, no dosage adjustment is necessary based on age, gender, BMI/weight, or race. However, due to the metformin component, treatment with the combination should generally be avoided in hepatically impaired patients since hepatic impairment has been associated rarely with lactic acidosis. The FDC is also contraindicated in patients with renal impairment (creatinine clearance <60 mL/min).

3.1 PHARMACOKINETICS

3.1.1 Linagliptin

The pharmacokinetics of linagliptin have been extensively characterised in healthy subjects and patients with T2DM. After oral administration of a single 5 mg dose to healthy subjects, linagliptin was rapidly absorbed, with peak plasma concentrations (T_{max}) occurring 1.5 hour post dose; the gMean plasma AUC_{0-24} was 139 nmol•h/L (gCV 25.9%) and gMean C_{max} was 8.90 nmol/L (gCV 40.3%) [1218.25, U07-2003, Module 5.3.1.2].

Plasma concentrations of linagliptin decline in a triphasic manner with a long terminal half life (>100 h) related to the saturable, tight binding of linagliptin to DPP-4. The prolonged elimination phase does not contribute to the accumulation of the drug. The effective half-life for accumulation of linagliptin, as determined from oral administration of multiple doses of

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linagliptin 5 mg, is approximately 12 hours. After once-daily dosing, steady-state plasma concentrations of linagliptin 5 mg are reached by the third dose, and C_{max} and AUC increased by a factor of about 1.3 at steady-state compared to the first dose. The intra-subject and inter-subject coefficients of variation for linagliptin steady-state AUC were small (12.6% and 28.5%, respectively). Plasma AUC of linagliptin increased in a less than dose proportional manner. The pharmacokinetics of linagliptin were similar in healthy subjects and in patients with T2DM. Bioequivalence of 5 mg linagliptin once daily compared with 2.5 mg linagliptin twice daily was shown for 24-hour steady-state AUC in study 1218.45.

Absorption: The absolute bioavailability of linagliptin is approximately 30%. Because co-administration with a high-fat meal had no clinically relevant effect on the pharmacokinetics, linagliptin itself may be administered with or without food.

Distribution: The mean apparent volume of distribution at steady state following a single intravenous dose of linagliptin 5 mg to healthy subjects was approximately 1110 L, indicating that linagliptin extensively distributes into the tissues. Plasma protein binding of linagliptin is concentration dependent, decreasing from about 99% at 1 nmol/L to 75-89% at ≥30 nmol/L, reflecting saturation of binding to DPP-4 with increasing concentration of linagliptin. At high concentrations, where DPP-4 is fully saturated, 70-80% of linagliptin was bound to plasma proteins other than DPP-4; hence, 20-30% was unbound in plasma. Plasma protein binding was not altered in patients with renal or hepatic impairment.

Metabolism: Following oral administration, the majority (about 90%) of linagliptin was excreted unchanged, indicating that metabolism represents a minor elimination pathway. A small fraction of absorbed linagliptin is metabolized by CYP3A4 to a pharmacologically inactive metabolite (CD 1790), which shows a steady-state exposure of 13.3% relative to linagliptin and is predominantly eliminated non-renally.

Excretion: Linagliptin is primarily eliminated unchanged via the faeces. Following administration of an oral [^{14}C]-linagliptin dose to healthy subjects, approximately 85% of the administered radioactivity was eliminated via the faeces (80%) or urine (5%) within 4 days of dosing. Renal clearance at steady state was approximately 70 mL/min.

Special populations

Results of individual studies and population pharmacokinetic analyses of data from phase II trials indicate that no dosage adjustments are required because of renal or hepatic impairment, gender, weight/Body Mass Index (BMI), race, ethnicity, or age. Studies characterizing the pharmacokinetics of linagliptin in paediatric patients are ongoing (1218.56).

3.1.2 Metformin

No studies dedicated to the evaluation of pharmacokinetics of metformin were conducted in the context of the linagliptin/metformin FDC; therefore information of pharmacokinetics is derived from the SPC [R10-5537, R11-0761; Module 2.7.5].

Absorption and bioavailability: After an oral dose of metformin, T_{max} is reached in 2.5 hours. The absolute bioavailability is approximately 50% to 60%. After oral administration, metformin hydrochloride absorption is saturable and therefore metformin pharmacokinetics is non-linear. At the usual metformin doses and dosing schedules, steady state plasma concentrations are reached within 24 to 48 hours. Food decreases the extent of and slightly

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delays the absorption of metformin as shown by an approximately 40% lower C_{max}, a 25% lower AUC, and a 35-minute prolongation of T_{max} following administration of a single 850 mg tablet. However, the clinical relevance of these changes is unknown.

Distribution: Plasma protein binding is negligible. Metformin partitions into erythrocytes and the blood peak is lower than the plasma peak and appears at approximately the same time; the mean volume of distribution (Vd) was between 63-276 L.

Metabolism and elimination: Metformin is excreted unchanged in the urine; no metabolites have been identified in humans. The renal clearance of metformin is greater than 400 mL/min, indicating that metformin is eliminated by glomerular filtration and tubular secretion. Following an oral dose, the apparent terminal elimination half-life is approximately 6.5 hours.

Special populations

Renal impairment: In patients with decreased renal function, the plasma and blood half-life of metformin is prolonged and its renal clearance is decreased in proportion to the decrease in creatinine clearance, leading to higher levels of metformin in the plasma. Metformin is contraindicated in acute conditions that have a potential to alter renal function such as dehydration, severe infection, shock, or patients receiving an intravascular administration of iodinated contrast agents.

Hepatic impairment: No studies in patients with hepatic impairment were performed for the linagliptin/metformin FDC development because metformin is contraindicated in these patients.

Gender and race: Although dedicated studies to study the influence of gender or race are not available, there are no indications that the pharmacokinetics and clinical efficacy of metformin are substantially influenced by these factors.

Contraindications: Metformin is contraindicated in patients with renal impairment or hepatic insufficiency, patients who suffer from acute alcohol intoxication, or patients with alcoholism. Metformin is also contraindicated in patients with an acute or chronic disease which may cause tissue hypoxia such as cardiac or respiratory failure, recent myocardial infarction, or shock and also during lactation. Patients with a history of lactic acidosis must not take metformin.

Drug interactions with metformin

Drug interactions of metformin were reported for glyburide, furosemide, nifedipine, and are likely for cationic drugs that are eliminated by renal tubular secretion (e.g. amiloride, cimetidine, digoxin, morphine, procainamide, quinidine, quinine, ranitidine, triamterene, trimethoprim, or vancomycin). Certain drugs tend to produce hyperglycaemia and may cause loss of glycaemic control. These drugs include the thiazides and other diuretics, corticosteroids, phenothiazines, thyroid products, estrogens, oral contraceptives, phenytoin, nicotinic acid, sympathomimetics, calcium-channel-blocking drugs, and isoniazid. In healthy volunteers, the pharmacokinetics of metformin and propranolol, and metformin and ibuprofen were not affected when co-administered in single-dose interaction studies. Metformin is negligibly bound to plasma proteins and is, therefore, less likely to interact with highly protein-bound drugs such as salicylates, sulfonamides, chloramphenicol, and probenecid, as compared to the sulfonylureas, which are extensively bound to serum proteins. Detailed

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information on the magnitude of the interaction effects is available in the scientific literature [P10-09483, R10-2970, Module 2.7.5].

4. OVERVIEW OF EFFICACY

4.1 EFFICACY OF LINAGLIPTIN MONOTHERAPY

4.1.1 Clinical trial programme

The clinical programme for linagliptin as monotherapy started in September 2004 and comprised 14 phase II/III trials; among them 4 studies that investigated the efficacy of linagliptin as monotherapy versus placebo medication. The pivotal trial for linagliptin monotherapy was study 1218.16, a phase III-trial that investigated the efficacy of linagliptin 5 mg once daily versus placebo over 24 weeks. Supportive evidence is derived from a 12-week phase II-trial (1218.5), a 12-week phase III-trial in Japanese patients (1218.23), and an 18-week phase III-trial in patients ineligible for treatment with metformin (1218.50). These 4 studies formed the EFF-4 grouping whose data were submitted in June 2010 to support a linagliptin monotherapy registration.

Further evidence is available from study 1218.46, which included a comparison of linagliptin 5 mg qd with placebo over 24 weeks in a partial first-line therapy setting. These data are presented in the section on the trials that support linagliptin/metformin combination (Section 4.1.3). Long-term efficacy of linagliptin monotherapy was investigated by combining data from patients who participated in study 1218.16 and continued treatment in the open-label extension study 1218.40. The data of the 2 trials are combined to form efficacy grouping EFF-M1 (“M” denotes linagliptin monotherapy).

Table 4.1.1: 1 Studies for the evaluation of the efficacy of linagliptin monotherapy

Studies	Study characteristics	Treatment duration	Number of patients [a]	Shorthand	Data status [b]
1218.16 1218.5 1218.23 1218.50	Double-blind, placebo-controlled studies without background medication	24 weeks 12 weeks 12 weeks [c] 18 weeks [d]	496 117 239 220	EFF-4 (mono submission)	as previously submitted
1218.46	Double-blind, placebo-controlled study in a first-line therapy setting	24 weeks	200[e]		new data
1218.16 + 1218.40	Studies to assess long-term efficacy of linagliptin monotherapy (combined) [f]	24 weeks + 78 weeks [g]	443	EFF-M1	updated data

[a] Number of patients in the Full Analysis Set (FAS), the patient set used for the efficacy analyses.
[b] Referring to the data included in the linagliptin mono SCE.
[c] Duration of placebo-controlled treatment (was followed by extension treatment phase).
[d] Duration of placebo-controlled treatment (followed by 34 weeks active-controlled treatment with linagliptin vs. glimepiride).
[e] Only patients in the FAS receiving linagliptin monotherapy or placebo are included in this number.
[f] Combined patient data from the initial study and the extension study.
[g] The planned total treatment duration of study 1218.40 is 78 weeks. Since interim data from this study are included in this submission, not all patients have completed the treatment period.

Source data: [U10-1608, Module 5.3.5.3, Table 1.2 and U11-1151, Module 5.3.5.3, Table 1.2]

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4.1.2 Disposition, demographics, and baseline characteristics

The durations of the trials included in EFF-4 varied from 12 to 24 weeks (see above). The percentage of patients who completed the planned treatment duration was above 90% in all studies but trial 1218.5 (74.4%). The frequencies of premature discontinuations were lower with linagliptin than with placebo in all studies. The most frequent reasons for premature discontinuation were either lack of efficacy, occurrence of adverse events, or administrative issues.

Of the patients who had completed the initial study 1218.16, almost all (94.3%) continued treatment in the open-label extension study 1218.40. In study 1218.40, all patients received linagliptin 5 mg, i.e. patients who had received placebo in study 1218.16 were switched to linagliptin 5 mg. Overall, 90.1% of the patients from study 1218.16 did not prematurely discontinue treatment in study 1218.40, with similar frequencies in those patients who had previously received linagliptin (88.9%) or placebo (92.5%). Premature discontinuations occurred most frequently for administrative reasons (3.4%) and the occurrence of adverse events (2.9%). Further details on the disposition of patients in EFF-M1 are available in the SCE [Module 2.7.3, Section 3.1.1.1].

The mean age was similar in the 4 studies in EFF-4 (55 to 60 years); the proportion of male patients ranged from about 36% (1218.50) to about 70% (1218.23). Due to the different locations of the studies, the percentages of White, Black and Asian patients varied. The majority of patients were White in all studies except study 1218.23, which comprised only Asian patients. Consequently, the mean BMI in study 1218.23 was lower than in the other studies in EFF-4.

Mean baseline HbA_{1c} and mean FPG concentrations were similar across studies in EFF-4. The percentage of patients diagnosed with diabetes for more than 5 years varied between trials (20% to 50%), but was similar between treatment groups in all studies. About half of the patients in all 4 studies had not been pre-treated with antidiabetic drugs. Most of those with pre-treatment, had previously received 1 OAD; 10% or less in all trials had received 2 or more OADs. The baseline characteristics of the patients from study 1218.16 who continued treatment in the open-label extension study 1218.40 (EFF-M1) were very similar to the baseline characteristics of all patients in study 1218.16. Further details on disposition, demographics and baseline characteristics are available in the SCE [Module 2.7.3, Section 3.1].

4.1.3 Efficacy results of linagliptin monotherapy

In all studies linagliptin monotherapy was shown to provide superior and clinically meaningful improvements in glycaemic control compared with placebo. The placebo-corrected HbA_{1c} reduction in the pivotal trial 1218.16 was -0.69% and the magnitude of the treatment effect was similar in the other trials (Table 4.1.3: 1). This consistency in effect size corroborates the findings of the pivotal trial.

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Table 4.1.3: 1 Change from baseline in HbA_{1c} [%] in the studies that investigated linagliptin monotherapy (EFF-4 and 1218.46) - FAS (LOCF)

Study/ treatment group	Number of patients	Baseline HbA_{1c}, mean (SD)	Change from baseline in HbA_{1c}		Difference to placebo		
			Mean (SD)	Adjusted mean (SE)	Adjusted mean (SE)	95% CI	p-value
1218.5/ [a]				**Endpoint assessed after 12 weeks**			
Placebo	63	8.27 (0.63)	0.19 (0.81)	0.18 (0.09)			
Lina5 [b]	54	8.38 (0.79)	-0.29 (0.67)	-0.28 (0.10)	-0.47 (0.14)	(-0.74, -0.19)	0.0011
1218.23/ [c]				**Endpoint assessed after 12 weeks**			
Placebo	80	7.95 (0.67)	0.39 (0.92)	0.66 (0.08)			
Lina5 [b]	159	8.07 (0.66)	-0.49 (0.72)	-0.22 (0.07)	-0.88 (0.09)	(-1.06, -0.69)	<0.0001
1218.50/ [d]				**Endpoint assessed after 18 weeks**			
Placebo	73	8.06 (0.89)	0.25 (1.06)	0.06 (0.22)			
Lina5	147	8.11 (0.95)	-0.33 (1.03)	-0.49 (0.21)	-0.55 (0.14)	(-0.83, -0.27)	0.0001
1218.16/ [c]				**Endpoint assessed after 24 weeks**			
Placebo	163	8.00 (0.86)	0.22 (1.07)	0.25 (0.07)			
Lina5	333	8.00 (0.87)	-0.46 (0.81)	-0.44 (0.05)	-0.69 (0.08)	(-0.85, -0.53)	<0.0001
1218.46/ [e]				**Endpoint assessed after 24 weeks**			
Placebo	65	8.67 (0.95)	0.13 (1.17)	0.13 (0.11)			
Lina5	135	8.70 (0.97)	-0.43 (1.05)	-0.45 (0.08)	-0.57 (0.14)	(-0.85, -0.30)	<0.0001

Lina5 = linagliptin 5 mg qd
[a] Model includes baseline HbA_{1c} and treatment.
[b] Patients receiving linagliptin 5 mg only
[c] Model includes baseline HbA_{1c}, number of prior OADs, and treatment
[d] Model includes baseline HbA_{1c}, number of prior OADs, reason for ineligibility for metformin, and treatment
[e] Model includes continuous baseline HbA_{1c}, prior use of antidiabetic drugs, and treatment

Source data: [U10-1608, Module 5.3.5.3, Table 4.5.1.1, U10-2372, Module 5.3.5.1, Tables 15.2.1.2.1: 4 and 15.2.1.2.2.1: 2]

Linagliptin was also superior to placebo in reducing FPG levels from baseline in all studies conducted with linagliptin as monotherapy. The placebo-corrected treatment differences ranged from -15.12 mg/dL in the 12-week study 1218.5 to -23.34 mg/dL in the 24-week phase III-study 1218.16 (see Table 4.1.3: 2). These results indicate that the treatment effect of linagliptin monotherapy was consistent across the studies.

Table 4.1.3: 2 Change from baseline in FPG [mg/dL] in the studies that investigated linagliptin monotherapy (EFF-4 and 1218.46) - FAS (LOCF)

Study/ treatment group	Number of patients [a]	Baseline FPG, mean (SD)	Change from baseline in FPG		Difference to placebo		
			Mean (SD)	Adjusted mean (SE)	Adjusted mean (SE)	95% CI	p-value
EFF-4 /							
1218.5/ [b]				**Endpoint assessed after 12 weeks**			
Placebo	63	181.22 (33.93)	6.90 (34.13)	5.92 (4.30)			
Lina5 [c]	54	190.81 (40.64)	-10.35 (39.99)	-9.20 (4.65)	-15.12 (6.36)	(-27.72, -2.51)	0.0192
1218.23/ [d]				**Endpoint assessed after 12 weeks**			
Placebo	80	161.71 (30.65)	5.45 (24.40)	8.68 (2.86)			
Lina5 [c]	159	163.33 (31.80)	-14.86 (24.38)	-11.51 (2.26)	-20.19 (3.16)	(-26.41, -13.96)	<0.0001

(Table continued on next page)

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Table 4.1.3: 2 Change from baseline in FPG [mg/dL] in the studies that investigated linagliptin monotherapy (EFF-4 and 1218.46) - FAS (LOCF) - continued

Study/ treatment group	Number of patients [a]	Baseline FPG, mean (SD)	Change from baseline in FPG		Difference to placebo		
			Mean (SD)	Adjusted mean (SE)	Adjusted mean (SE)	95% CI	p-value
1218.50/ [e]				**Endpoint assessed after 18 weeks**			
Placebo	66	175.64 (36.65)	7.21 (35.55)	8.58 (8.22)			
Lina5	138	178.41 (42.26)	-14.30 (42.00)	-12.18 (7.84)	-20.76 (5.33)	(-31.27, -10.25)	0.0001
1218.16/ [d]				**Endpoint assessed after 24 weeks**			
Placebo	149	165.72 (37.69)	13.28 (42.84)	14.86 (2.98)			
Lina5	318	163.86 (41.58)	-8.53 (37.43)	-8.48 (2.03)	-23.34 (3.59)	(-30.40, -16.29)	<0.0001
1218.46/ [f]				**Endpoint assessed after 24 weeks**			
Placebo	61	203.30 (51.53)	6.02 (55.80)	10.16 (5.26)			
Lina5	134	195.25 (50.23)	-7.98 (46.83)	-8.58 (3.55)	-18.74 (6.35)	(-31.21, -6.27)	0.0033

[a] Patients in the FAS with a baseline FPG value and at least 1 on-treatment FPG value
[b] Model includes baseline HbA_{1c}, baseline FPG, and treatment
[c] Patients receiving linagliptin 5 mg only
[d] Model includes baseline HbA_{1c}, baseline FPG, number of prior OADs, and treatment
[e] Model includes baseline HbA_{1c}, baseline FPG, number of prior OADs, reason for ineligibility for metformin, and treatment
[f] Model includes baseline $HbA_{1c,}$ baseline FPG, prior antidiabetic drugs, and treatment

Source data: [U10-1608, Module 5.3.5.3, Table 4.5.2.1; U10-2372, Module 5.3.5.1, Tables 15.2.2.1.1: 2 and 15.2.2.1.1: 7]

4.2 EFFICACY OF METFORMIN MONOTHERAPY

Antidiabetic therapy with metformin is associated with dose-related reductions in HbA_{1c} and FPG. According to literature data, treatment with metformin 500 to 2000 mg per day decreases HbA_{1c} levels by 0.4% to 2.0% and decreases FPG levels by 5.4 to 68.4 mg/dL [P97-1130, R10-0861, Module 2.7.5].

During clinical development of the combination the linagliptin/metformin FDC, no dedicated studies investigating the efficacy of metformin monotherapy were conducted. However, efficacy data for metformin monotherapy are available from the placebo-controlled trials for the combination therapy with linagliptin+metformin, where metformin was taken either as background therapy as in studies 1218.17 and 1218.62 or as study medication as in study 1218.46 (metformin 500 mg bid or 1000 mg bid). Placebo-controlled efficacy data for metformin monotherapy from study 1218.46 are shown in the Table 4.2: 1, below. After 24 weeks of treatment, both metformin doses led to reductions in HbA_{1c} and FPG that were consistent with literature data.

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Table 4.2: 1 Change from baseline in HbA_{1c} [%] and FPG [mg/dL] for metformin groups in study 1218.46 - FAS (LOCF)

Study/ treatment group	Number of patients	Baseline, mean (SD)	Change from baseline		Difference to placebo		
			Mean (SD)	Adjusted mean (SE)	Adjusted mean (SE)	95% CI	p-value
1218.46/ [a]			**Endpoint assessed after 24 weeks**				
HbA_{1c}							
Placebo	65	8.67 (0.95)	0.13 (1.17)	0.13 (0.11)			
Met500	141	8.66 (0.90)	-0.63 (1.05)	-0.64 (0.08)	-0.77 (0.14)	(-1.04, -0.50)	<0.0001
Met 1000	138	8.52 (0.87)	-1.02 (1.01)	-1.07 (0.08)	-1.20 (0.14)	(-1.47, -0.93)	<0.0001
FPG							
Placebo	61	203.3 (51.5)	6.0 (55.8)	10.2 (5.26)			
Met500	136	190.6 (46.6)	-13.4 (52.5)	-15.8 (3.52)	-26.0 (6.34)	(-38.4, -13.5)	<0.0001
Met 1000	132	190.6 (52.2)	-30.2 (42.9)	-32.2 (3.58)	-42.3 (6.37)	(-54.8, -29.8)	<0.0001

[a] Models include baseline FPG, continuous baseline HbA_{1c}, prior antidiabetic drugs, and treatment

Source data: [U11-1151, Module 5.3.5.3, Tables 4.3.1.4, 4.3.2.2]

4.3 EFFICACY OF COMBINATION THERAPY WITH LINAGLIPTIN AND METFORMIN

4.3.1 Clinical trial programme for combination therapy with linagliptin and metformin

Overall, 7 trials contributed to the demonstration of efficacy of the combination therapy with linagliptin and metformin. The pivotal placebo-controlled phase III-trial 1218.17 investigated the efficacy of linagliptin as add-on therapy to metformin over 24 weeks. Additional key trials for the establishment of efficacy were studies 1218.62 and 1218.20. The purpose of trial 1218.62, a placebo-controlled bridging study over 12 weeks, was to show comparable efficacy of the linagliptin 2.5 mg twice daily (bid) regimen and the linagliptin 5 mg qd regimen when added to metformin. Trial 1218.20 was an active-controlled phase III-trial that compared linagliptin 5 mg qd with glimepiride as add-on therapy to metformin over 2 years. These 3 key trials were supported by the phase III-study 1218.46, which tested the combination of linagliptin and metformin in a partial first-line setting, i.e. patients were either treatment-naïve (47.5%) or washed out one previously taken oral antidiabetic agent before randomisation.

Efficacy of linagliptin in combination with metformin and an SU over 24 weeks was investigated in the phase III-study 1218.18, where linagliptin 5 mg qd or placebo were added to ongoing therapy with metformin and an SU.

Long-term efficacy data are available from 2 extension trials. Study 1218.40 is an open-label extension trial in patients who completed the treatment periods of the 4 pivotal placebo/ background-controlled studies from the linagliptin mono programme. To show long-term efficacy of the combination therapy with linagliptin and metformin (and SU), data from the patients who completed studies 1218.17 or 1218.18 and continued in study 1218.40 are presented. Study 1218.52 is a double-blind metformin monotherapy-controlled extension trial in patients who completed the treatment period of study 1218.46 without requiring rescue medication. From studies 1218.20 and 1218.40, interim data were presented for the

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linagliptin mono submission. In this FDC submission, final (i.e. 2-year) data from study 1218.20 are included, and a new interim data cut-off was performed for study 1218.40. Not included in the evaluation of efficacy is the phase II study 1218.6, a 12-week dose-finding trial, which tested linagliptin qd doses of 1 mg, 5 mg, and 10 mg versus placebo as add-on therapy to metformin. The trial included 66 patients treated with linagliptin 5 mg qd and it was felt that the contribution to the overall evidence would be small, particular since the treatment duration was just 12 weeks. However, the results were consistent with other trials. The difference to placebo in adjusted mean change from baseline (SE) in HbA_{1c} was -0.73% (0.14), p<0.0001; for FPG the adjusted mean change was -34.7 mg/dL (6.2); further details are in the study report [U08-1056, Module 5.3.5.1].

Table 4.3.1: 1 Studies investigating the efficacy of combination therapy with linagliptin and metformin

Study number	Study characteristics	Comparison	Treatment duration	Number of patients [a]
1218.17	Double-blind, placebo-controlled	Linagliptin 5 mg qd versus placebo as add-on to metformin	24 weeks	688
1218.62	Double-blind, placebo-controlled	Linagliptin 2.5 mg bid versus linagliptin 5 mg qd versus placebo as add-on to metformin	12 weeks	478
1218.20	Double-blind, active-controlled	Linagliptin 5 mg qd versus glimepiride as add-on to metformin	104 weeks	1519
1218.46	Double-blind, placebo-controlled (additional open-label arm)	Linagliptin 2.5 mg bid + metformin 500 mg bid or linagliptin 2.5 mg bid + metformin 1000 mg bid versus metformin 500 mg bid or metformin 1000 mg bid or linagliptin 5 mg qd or placebo	24 weeks	756 [b]
1218.18	Double-blind, placebo-controlled	Linagliptin 5 mg qd versus placebo as add-on to metformin and SU	24 weeks	1040
1218.40	Open-label extension, uncontrolled	Linagliptin 5 mg qd	78 weeks [c]	1336 [d]
1218.52	Double-blind extension	Linagliptin 2.5 mg bid + metformin 500 mg bid versus linagliptin 2.5 mg bid + metformin 1000 mg bid versus metformin 1000 mg bid	54 weeks [c]	565 [e]

[a] Number of patients in the Full Analysis Set (FAS), the patient set used for the efficacy analyses
[b] An additional 66 patients were treated in the open-label arm of the study. These patients are not included in the FAS.
[c] Planned treatment duration. Interim data from this study are included in the submission.
[d] Patients from the initial studies 1218.17 and 1218.18 who continued treatment in study 1218.40
[e] Patients from initial trial who continued treatment in 1218.52

Source data: [U11-1151, Module 5.3.5.3, Table 1.2]

The 7 studies were arranged into 10 different efficacy groupings to provide the most appropriate basis for the analysis of efficacy (the letter ‘C’ in the name of the grouping denotes combination therapy with linagliptin and metformin). The groupings are summarised in Table 4.3.1: 2, below.

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Table 4.3.1: 2 Grouping of studies for the evaluation of the efficacy of combination therapy with linagliptin and metformin

Shorthand	Study characteristics and objectives	Studies	Treatment duration	Data status [a]
EFF-C1	Pivotal double-blind placebo-controlled efficacy study with metformin background	1218.17	24 weeks	as previously submitted
EFF-C2	Double-blind active-controlled 2-year efficacy study with metformin background	1218.20	104 weeks	updated data
EFF-C3	Supportive double-blind placebo-controlled efficacy study with combination of linagliptin and metformin in a first-line therapy setting	1218.46	24 weeks	new data
EFF-C4	Double-blind placebo-controlled efficacy study with metformin and SU background	1218.18	24 weeks	as previously submitted
EFF-C5	Double-blind placebo-controlled efficacy studies (pooled to increase sample size for subgroup analyses)	1218.17 1218.46	24 weeks	new data
	Additional subgroup analyses by metformin background dose:			
EFF-C6a	Double-blind placebo-controlled efficacy study with metformin background (metformin 850 mg and 1000 mg bid)	1218.17	24 weeks	based on previously submitted
EFF-C6b	Double-blind active-controlled efficacy study with metformin background (metformin 850 mg and 1000 mg bid)	1218.20	104 weeks	data (EFF-C6b: based on updated
EFF-C6c	Double-blind placebo-controlled efficacy study with metformin (850 mg and 1000 mg bid) and SU background	1218.18	24 weeks	data)
EFF-C7a	Double-blind placebo-controlled efficacy study with metformin background	1218.62	12 weeks	new data
	Additional subgroup analyses by metformin background dose:			
EFF-C7b	Double-blind placebo-controlled efficacy study with metformin background (metformin 850 mg and 1000 mg bid)	1218.62	12 weeks	
EFF-C8	Studies to assess long-term efficacy of linagliptin with metformin background (combined) [b]	1218.17 + 1218.40	24 weeks + 78 weeks [c]	updated data
EFF-C9	Studies to assess long-term efficacy of linagliptin plus metformin (combined) [b]	1218.46 + 1218.52	24 weeks + 52 weeks [d]	new data
EFF-C10	Studies to assess long-term efficacy of linagliptin with metformin and SU background (combined) [b]	1218.18 + 1218.40	24 weeks + 78 weeks [c]	updated data

[a] Referring to the data included in the linagliptin mono SCE
[b] Combined patient data from the initial study and the extension study
[c] The planned total treatment duration of study 1218.40 is 78 weeks. Since interim data from this study are included in this submission, not all patients have completed the treatment period.
[d] The planned total treatment duration of study 1218.52 is 52 weeks. Since interim data from this study are included in this submission, not all patients have completed the treatment period.

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4.3.2 Disposition, demographics, and baseline characteristics

Studies conducted for the linagliptin/metformin FDC programme as well as studies conducted as part of the linagliptin monotherapy programme were considered for the evaluation of the efficacy of combination therapy with linagliptin and metformin. Long-term efficacy was assessed by combining the parent studies 1218.17 and 1218.18 with the open-label extension trial 1218.40 to reach treatment durations of up to 102 weeks (EFF-C8 and EFF-C10). Additional information on long-term treatment is available from study 1218.46 combined with its double-blind extension study 1218.52 for treatments of up to 78 weeks (EFF-C9). In the following section disposition information is based on the FAS sets.

The durations of the combination therapy trials ranged from 12 weeks (1218.62) to 104 weeks (1218.20). The percentage of randomised patients who completed the planned treatment time was above 90% in all trials except study 1218.20 (78.4%), possibly because of the longer duration of the trial. The percentages of premature discontinuation in the Lina+Met groups were below 7.5% in all placebo-controlled trials and the incidences were generally lower in the Lina+Met groups than in the Met groups except for study 1218.62. The most frequent reasons for premature discontinuations in the Lina+Met groups were administrative reasons (1.4% to 4.4% across trials) and adverse events (0.9% to 2.5% across trials).

In the combined studies for long-term efficacy assessment, 85.0% or more of the patients did not prematurely discontinue their study treatment. In the extensions of trials 1218.17 (EFF-C8) and 1218.18 (EFF-C10), the frequencies of premature discontinuations were between 10.5% and 12.9% across treatment groups. The discontinuation rates were comparable between those patients who had received linagliptin already in the initial trial and those patients who had switched to linagliptin with the entry in the follow-up trial. In the combined 1218.46 and 1218.52 studies (EFF-C9), the percentage of patients who discontinued prematurely were in a similar range with 7.3% (Met1000 bid), 10.8% (Lina2.5 mg+Met1000 bid), and 15.0% (Lina2.5 mg+Met 500).

The key demographic characteristics of patients were balanced between treatment groups for the studies that investigated the efficacy of linagliptin and metformin combination therapy. Across these studies, the mean age of patients was similar (range 55.2 to 59.9 years). The percentage of male patients varied between 47.2% and 60.1%. In all studies but 1218.18, the majority of patients were of White race, with the greatest percentage of White patients in study 1218.20 (84.7%). In study 1218.18, 52.7% of patients were of Asian race. As a consequence of the different race composition, the total mean BMI was lowest in study 1218.18 (28.3 kg/m^2) and highest in study 1218.20 (30.3 kg/m^2).

Mean baseline HbA_{1c} was in the range of 7.7% to 8.7%; mean FPG at baseline was between 158.7 mg/dL and 203.7 mg/dL. The percentages of patients diagnosed with diabetes for more than 5 years differed between trials (19.6% to 73.9%). In studies 1218.17 (EFF-C1), 1218.20 (EFF-C2), and 1218.62 (EFF-C7a), about 50% of patients had been diagnosed with diabetes for more than 5 years. All studies except the 1218.46 recruited only patients who had previously been treated with OADs: Study 1218.46 was performed in a first-line setting and 47.5% of patients were treatment-naïve. Patients with 1 prior OAD were not eligible for the triple combination study 1218.18; in the other studies, the percentage of patients who had received 1 prior OAD ranged between 50.4% and 74.4%. Within the individual studies, the

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key baseline characteristics were balanced between the treatment groups except for the lower proportion of patients diagnosed with diabetes for more than 5 years in study 1218.46.

Table 4.3.2: 1 Key baseline characteristics of patients in the study groupings to assess efficacy of combination therapy with linagliptin and metformin - FAS

Study (grouping)/ treatment groups	Number of patients	Baseline HbA_{1c}, mean (SD) [%]	Baseline FPG, mean (SD) [mg/dL]	Time since diagnosis of diabetes, N (%)	Number of antidiabetic drugs at enrolment, N (%)		
				>5 years	None	One	Two or more
1218.17 (EFF-C1)/							
Met	175	8.0 (0.9)	166.4 (41.9)	93 (53.1)	0 (0.0)	121 (69.1)	54 (30.9)
Lina5+Met	513	8.1 (0.9)	169.6 (43.5)	285 (55.6)	0 (0.0)	351 (68.4)	162 (31.6)
1218.20 (EFF-C2)/							
Lina5+Met	764	7.7 (0.9)	164.5 (42.8)	398 (52.1)	0 (0.0)	535 (70.0)	229 (30.0)
Glim+Met	755	7.7 (0.9)	166.1 (42.0)	406 (53.8)	0 (0.0)	532 (70.5)	223 (29.5)
1218.46 (EFF-C3)/ [a]							
Placebo	65	8.7 (0.9)	203.7 (51.2)	22 (33.8)	32 (49.2)	33 (50.8)	0 (0.0)
Lina5	135	8.7 (1.0)	195.3 (50.2)	34 (25.2)	61 (45.2)	74 (54.8)	0 (0.0)
M500 bid	141	8.7 (0.9)	191.2 (46.9)	33 (23.4)	69 (48.9)	71 (50.4)	1 (0.7)
M1000 bid	138	8.5 (0.9)	192.3 (52.8)	27 (19.6)	67 (48.6)	71 (51.4)	0 (0.0)
L2.5+M500 bid	137	8.7 (1.0)	198.6 (60.2)	39 (28.5)	65 (47.4)	72 (52.6)	0 (0.0)
L2.5+M1000 bid	140	8.7 (1.0)	196.9 (51.2)	39 (27.9)	65 (46.4)	75 (53.6)	0 (0.0)
1218.18 (EFF-C4)/							
Met+SU	262	8.1 (0.8)	162.6 (37.1)	193 (73.7)	0 (0.0)	0 (0.0)	262 (100.0)
Lina5+Met+SU	778	8.1 (0.8)	159.3 (36.5)	569 (73.1)	0 (0.0)	0 (0.0)	778 (100.0)
1218.62 (EFF-C7a)/ [b]							
Met	43	7.9 (0.7)	166.2 (40.3)	24 (55.8)	0 (0.0)	32 (74.4)	11 (25.6)
Lina2.5+Met	214	8.0 (0.8)	164.1 (41.8)	112 (52.3)	0 (0.0)	152 (71.0)	62 (29.0)
Lina5+Met	221	8.0 (0.7)	165.8 (35.1)	115 (52.0)	0 (0.0)	157 (71.0)	64 (29.0)
1218.17 + .40 (EFF-C8)/ [c]							
Met	153	8.0 (0.9)	167.3 (41.9)	78 (51.0)	0 (0.0)	110 (71.9)	43 (28.1)
Lina5+Met	457	8.1 (0.9)	169.6 (43.8)	255 (55.8)	0 (0.0)	311 (68.1)	146 (31.9)
1218.46 + .52 (EFF-C9)/							
M1000 bid	109	8.5 (0.9)	185.5 (46.3)	18 (16.5)	57 (52.3)	52 (47.7)	0 (0.0)
L2.5+M500 bid	113	8.6 (0.9)	193.5 (55.2)	30 (26.5)	55 (48.7)	58 (51.3)	0 (0.0)
L2.5+M1000 bid	111	8.6 (1.0)	191.1 (44.4)	26 (23.4)	54 (48.6)	57 (51.4)	0 (0.0)
1218.18 + .40 (EFF-C10)/ [b]							
Met+SU	182	8.1 (0.8)	159.9 (34.7)	130 (71.4)	0 (0.0)	0 (0.0)	182 (100.0)
Lina5+Met+SU	544	8.1 (0.8)	158.7 (35.8)	402 (73.9)	0 (0.0)	0 (0.0)	544 (100.0)

Met = metformin as background therapy; SU = sulphonylurea as background therapy; Lina5 = linagliptin 5 mg qd; L2.5 = Linagliptin 2.5 mg bid; Glim = glimepiride; M 500/1000 bid = Metformin 500 mg/100 mg bid

EFF-C5 is not displayed as this pooling of EFF-C1 and EFF-C3 was created to provide the basis for subgroup analysis.

EFFC6 a) to c) are not displayed because they represent subgroup analysis of EFF-C1, EFF-C2, and EFF-C4 by metformin posology of 850 mg bid and 1000 mg bid.

[a] Baseline characteristics of the patients who received open-label treatment with linagliptin 2.5 mg bid + metformin 1000 mg bid are provided in [U11-1151, Module 5.3.5.3, Table 3.2.3.2].

[b] Comprises all patients irrespective of the metformin dose they received.

[c] All patients received linagliptin in study 1218.40. The treatment groups displayed refer to the treatment in the initial trials.

Source data: [U11-1151, Module 5.3.5.3, Tables 3.2.1.2, 3.2.2.11, 3.2.3.3, 3.2.4.2, 3.2.7.11, 3.2.9.1, 3.2.10.1, and 3.2.11.1]

Since the study groupings EFF-C8, EFF-C9, and EFF-C10 are based on data from combined studies, the baseline characteristics described represent the baseline values from the initial

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trials (1218.17, 1218.46, and 1218.18), and not the baseline values from the extension studies (1218.40 and 1218.52). In general, the baseline characteristics of the patients who later continued treatment in the extension studies 1218.40 and 1218.52 were comparable with the overall population of the initial trials.

4.3.3 Efficacy results of combination therapy with linagliptin and metformin

4.3.3.1 Changes from baseline in HbA_{1c}

Changes from baseline in HbA_{1c} in the pivotal study 1218.17 (EFF-C1)

Linagliptin ('Lina') added to ongoing metformin ('Met') background therapy was superior to metformin alone in regard to the HbA_{1c} change from baseline after 24 weeks, with a clinically meaningful adjusted mean treatment difference of -0.64%, p<0.0001. This result was corroborated by sensitivity analyses using different imputation methods for missing data. The primary analysis included all metformin doses and posologies; an analysis by metformin dose is presented in Section 4.4.3.

Table 4.3.3.1: 1 Change from baseline in HbA_{1c} [%] after 24 weeks in study 1218.17 (EFF-C1) - FAS (LOCF)

Treatment group	Number of patients, N (%)	HbA_{1c} baseline, mean (SD)	Change in HbA_{1c} from baseline		Difference to Met		
			Mean (SD)	Adjusted [a] mean (SE)	Adjusted [a] mean (SE)	95% CI	p-value
Met	175 (25.4)	8.02 (0.88)	0.10 (1.00)	0.15 (0.06)			
Lina+Met	513 (74.6)	8.09 (0.86)	-0.56 (0.83)	-0.49 (0.04)	-0.64 (0.07)	(-0.78, -0.50)	<0.0001

Met = metformin background therapy; Lina5 = linagliptin 5 mg qd
a Model includes continuous baseline HbA_{1c}, prior use of antidiabetic drugs, and treatment

Source data: [U11-1151, Module 5.3.5.3, Table 4.1.1.1]

Additional evidence for the efficacy of combination therapy with linagliptin and metformin

Additional evidence for the efficacy of combination therapy with Lina+Met is derived from 3 large trials: study 1218.62 compared Lina 2.5 mg bid with Lina 5 mg qd and with placebo as add-on to metformin over 12 weeks, study 1218.46 evaluated 2 different dose combinations of Lina and Met over 24 weeks, and study 1218.20 tested Lina 5 mg qd versus glimepiride ('Glim') 1 to 4 mg qd over 104 weeks on a background of metformin. The analyses presented in this section were not stratified for metformin dose or for metformin posology; analyses by metformin dose are available in Section 4.4.3.

Study 1218.62 had a non-inferiority design that employed a hierarchical testing procedure. Step 1 was a test for superiority of Lina 2.5 mg bid over placebo; if successful, testing continued (Step 2) with a non-inferiority test of Lina 2.5 mg bid versus Lina 5 mg qd using a pre-specified non-inferiority margin of 0.35%. If successful, testing concluded in Step 3 with a test for superiority of Lina 5 mg qd over placebo. The adjusted mean treatment difference in HbA_{1c} change from baseline to Week 12 for Lina 2.5 bid over placebo was -0.74% (95% CI -0.97, -0.52), p<0.0001. The adjusted mean treatment difference for Lina 2.5 mg bid vs. Lina 5 mg qd was 0.06% (95% CI -0.07, 0.19), indicating non-inferiority of Lina 2.5 mg

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bid to Lina 5 mg qd as the upper bound of the 95% confidence interval was below the prespecified margin of 0.35%. For Lina 5 mg over placebo the adjusted mean difference was -0.80% (95% CI -1.02, -0.58), p<0.0001.

Study 1218.46 was conducted in a first-line setting (47.5% treatment-naïve patients) and yielded qualitatively similar results to the pivotal trial. The Lina+Met groups were superior to both Met monotherapy groups and also to linagliptin monotherapy with p-values below 0.0001.

Across all studies, including the pivotal trial 1218.17, the additional reductions in HbA_{1c} achieved with combination therapies with linagliptin and metformin over metformin monotherapy (as background or as study treatment) were in the range of 0.51% to 0.64%. Hence all trials demonstrated a clinically meaningful reduction in HbA_{1c} with combination therapy of linagliptin and metformin.

Study 1218.20 was also designed as a non-inferiority trial. After 104 weeks of treatment, 4 confirmatory hypotheses were to be tested sequentially at α=0.025 (2-sided; alpha-adjustment due to interim analysis after 52 weeks): (1) non-inferiority of Lina versus glimepiride in the change in HbA_{1c} from baseline (pre-defined non-inferiority margin: 0.35%), (2) superiority of Lina over glimepiride for the change in HbA_{1c} from baseline, (3) superiority of Lina over glimepiride for the change in body weight from baseline, and (4) superiority of Lina over glimepiride in the occurrence of hypoglycaemic events. After 104 weeks of treatment, a mean HbA_{1c} decrease from baseline was observed in both treatment groups (Lina+Met -0.16% vs. Glim+Met -0.36%; adjusted mean difference 0.20%). The 2-sided 97.5% CI (0.09%, 0.299%) for the treatment difference excluded both the pre-specified non-inferiority margin of 0.35% and the stricter margin of 0.30%. Thus, Lina+Met was non-inferior to Glim+Met. This notwithstanding, superiority of linagliptin over glimepiride could not be shown. Therefore, the remaining 2 hypotheses of the sequential testing procedure could not be tested in a confirmatory way. A number of sensitivity analyses were performed for the primary endpoint of HbA_{1c} change from baseline. The results of the sensitivity analyses were concordant with the results for the primary analysis as the 97.5% CIs excluded the non-inferiority margin of 0.35%. Only in the PPS-LOCF-analysis (n=1160), which excluded patients with protocol violations that possibly impacted on the efficacy outcomes, the upper bound of the 97.5% CI was above the margin (0.40%). However, a more pertinent analysis concentrating on long-term efficacy and therefore based on the PPS-completers (OC, n=905)), i.e. those patients in the PPS that were treated with study medication for at least 684 days, resulted in a 97.5% CI (0.07, 0.28) with the upper bound being well below the non-inferiority margin of 0.35% as well as below 0.30%. (Details on the definiton of analysis sets are available in the SCE [Module 2.7.3, Section 1.3.4])

The HbA_{1c} changes from baseline in the combination therapy trials are summarised in Table 4.3.3.1: 2 below.

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Table 4.3.3.1: 2 Changes from baseline in HbA_{1c} [%] in trials 1218.62, 1218.46. 1218.20 - FAS (LOCF)

	Change from baseline	Difference		
Study (Grouping) Treatment groups	Adjusted [a] mean (SE)	Adjusted [a] mean (SE)	95% CI	p-value
1218.62 (EFF-7a)		Difference between treatments		
		12 weeks treatment		
Met	0.28 (0.11)			
Lina 5 mg qd + Met vs Met	-0.52 (0.05)	-0.80 (0.11)	(-1.02, -0.58)	<0.0001
Lina 2.5 mg bid + Met vs Met	-0.46 (0.05)	-0.74 (0.11)	(-0.97, -0.52)	<0.0001
Lina 2.5 mg bid + Met vs Lina 5mg qd + Met		0.06 (0.07)	(-0.07, 0.19)	0.3834
1218.46 (EFF-C3)		Difference between combination and components		
		24 weeks treatment		
Placebo	0.13 (0.11)			
Lina 2.5 mg bid + Met 500 mg bid	-1.22 (0.08)			
Met 500 mg bid	-0.64 (0.08)	-0.58 (0.11)[b]	(-0.79, -0.36)	<0.0001
Lina 5 mg qd	-0.45 (0.08)	-0.77 (0.11)[c]	(-0.99, -0.55)	<0.0001
Lina 2.5 mg bid + Met 1000 mg bid	-1.59 (0.08)			
Met 1000 mg bid	-1.07 (0.08)	-0.51 (0.11)[d]	(-0.73, -0.30)	<0.0001
Lina 5 mg qd	-0.45 (0.08)	-1.14 (0.11)[e]	(-1.36, -0.92)	<0.0001
1218.20 (EFF-C2)		Difference between treatments		
		104 weeks of treatment		
Lina 5 mg + Met vs Glim+Met	-0.16 (0.03)	0.20 (0.05)	(0.09, 0.30)[g]	0.0004 [f]
Glim + Met	-0.36 (0.03)			

Glim = glimepiride; Met = metformin; Lina 5 mg, Lina = linagliptin 5 mg; Lina 2.5 mg = linagliptin 2.5 mg
[a] Model includes continuous baseline HbA_{1c}, prior use of antidiabetic drugs, and treatment
[b] Difference between Lina 2.5 mg + Met 500 mg bid and Met 500 mg bid;
[c] Difference between Lina 2.5 mg + Met 500 mg bid and Lina 5 mg qd
[d] Difference between Lina 2.5 mg + Met 1000 mg bid and Met 1000 mg bid
[e] Difference between Lina 2.5 mg + Met 1000 mg bid and Lina 5 mg qd
[f] 1-sided p-value (non-inferiority test at margin 0.35%).
[g] For study 1218.20 the 97.5% confidence intervals were calculated.

Source data: [U11-1151, Module 5.3.5.3, Tables 4.2.1.1, 4.3.1.1, 4.7.1.1]

4.3.3.2 Changes from baseline in FPG

The change from baseline in FPG was a secondary endpoint in all trials. Combination therapy with Lina+Met was more efficacious in the reduction of FPG than metformin monotherapy (background or study treatment) and linagliptin monotherapy. In the pivotal study 1218.17, the addition of linagliptin to ongoing metformin therapy resulted in an adjusted mean difference in the change from baseline in FPG of -21.1 mg/dL after 24 weeks of treatment. Similarly, in study 1218.62, the Lina+Met groups showed a clinically meaningful reduction in FPG with adjusted mean differences to Met of -17.8 mg/dL (Lina 5 mg qd+Met) and -13.7 mg/dL (Lina 2.5 mg bid+Met) after 12 weeks of treatment.
In study 1218.46, the reductions in FPG were similar in both Lina+Met combination therapy groups and were larger than in the respective metformin groups. Also, as for the HbA_{1c} change from baseline, both combination therapies were more efficacious in lowering FPG

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than Lina 5 mg once daily. Thus the analysis of the mean change from baseline in FPG corroborates the results observed for HbA_{1c}, providing further evidence for the efficacy of combination therapies of linagliptin + metformin.

In study 1218.20, consistent with the HbA_{1c} decrease from baseline, a mean reduction in FPG after 104 weeks was observed in both treatment groups. The mean decrease in FPG was less pronounced for Lina+Met than for Glim+Met, with a treatment difference in the adjusted mean FPG change from baseline of 6.4 mg/dL (p=0.0012). Lina+Met provided a smaller reduction in FPG levels than Glim+Met after 104 weeks of treatment. However, the analysis of 2hPPG over time showed that the mean change from baseline in 2hPPG tended to be larger for Lina+Met than for Glim+Met after 104 weeks of treatment.

Table 4.3.3.2: 1 Changes from baseline in FPG [mg/dL] in trials 1218.17, 1218.62, 1218.46, 1218.20 - FAS (LOCF)

	Change from baseline	Difference		
Study (Grouping) Treatment groups	Adjusted [a] mean (SE)	Adjusted [a] mean (SE)	95% CI	p-value
1218.17 (EFF-C1)		Difference between treatments **24 weeks treatment**		
Metformin	10.5 (2.80)			
Lina 5 mg qd + Met vs Met	-10.7 (1.65)	-21.1 (3.14)	(-27.3, -15.0)	<0.0001
1218.62 (EFF-C7a)		Difference between treatments **12 weeks treatment**		
Metformin	-3.5 (4.24)			
Lina 5 mg qd + Met vs Met	-21.3 (1.92)	-17.8 (4.55)	(-26.7, -8.8)	0.0001
Lina 2.5 mg bid + Met vs Met	-17.2 (1.94)	-13.7 (4.57)	(-22.7, -4.7)	0.0029
Lina 2.5 mg bid + Met vs Lina 5mg qd + Met		4.1 (2.59)	(-1.0, 9.2)	0.1142
1218.46 (EFF-C3)		Difference between combination and components **24 weeeks treatment**		
Placebo	10.2 (5.26)			
Lina 2.5 mg bid + Met 500 mg bid	-33.2 (3.54)			
Met 500 mg bid	-15.8 (3.52)	-17.4 (4.99) [b]	(-27.2, -7.6)	0.0005
Lina 5 mg qd	-8.6 (3.55)	-24.6 (5.01) [c]	(-34.4, -14.8)	<0.0001
Lina 2.5 mg bid + Met 1000 mg bid	-49.4 (3.52)			
Met 1000 mg bid	-32.2 (3.58)	-17.2 (5.02) [d]	(-27.1, -7.3)	0.0006
Lina 5 mg qd	-8.6 (3.55)	-40.8 (5.00) [e]	(-50.6, -31.0)	<0.0001
1218.20 (EFF-C2)		Difference between treatments **104 weeks treatment**		
Lina5+Met vs Glim+Met	-2.4 (1.46)	6.4 (1.97)	(2.5, 10.3)	0.0012
Glim +Met	-8.8 (1.47)			

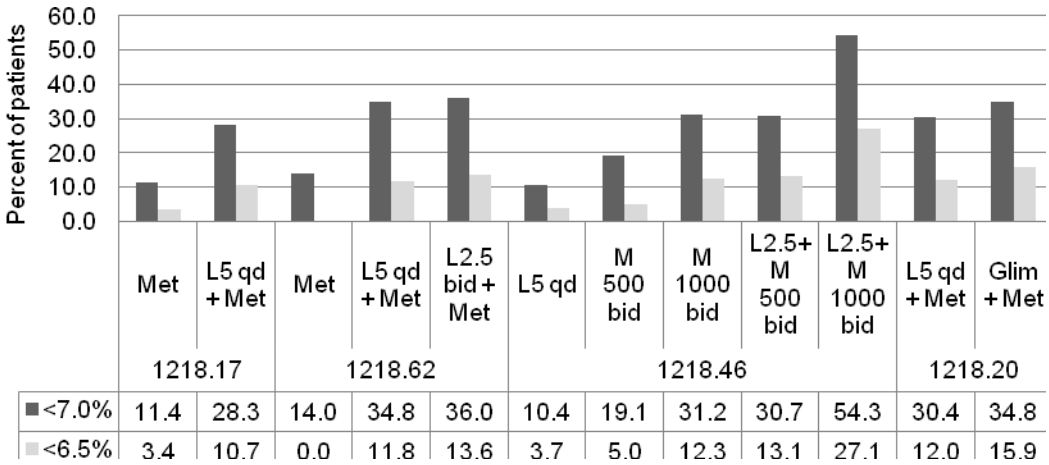
Glim=glimepiride; Met = metformin; Lina 5 mg = linagliptin 5 mg; Lina 2.5 mg = linagliptin 2.5 mg
[a] Model includes continuous baseline HbA_{1c},continuous baseline FPG, prior use of antidiabetic drugs, and treatment
[b] Difference between Lina 2.5 mg + Met 500 mg bid and Met 500 mg bid
[c] Difference between Lina 2.5 mg + Met 500 mg bid and Lina 5 mg qd
[d] Difference between Lina 2.5 mg + Met 1000 mg bid and Met 1000 mg bid
[e] Difference between Lina 2.5 mg + Met 1000 mg bid and Lina 5 mg qd

Source: [U11-1151, Module 5.3.5.3, Tables 4.1.2.1, 4.2.2.1, 4.3.2.1, 4.7.2.1]

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4.3.3.3 'Treat-to-target' analyses: proportions of patients with HbA_{1c} below 7.0% or 6.5%

The proportion of patients who achieved HbA_{1c} levels below 7.0% or below 6.5% was an additional endpoint in all studies. For this analysis, missing values were imputed using the non-completers considered failure (NCF) approach, i.e. missing data due to premature discontinuation or values obtained after the introduction of rescue therapy were considered as failure to attain the pre-specified criterion up to the last planned visit. In all placebo-controlled trials, the proportion of patients who achieved glycemic control (HbA_{1c} <7% or <6.5%) was higher with the combination of Lina+Met than with metformin mono therapy. In study 1218.20, the proportions of patients achieving glycaemic control was slightly higher in the Glim+Met group.



Met = metformin, L5 = linagliptin 5 mg; L2.5 = linagliptin 2.5 mg; Glim = glimepiride

Figure 4.3.3.3: 1: Percentage of patients achieving target HbA_{1c} levels below 7% and below 6.5% in studies 1218.17, 1218.62, 1218.46, and 1218.20 - FAS-NCF

Source: [U11-1151, Module 5.3.5.3, Tables 4.1.7.1, 4.3.7.1, 4.7.3.1]

4.3.3.4 Other measures of efficacy: Post-prandial glucose (2hPPG), use of rescue therapy, body weight, and HOMA indices

For the pivotal trial 1218.17, the reduction in post-prandial glucose levels (2hPPG) showed a significant difference in favour of combination therapy of Lina+Met over Met background therapy (adjusted mean 2hPPG change -67.1 mg/dL; 95% CI -94.7, -39.6; p <0.0001). Post-prandial glucose levels were also measured in trial 1218.46 but due to the small number of patients in the placebo group in the MTT-set in this study, the data is inconclusive.

In study 1218.20, the adjusted mean change from baseline in 2hPPG at Week 104 was numerically larger for patients receiving Lina+Met (-28.5 mg/dL) than for patients receiving Glim+Met (-18.7 mg/dL), even though the treatment difference of -9.7 mg/dL was not statistically significant (95% CI -21.1, 1.6; p=0.0918).

In the pivotal trial 1218.17, the use of rescue therapy was substantially lower in the Lina+Met group than in the Met group (7.8% vs. 18.9%). This was also consistently observed in the other trials: the use of rescue medication was always lower in the combination therapy groups

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than in monotherapy groups. This agrees with the analysis of patients achieving glycaemic control where the combination therapy groups exhibited higher proportions of patients achieving HbA_{1c} levels below 7% or 6.5% (Figure 4.3.3.3: 1, above).

In study 1218.17, the effect of combination therapy was shown to be neutral in regard to body weight, since the mean body weight decreased by 0.4 kg in both treatment groups over the 24 weeks of treatment. Also in the other trials only minor differences between the combination therapy groups and metformin mono therapy were observed.

In the long-term active-controlled study 1218.20, after 104 weeks of treatment, a decrease in body weight was noted in the Lina+Met group as opposed to a weight gain in the Glim+Met group (adjusted mean changes -1.4 kg vs. +1.3 kg). Thus, a clinically meaningful treatment difference of -2.7 kg ($p < 0.0001$) was observed between the Lina+Met and Glim+Met groups.

There was no difference in study 1218.17 between Lina+Met and Met in regards to the change in insulin resistance (HOMA-IR). The evaluation of HOMA-%B indicated an improvement in ß-cell function in patients receiving Lina+Met (mean change from baseline 1.21 mU/mmol) but not in those receiving metformin (-10.67 mU/mmol); p=0.1177.

4.3.3.5 Efficacy in subpopulations

To obtain reasonable group sizes for the evaluation of subgroups, the 24-week trials 1218.17 and 1218.46 were pooled (EFF-C5, n=1244); the analysis was confined to the change from baseline in HbA_{1c}. Data on disposition, demographics and baseline characteristics in subgroups are available in the SCE [Module 2.7.3, Section 3.1.3].

For most subgroups investigated, the treatment effect was consistent and the achieved changes from baseline in HbA_{1c} were comparable across all subcategories (Table 4.3.3.5: 1). Thus, the factors age, gender, ethnicity, geographical region, baseline BMI, and baseline HbA_{1c} did not have an influence on the efficacy of combination therapy with Lina+Met. The treatment effect of Lina+Met compared with Met was almost identical in Europe (n=442) and in Asia (n=446) with HbA_{1c} changes of -0.56% (Europe) and -0.59% (Asia), and p-values below 0.0001. Likewise, for race there was no difference between White (-0.61%) and Asian (-0.57%) patients in the efficacy of Lina+Met compared with Met ($p<0.0001$ for both races).

Table 4.3.3.5: 1 Summary of change in HbA_{1c} in subgroups of EFF-C5

Subgroup name	Subcategories analysed	Possible differential treatment effect of Lina+Met across subcategories
Age	≤50 years vs. 51 to 64 years vs. 65 to 74 years vs. ≥75 years	no
Gender	male vs. female	no
Race	White vs. Black or African American vs. Asian	no
Ethnicity	Hispanic/Latino vs. Not Hispanic/Latino	no
Geographical region	Europe vs. North America vs. South America vs. Asia	no
Baseline BMI	<30 kg/m^2 vs. ≥30 kg/m^2	no
Baseline HbA_{1c}	<7% vs. 7.0% to <8.0% vs. 8.0% to <9.0% vs. ≥9.0%	no
Washout of prior OADs	yes vs. no	yes
Time since diagnosis of diabetes	≤1 year vs. >1 to 5 years vs. >5 years	yes

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For some features more closely associated with diabetes, such as presence of a washout of previously taken OADs and time since diagnosis of diabetes, apparent treatment-by-subgroup interactions were observed. The adjusted mean differences between the Lina+Met and the metformin groups was greater for those patients who did wash out prior OADs than for those without a washout (-0.70% vs. -0.52%), and also the p-value of 0.0782 indicated a subgroup-by-treatment interaction. For the time since diagnosis of diabetes, the adjusted mean changes from baseline in HbA_{1c} in the Lina+Met groups were between -1.07% and -1.18% across the categories analysed (≤1, >1 to ≤5, >5 years) while metformin mono treatment resulted in HbA_{1c} changes of only -0.74% (≤1 year), -0.56% (>1 to ≤5 years), and -0.33% (>5 years). Hence the differences between the treatments increased with the duration of the diagnosis and were largest in patients who were diagnosed for 5 years or longer (adjusted mean difference Lina+Met vs. Met -0.79%). Also the p-value of 0.011 indicated a subgroup-by-treatment effect. However, the analyses of the influence of washout of prior OADs and the duration of diagnosis are linked since the likelihood that a patient has received pre-treatment with OADs increases with the duration of diabetes. And indeed 64.3% of patients who were diagnosed for 5 years or longer received 1 OAD and 31.4% had received 2 OADs, compared with only 38.2% of patients with 1 OAD and 3.2% with 2 OADs among those patients diagnosed for up to 1 year. The majority of the patients who were pretreated had taken either metformin (54.3%) or a combination of Met+SU (15.0%). It is therefore not surprising that the efficacy of continued metformin therapy was lower in the group of patients who had been diagnosed for 5 years or longer but not in those patients with a more recent diagnosis. Taken together, it appears that the interactions between treatment, washout of OADs, and the duration of diagnosis are a consequence of the current antidiabetic treatment algorithms but are not linked to the combination therapy with linagliptin and metformin. Subgroup analyses were also performed for study 1218.62 and study 1218.20; the findings largely support the analysis based on EFF-C5. Details are available in the SCE [Module 2.7.3, Sections 3.3.2 and 3.3.3].

4.3.4 Combination therapy with linagliptin, metformin, and SU

The placebo-controlled phase III-trial 1218.18 (EFF-C4) compared the efficacy of linagliptin 5 mg once daily with placebo added to ongoing combination therapy with metformin and an SU over 24 weeks. Starting from nearly idendical baseline values, after 24 weeks, treatment with Lina+Met+SU provided significant and clinically meaningful reductions in HbA_{1c} from baseline compared with Met+SU, with an adjusted mean treatment difference of -0.62% (Table 4.3.4: 1). This finding by was corroborated by sensitivty analyses based on different methods of imputing missing data. Details are available in the SCE [Module 2.7.3, Section 3.2.4.1].

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Table 4.3.4: 1 Change from baseline in HbA_{1c} [%] after 24 weeks in study 1218.18 (EFF-C4) - FAS (LOCF)

Treatment group	Number of patients, N (%)	HbA_{1c} baseline, mean (SD)	Change in HbA_{1c} from baseline		Difference from Met + SU		
			Mean (SD)	Adjusted [a] mean (SE)	Adjusted [a] mean (SE)	95% CI	p-value
Met+SU	262 (25.2)	8.14 (0.84)	-0.10 (0.87)	-0.10 (0.05)			
Lina5 +Met+SU	778 (74.8)	8.15 (0.80)	-0.72 (0.86)	-0.72 (0.03)	-0.62 (0.06)	(-0.73, -0.50)	<0.0001

Met+SU = metformin and sulfonylurea background therapy; Lina5 = linagliptin 5 mg qd
a Model includes continuous baseline HbA_{1c} and treatment

Source data: [U11-1151, Module 5.3.5.3, Table 4.4.1.1]

For all secondary endpoints, the triple combination of Lina 5 mg qd+Met+SU was superior to the dual therapy of Met+SU. With the triple combination a reduction in FPG was observed and a substantially higher proportion of patients achieved glycaemic control defined as HbA_{1c} below 7% or 6.5%. The use of rescue therapy was also less frequent with the triple combination. In regard to body weight, there was a slight increase of 0.3 kg in the triple combination group; however, the p-value did not suggest a true difference between the treatment groups. Overall, these data demonstrate the superiority of the triple combination of linagliptin + metformin + SU over metformin + SU.

Table 4.3.4: 2 Secondary endpoints in trial 1218.18 (EFF-C4)

Endpoint	Met+SU		Lina+Met+SU	
FPG [mg/dL] (FAS-LOCF)	Baseline Mean (SD) 162.6 (37.2)	Adjusted Mean change (SE) 8.1 (2.38)	Baseline Mean (SD) 159.2 (36.5)	Adjusted mean change (SE) -4.6 (1.38)
Difference to Met+SU (SE); p-value			-12.7 (2.75); p<0.0001	
Patients achieving HbA_{1c} target [%] (FAS-NCF)				
<7%	9.2%		31.2%	
<6.5%	4.2%		13.1%	
Patients using rescue therapy [%] (FAS-OR)	13.0%		5.4%	
Body weight [kg] (FAS-OC)	Baseline (Mean) 77.4	Adjusted mean change -0.1	Baseline (Mean) 76.6	Adjusted mean change 0.3
Difference to Met+SU (SE); p-value			0.3 (0.19); p=0.0803	

Met+SU = metformin and sulfonylurea background therapy; Lina5 = linagliptin 5 mg qd

Source: [U11-1151, Module 5.3.5.3, Table 4.4.6.1, Table 4.4.7.1 and Figure 4.4.7.2, Table 4.4.3.1, Table 4.4.2.1].

4.4 DOSE RESPONSE AND DOSING RATIONALE

4.4.1 Dosing rationale for linagliptin monotherapy

The optimum daily linagliptin dose of 5 mg was determined in the linagliptin mono development programme and was primarily based on HbA_{1c} reduction and DPP-4 inhibition

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in the 2 dose-finding trials 1218.5 [U08-3761, Module 5.3.5.1] and 1218.6 [U08-1056, Module 5.3.5.1]. Tolerability was excellent on all dose levels tested in these 2 trials (0.5 mg, 1 mg, 2.5 mg, 5 mg, and 10 mg), and no dose-dependent increase in adverse events was observed. A median DPP-4 inhibition of at least 80% was only reached with the 5 mg and 10 mg linagliptin doses, but not with lower doses. The 10 mg dose did not result in a substantially greater median DPP-4 inhibition or HbA_{1c} reduction than the 5 mg dose, and hence, the 5 mg dose was used in phase III. In the entire phase III-programme, linagliptin 5 mg once daily was shown to provide consistent and clinically meaningful improvements in glycaemic control, as asssessed by HbA_{1c}, FPG, or postprandial glucose levels. Together with the favourable safety profile observed, the risk-benefit ratio of linagliptin is considered optimal for a daily dose of 5 mg linagliptin.

Because the pharmacokinetics of metformin require an at least twice daily dosing, for the development of the FDC the once daily dosing of 5 mg linagliptin was split into 2 daily doses of 2.5 mg. In trial 1218.45, a phase I-study comparing once daily and twice daily dosing in healthy subjects, the 2 linagliptin dosing regimens were shown to be bioequivalent in regard to AUC at steady state; the DPP-4 inhibition levels reached with both posologies were also similar [U08-2132, Module 5.3.1.2]. The clinical equivalence of both posologies was demonstrated in a large, placebo-controlled, 12-week trial 1218.62 that investigated Lina 2.5 mg bid+Met vs. Lina 5 mg qd + Met. Details on trial 1218.62 are available in Section 4.3 above.

4.4.2 Dosing of metformin

Metformin in immediate release formulation is approved as tablet for oral administration in the dose strengths of 500 mg, 850 mg, and 1000 mg [R10-5341, Module 2.7.5]. In case of tolerability issues, the metformin dose should be reduced. It is therefore proposed to provide FDC tablets with linagliptin and all 3 metformin dose strenghts to offer patients maximal flexibility in attaining their individual daily metformin dose.

For the key trials in this submission (1218.17, 1218.20, 1218.62, and 1218.18) the protocols specified that the daily metformin dose should be 1500 mg or above. The actual unit dose strength and the actual posology was at the investigator's descretion and depending on the patient's need and tolerability of metformin. The majority of patients in the 4 studies received a daily dose of 1500 mg to 2000 mg metformin at baseline; a considerable proportion of patients took took less than 1500 mg metformin per day (range 6.3% to 8.6%). In trial 1218.46, two treatment groups with metformin mono therapy of 500 mg and 1000 mg bid were evaluated.

Table 4.6.2: 1 Metformin background dose categories at baseline in studies 1218.17 (EFF-C1), 1218.20 (EFF-C2), 1218.18 (EFF-C4), 1218.62 (EFF-C7a) - FAS

Metformin background dose category	Study 1218.17 (EFF-C1) N (%)	Study 1218.20 (EFF-C2) N (%)	Study 1218.18 (EFF-C4) N (%)	Study 1218.62 (EFF-C7a) N (%)
<1500 mg	49 (7.1)	102 (6.7)	89 (8.6)	30 (6.3)
1500 mg to 2000 mg	457 (66.4)	1054 (69.4)	803 (77.2)	364 (76.2)
>2000 mg	182 (26.5)	363 (23.9)	148 (14.2)	84 (17.6)

Source data: [U11-1151, Module 5.3.5.3, Tables 3.1.1.1, 3.1.2.1, 3.1.4.1, and 3.1.7.1]

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4.4.3 Dosing of combination therapy with linagliptin and metformin

The applicant proposes 3 FDC dose strengths, namely Lina 2.5 mg/Met 500 mg, Lina 2.5 mg /Met 850 mg, and Lina 2.5 mg/Met 1000 mg. To substantiate the effectiveness of these 3 dose strengths, the placebo-controlled trials (1218.17, .62, .46, .18) were analysed by metformin dose and posology. Emphasis was placed on the posologies of 500 mg bid, 850 mg bid and 1000 mg bid. However, except in study 1218.46, in which all patients took either metformin 500 mg bid or 1000 mg bid, the proportions of patients who received other doses or other posologies were between 42.1% (1218.62) and 58.9% (1218.17). For each of the studies (1218.17, .18, .20, .62), the analyses that comprised all 'other' metformin doses demonstrated that the linagliptin treatment effect was similar to the efficacy observed for the planned FDC combinations.

Independent of metformin dose, the addition of linagliptin either once daily or twice daily led to a substantial and clinically meaningful reduction in HbA_{1c}. The magnitude of the combination treatment effect was largely consistent across the different metformin doses with adjusted mean changes from baseline of -0.51% to -0.70% after 24 weeks of treatment. Only for the 850 mg group in study 1218.62 no difference between Met 850 mg and Lina+Met 850 mg could be established. This was likely a consequence of the small number of patients in this analysis (Met 850 mg n=9; L5 mg qd+Met 850 mg n=55, L2.5 mg bid+Met 850 mg, n=57). In addition, study 1218.46 showed that the combination of Lina 2.5 mg bid+Met 500 mg bid was at least as effective in reducing HbA_{1c} as M1000 mg bid (-1.22% vs. -1.07%), adjusted mean difference -0.14%, p=0.1903. Thus, the Lina 2.5 mg/ Met 500 mg FDC could be considered a useful alternative to metformin monotherapy for patients intolerant to the recommended daily metformin dose.

For all placebo-controlled trials, ANCOVA analyses demonstrated that there was no treatment-by-metformin dose effect for the reduction of HbA_{1c} (treatment-by-dose p-values 0.2172 to 0.9846). In the active-controlled study 1218.20, the adjusted mean changes (SE) from baseline in HbA_{1c} after 104 weeks of treatment were -0.10% (0.08) for Lina+Met 850 mg bid and -0.12% (0.06) for Lina+Met 1000 mg bid; the treatment-by-dose interaction p-value was 0.8056. Thus, it can be concluded that each of the proposed posologies of Lina+Met combinations are efficacious in the reduction of HbA_{1c}. A summary of the results by metformin dose is shown below.

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Table 4.4.3: 1 Change from baseline in HbA_{1c} in placebo-controlled trials with linagliptin+metformin combinations analysed by daily metformin doses 500 mg, 850 mg and 1000 mg - FAS (LOCF)

	Change from baseline	Difference between treatments		
Study (Groupings) Treatment groups	Adjusted [a] mean (SE)	Adjusted [a] mean (SE)	95% CI	p-value
1218.17 (EFF-C6a)		**24 weeks treatment**		
850 mg Metformin				
Metformin	0.13 (0.14)			
L5 mg qd+Met vs Met	-0.56 (0.07)	-0.70 (0.15)	(-1.00, -0.39)	<0.0001
1000 mg Metformin				
Metformin	0.31 (0.16)			
L5 mg qd+Met vs Met	-0.36 (0.10)	-0.67 (0.19)	(-1.05, -0.29)	0.0005
1218.62 (EFF-C7b)		**12 weeks treatment**		
850 mg Metformin				
Metformin	-0.13 (0.23)			
L5 mg qd+Met vs Met	-0.50 (0.09)	-0.37 (0.24)	(-0.85, 0.11)	0.1283
L2.5 mg bid+Met vs Met	-0.51 (0.09)	-0.38 (0.24)	(-0.86, 0.10)	0.1191
L2.5 mg bid+Met vs L5 mg qd+Met		0.01 (0.13)	(-0.26, 0.24)	0.9502
1000 mg Metformin				
Metformin	0.19 (0.19)			
L5 mg qd+Met vs Met	-0.50 (0.09)	-0.69 (0.21)	(-1.10, -0.28)	0.0009
L2.5 mg bid+Met vs Met	-0.48 (0.09)	-0.67 (0.21)	(-1.08, -0.26)	0.0013
L2.5 mg bid+Met vs L5mg qd + Met		0.02 (0.12)	(-0.22, 0.26)	0.8855
1218.46 (EFF-C3)		**24 weeeks treatment**		
500 mg Metformin				
Met bid	-0.64 (0.08)			
L2.5 mg bid+Met bid vs. Met bid	-1.22 (0.08)	-0.58 (0.11)	(-0.79, -0.36)	<0.0001
1000 mg Metformin				
Met bid	-1.07 (0.08)			
L2.5 mg bid+Met bid vs Met bid	-1.59 (0.08)	-0.51 (0.11)	(-0.73, -0.30)	<0.0001
1218.18 (EFF-C6c)		**24 weeks treatment**		
850 mg Metformin+SU				
Met+SU	-0.17 (0.12)			
L5 mg qd+Met+SU vs Met+SU	-0.79 (0.07)	-0.63 (0.14)	(-0.90, -0.36)	<0.0001
1000 mg Metformin+SU				
Met+SU	-0.05 (0.10)			
L5 mg qd+Met+SU vs Met+SU	-0.65 (0.06)	-0.60 (0.12)	(-0.83, -0.37)	<0.0001

Met = metformin; Lina 5 mg = linagliptin 5 mg; Lina 2.5 mg = linagliptin 2.5 mg
[a] Model includes continuous baseline HbA_{1c}, number of prior OADs, dose (categories: 850 mg bid, 1000 mg bid, other), treatment, and treatment-by-dose interaction

Source: [U11-1151, Module 5.3.5.3, Tables 4.6.1.1, 4.8.1.1, 4.3.1.1, and 4.6.1.5]

4.5 PERSISTENCE OF EFFICACY

This section seperately describes the persistence of efficacy of linagliptin monotherapy, metformin monotherapy, and combination therapy with linagliptin and metformin. The key evidence for persistent efficacy is derived by combining data from the 24-week trials 1218.16 (linagliptin monotherapy) and 1218.17 (linagliptin + metformin combination) with their open-label extension 1218.40. Note that these analyses were based on observed cases (OC),

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i.e. missing values were not imputed, and values measured after a patient had received rescue medication were set to missing.

Furthermore, persistence of the initial combination of linagliptin + metformin is derived from a combination of data from the 24-week study 1218.46 and its double-blind 54-week extension 1218.52. The 2-year data of the double-blind active-controlled trial 1218.20 provides evidence of the effectiveness of linagliptin + metformin for 102 weeks. Persistent efficacy for the triple combination of linagliptin + metformin + SU is demonstrated by combining the data of study 1218.18 with the open-label extension 1218.40.

Generally, the study protocols of the initial studies 1218.16, .17, and .18 and their follow-up trial 1218.40 made identical provisions in regard to inclusion/exclusion criteria, compliance with study medication, reporting of safety data etc. However, the criteria for the use of rescue differed slightly. In the last 12 weeks of the study, patients were to receive rescue medication when their FPG levels were above 200 mg/dL; during the last 72 weeks of the follow-up trial, patients were to receive rescue medication if their FPG was above 180 mg/dL. This lower threshold for initiation of rescue medication might have inflated the proportion of patients receiving rescue medication in the follow-up trial. For the extension study 1218.40, all patients completing the initial trials were eligible; however, study 1218.52 included only patients completing the 1218.46 trial without requiring rescue medication. Therefore, a comparison between the 2 extension trials in regard to the use of rescue therapy has to be done with caution.

4.5.1 Persistence of efficacy of linagliptin monotherapy

To evaluate the persistence of linagliptin monotherapy, data from the placebo-controlled 24-week study 1218.16 were combined with interim data from its 78-week open-label extension study 1218.40 to provide data over up to 102 weeks of treatment (study grouping EFF-M1).

Patients who had received placebo in the initial trial received linagliptin 5 mg qd in the follow-up trial (‘Pbo/Lina5’), patients who had received linagliptin in the initial trial continued with this treatment (‘Lina 5 mg’). The frequency of premature discontinuations was low in both arms (Pbo/Lina5: 7.5%; Lina 5 mg: 11.1%) and the median treatment duration was 721 and 720 days, respectively. Overall exposure was 279.7 patient years in the Pbo/Lina5 group and 552.3 patient years in the Lina 5 mg group. The demographic characteristics and baseline conditions were similar in the initial trial and the follow-up study.

Change from baseline in HbA_{1c} and FPG over time in the combined studies 1218.16 and 1218.40 (EFF-M1)

At the end of the 24-week treatment period of study 1218.16, the mean change from baseline in HbA_{1c} in the patients who had received linagliptin in study 1218.16 was -0.55%; after 102 weeks, the mean change from baseline was -0.48%. Likewise for FPG, patients who had received linagliptin in study 1218.16 had a mean FPG change of -10.3 mg/dL after 24 weeks and of -9.0 mg/dL after 102 weeks. Thus, the analysis of the FPG change over time in the combined studies 1218.16/1218.40 confirmed the results obtained for the HbA_{1c} change, suggesting that the glucose-lowering effect of linagliptin monotherapy was sustained over up to 102 weeks.

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Table 4.5.1: 1 Mean changes from baseline in HbA_{1c} [%] and FPG [mg/dL] in EFF-M1 over 102 weeks FAS-OC

	HbA_{1c}				FPG			
	Pbo/Lina5		Lina5		Pbo/Lina 5		Lina5	
	N	Mean HbA_{1c}	N	Mean HbA_{1c}	N	Mean FPG	N	Mean FPG
Baseline	147	7.97	296	7.98	143	167.4	289	165.0
Week 6	140	0.05	284	−0.41	130	5.9	265	−11.9
Week 12	126	0.01	278	−0.53	121	9.2	266	−11.4
Week 18	122	0.08	275	−0.51	116	9.3	265	−8.2
Week 24	115	0.09	261	−0.55	102	8.5	238	−10.3
Week 30	102	−0.42	237	−0.51	105	−5.0	244	−6.2
Week 42	99	−0.59	220	−0.48	96	−5.9	215	−7.0
Week 54	88	−0.64	191	−0.55	87	−9.0	186	−10.0
Week 66	82	−0.52	187	−0.44	79	−4.9	182	−6.3
Week 78	74	−0.51	173	−0.39	70	−9.7	165	−6.8
Week 90	71	−0.55	161	−0.44	68	−10.6	156	−10.0
Week 102	61	−0.63	138	−0.48	59	−11.4	131	−9.0

Source: [U11-1151, Module 5.3.4.3, Tables 5.4.1.1 and 5.4.2.1]

Other indicators of persistence of efficacy: HbA_{1c} response (<6.5%, <7%), use of rescue medication, body weight, (EFF-M1)

Among those who received linagliptin both in study 1218.16 and 1218.40, the proportion of patients achieving HbA_{1c} levels below 6.5% or below 7% remained essentially unchanged until Week 102. The proportion of patients with HbA_{1c} below 6.5% had been 12.3% after 24 weeks of treatment in study 1218.16; , and 16.7% after further treatment with linagliptin for 78 weeks in study 1218.40. HbA_{1c} levels below 7% were achieved by 33.7% after 24 weeks 38.4% (after 102 weeks.

Overall, 40.5% of the patients who received linagliptin monotherapy both in study 1218.16 and in study 1218.40 took rescue therapy at or before Week 102. This proportion is substantially higher than the overall proportion of patients taking linagliptin and requiring rescue medication in the course of study 1218.16 (10.2% [U10-1103, Module 5.3.5.1, Table 15.2.3.1: 2]). However, it should be noted that longer study duration and the stricter criteria for initiating rescue therapy during study 1218.40 have increased the proportion of patients with rescue therapy in the follow-up trial (for details see SCE [Module 2.7.3, Section 5.1.4]).

The mean change in body weight after 102 weeks of treatment with linagliptin monotherapy was -0.55 kg. Hence linagliptin monotherapy given for up to 102 weeks did not seem to have a relevant effect on body weight.

4.5.2 Persistence of efficacy of metformin monotherapy

Initial improvements in HbA_{1c} achieved with metformin monotherapy are generally lost after approximately 5 years. Patients who initially respond well to metformin monotherapy often require combination therapy to maintain long-term glycaemic control. In some patients, initial metformin monotherapy fails to produce sufficient reductions in HbA_{1c} and FPG; in these cases, add-on therapy is necessary [R10-4896, Module 2.7.5].

From the clinical development programme for combination therapy with linagliptin and metformin, long-term efficacy data for metformin monotherapy are available from study grouping EFF-C9. For this study grouping, data from the 24-week study 1218.46 were

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combined with data from its 52-week extension trial 1218.52, providing efficacy data for up to 78 weeks. During this period, the patients who received metformin monotherapy (1000 mg bid) in both studies showed a sustained reduction in HbA_{1c} and FPG over time, even though the changes from baseline were smaller than those in patients who received combined treatment with linagliptin and metformin in both trials. For a detailed description of the efficacy results in the combined studies 1218.46 and 1218.52 see below.

4.5.3 Persistence of efficacy of combination therapy with linagliptin and metformin

The following section provides long-term efficacy data for combination therapy with linagliptin and metformin. Presented are combined data from studies 1218.17 and 1218.40 (EFF-C8) and from studies 1218.46 and 1218.52 (EFF-C9), providing evidence of the persistence of the efficacy of combined treatment with linagliptin and metformin over up to 102 weeks and 78 weeks, respectively. In addition, active-controlled 2-year efficacy data are available from study 1218.20.

4.5.3.1 Persistence of the efficacy of linagliptin as add-on therapy to metformin over 102 weeks in the combined studies 1218.17 and 1218.40 (EFF-C8)

Evidence of the persistence of the efficacy of linagliptin added to metformin background therapy over 102 weeks is available from study grouping EFF-C8 which includes data of the initial study 1218.17, a double-blind, placebo-controlled 24-week trial, and interim data of the 78-week open-label extension study 1218.40 (data cut-off date: 15 October 2010).

Change from baseline in HbA_{1c} and FPG over time in the combined studies 1218.17 and 1218.40 (EFF-C8)

In patients receiving linagliptin in both studies (‘non-switchers’), the mean HbA_{1c} change from baseline was -0.64% after 24 weeks in study 1218.17 and increased to -0.72% after an additional 78 weeks of treatment in study 1218.40. Also for mean FPG, a clinically meaningful reduction was observed with the combination therapy. After 24 weeks of treatment with linagliptin + metformin in study 1218.17, a mean FPG decrease from baseline of -14.7 mg/dL was noted; at Week 102 the mean change from baseline was -15.9 mg/dL. In conclusion, these results show sustained glucose-lowering efficacy of combination therapy with linagliptin and metformin for up to 102 weeks.

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Table 4.5.3.1: 1 Mean changes from baseline in HbA_{1c} [%] and FPG [mg/dL] in EFF-C8 (1218.17 + 1218.40) over 102 weeks - FAS-OC

	HbA_{1c}				FPG			
	Met/L5+Met ('switchers')		Lina5+Met ('non-switchers')		Met/L5+Met ('switchers')		Lina5+Met ('non-switchers')	
	N	Mean HbA_{1c}	N	Mean HbA_{1c}	N	Mean FPG	N	Mean FPG
Baseline	153	8.04	457	8.08	150	167.3	447	169.6
Week 6	141	−0.02	443	−0.45	136	3.4	428	−15.0
Week 12	133	−0.07	425	−0.65	131	−0.7	413	−17.0
Week 18	126	−0.08	419	−0.63	122	3.7	396	−14.4
Week 24	118	−0.04	413	−0.64	113	4.8	372	−14.7
Week 30	105	−0.51	384	−0.69	106	−10.2	387	−15.7
Week 42	102	−0.81	362	−0.69	100	−15.5	342	−12.1
Week 54	100	−0.70	336	−0.71	100	−11.8	322	−13.2
Week 66	91	−0.60	303	−0.62	89	−10.9	291	−12.3
Week 78	86	−0.61	281	−0.57	83	−10.9	274	−14.7
Week 90	79	−0.61	255	−0.69	78	−10.8	256	−15.0
Week 102	68	−0.83	232	−0.72	65	−16.4	220	−15.9

Met = metformin; L5+Met = linagliptin 5 mg qd + metformin; Lina5+Met = linagliptin 5 mg qd + metformin.
The '/' separates the treatment patients received in the initial study from the treatment in study 1218.40.

Source: [U11-1151, Module 5.3.5.3, Tables 4.9.1.1 and 4.9.2.1]

Other indicators of persistence of efficacy: HbA_{1c} response (<6.5%, <7%), use of rescue medication, body weight (EFF-C8: 1218.17+1218.40)

In patients who were treated with Lina+Met in both trials, the proportions of those achieving HbA_{1c} levels below 6.5% or below 7% remained constant until week 102. The proportions of patients with HbA_{1c} below 6.5% were 12.8% (after 24 weeks) and 13.4% (after 102 weeks). Similarly, HbA_{1c} levels below 7% were achieved by 33.4% (after 24 weeks) and 46.1% (after 102 weeks).

Before or at Week 102, 38.5% of the patients receiving combination therapy with linagliptin and metformin both in 1218.17 and 1218.40 had received rescue medication. This proportion is considerably higher than the overall proportion of patients taking linagliptin as add-on to metformin and requiring rescue medication in study 1218.17 (7.8% [U09-2533, Module 5.3.5.1, Table 15.2.3.1: 2]). However, as described above for the combined studies 1218.16 and 1218.40, this may partly be attributed to the lower threshold for initiation of rescue therapy in the extension trial. In regards to HbA_{1c} levels, the patients (n=67) who had not used rescue medication exhibited a mean change from baseline of -0.82% at Week 102. Only very few patients who had taken rescue medication continued after Week 54; at that point in time their mean HbA_{1c} change from baseline was -0.10% (n=26).

The mean change in body weight after 102 weeks of treatment with linagliptin + metformin was -0.99 kg.

4.5.3.2 Persistence of efficacy of the initial combination of linagliptin and metformin over up to 78 weeks in the combined studies 1218.46 and 1218.52 (EFF-C9)

Long-term efficacy data for the initial combination of linagliptin and metformin are available from study grouping EFF-C9 which combined data from the placebo-controlled 24-week trial 1218.46 and its double-blind 54-week extension 1218.52, providing data for up to 78 weeks. Only patients who completed the treatment period of study 1218.46 without requiring rescue

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medication were eligible for the extension. Three treatments were available in the extension study, i.e. linagliptin 2.5 mg bid + metformin 500 mg bid, linagliptin 2.5 mg bid + metformin 1000 mg bid, or metformin 1000 mg bid. To assess persistence, the following section presents efficacy results for the patients who received the same treatment in both studies ('non-switchers'). Data for those patients who were re-randomised after completion of the initial trial, are not presented here, but are available in the FDC SCE Appendix 1, 1 [U11-1151, Module 5.3.5.3, Section 1.4.10]. The description focuses on the results of up to 54 weeks (i.e. 30 weeks of treatment in study 1218.52) because only a small number of patients had completed the total treatment duration of 78 weeks at the time of the interim cut-off (15 October 2010).

Change from baseline in HbA_{1c} and FPG over time in the combined studies 1218.46 and 1218.52 (EFF-C9)

Mean HbA_{1c} steadily decreased from baseline until Week 24 in all 3 treatment groups. The largest mean change from baseline in HbA_{1c} of -1.70% was seen in the patients receiving linagliptin 2.5 mg bid + metformin 1000 mg bid. In the extension study, mean HbA_{1c} levels remained fairly stable in all 3 treatment groups, showing persistence of the glucose-lowering effect of the initial combination of linagliptin and metformin as well as of metformin monotherapy.

The analysis of the change from baseline in FPG in the combined studies 1218.46/1218.52 was in line with the results seen for the HbA_{1c} change; showing that the FPG-lowering efficacy of the initial combination of linagliptin and metformin was sustained over time. The mean change from baseline in HbA_{1c} and mean FPG over time for the 3 treatment groups is displayed in Table 4.5.3.2: 1.

Table 4.5.3.2: 1 Mean changes from baseline in HbA_{1c} [%] and FPG [mg/dL] in EFF-C9 (1218.46 + 1218.52) over 78 weeks, patients who stayed on the same treatment - FAS-OC

	HbA_{1c}						FPG					
	L2.5+M500		L2.5+M1000		M1000		L2.5+M500		L2.5+M1000		M1000	
	N	Mean HbA_{1c}	N	Mean HbA_{1c}	N	Mean HbA_{1c}	N	Mean FPG	N	Mean FPG	N	Mean FPG
Baseline	113	8.61	111	8.61	109	8.47	111	193.5	110	191.1	107	185.5
Week 6	113	−0.85	111	−1.06	107	−0.68	109	−39.9	106	−47.1	105	−32.1
Week 12	113	−1.15	108	−1.46	107	−1.06	108	−36.0	109	−48.6	103	−34.2
Week 18	105	−1.22	110	−1.64	107	−1.15	105	−34.8	107	−48.0	105	−31.9
Week 24	107	−1.28	108	−1.70	106	−1.16	100	−32.5	104	−49.1	103	−33.0
Week 30	100	−1.32	106	−1.74	105	−1.20	94	−39.5	105	−50.6	100	−34.9
Week 42	90	−1.28	96	−1.62	94	−1.15	86	−39.8	93	−48.9	91	−36.7
Week 54	46	−1.38	59	−1.69	50	−1.29	41	−47.2	56	−49.2	47	−38.7

L2.5+M500 = linagliptin 2.5 mg bid + metformin 500 mg bid; L2.5+M1000 = linagliptin 2.5 mg bid + metformin 1000 mg bid; M1000 = metformin 1000 mg bid

Source: [U11-1151, Module 5.3.5.3, Tables 4.10.1.1 and 4.10.2.1]

Other indicators of persistence of efficacy: HbA_{1c} response (<6.5%, <7%), use of rescue medication, body weight (EFF-C9: 1218.46+1218.52)

The proportions of patients achieving HbA_{1c} levels below 6.5% were 15.9% (L2.5+M500), 30.6% (L2.5+M1000), and 15.1% (M1000) after 24 weeks. These proportions remained

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stable until Week 54. In constrast, the proportions of patients achieving HbA_{1c} levels below 7% increased over time. At Week 54, 43.5% (L2.5+M500), 64.4% (L2.5+M1000), and 46.0% (M1000) had achieved HbA_{1c} <7%.

The use of rescue medication at or before Week 78 differed between the treatment groups. The proportion of patients who required rescue medication up to Week 78 was lower in the Lina2.5+Met1000 bid group (8.1%) than in the Lina2.5+Met500 bid group (23.1%) and the Met1000 bid group (19.3%). The mean HbA_{1c} changes from baseline at Week 54 for patients who did not use rescue medication were -1.57% (L2.5+M500), -1.72% (L2.5+M1000), and -1.29% (M1000). The number of patients who took rescue medication either in the initial trial or in the extension was too small to permit valid comparisons.

Until Week 42 in the extension trial, weight was essentially unchanged in the Lina2.5+ Met500 bid group (-0.01 kg), and small reductions were seen for patients in the Lina2.5 +Met1000 bid (-0.49 kg) and the Met1000 bid (-0.80 kg) groups. Weight was not assessed at Week 52 and thereafter there were too few patients to allow valid comparisons.

4.5.3.3 Persistence of efficacy of triple combination of linagliptin with metformin, and SU

Persistence of the efficacy of linagliptin added to a background medication of metformin and an SU over up to 102 weeks is shown by the results of study grouping EFF-C10. This grouping combined data from the placebo-controlled 24-week study 1218.18 with interim data from the 78-week open-label extension study 1218.40 (interim cut-off date: 15 October 2010). Patients in the 'M+SU/L5+M+SU' treatment group had received placebo in addition to metformin and SU background therapy in study 1218.18 and were switched to linagliptin as add-on to metformin and SU in the extension period; the patients in the 'Lina5+Met+SU' treatment group took linagliptin added to metformin and SU background medication both in study 1218.18 and in study 1218.40.

Change from baseline in HbA_{1c} and FPG over time in the combined studies 1218.18 and 1218.40 (EFF-C10)

For the patients who had received linagliptin + metformin + SU (Lina+Met+SU) in study 1218.18, a decrease in mean HbA_{1c} of -0.72% from baseline to Week 24 was observed. When these patients continued with the same treatment in study 1218.40, HbA_{1c} levels remained reasonably stable, with a mean change from baseline in HbA_{1c} of -0.64% after 102 weeks. Hence the analysis demonstrated persistent glucose-lowering efficacy of the triple combination of Lina+Met+SU.

While the analyses of the changes from baseline in HbA_{1c} were in line with the long-term results seen for linagliptin mono therapy (1218.16+1218.40, Section 4.5.1) and the dual therapy of Lina+Met (EFF-C8), the long-term data for mean changes in FPG were different and seem to indicate dwindling efficacy. The mean changes in FPG were small, particularly in patients who had received the triple combination for 102 weeks. This result differs substantially from the FPG changes observed for the Lina+Met therapy (-15.9 mg/dL at 102 weeks). This difference might originate from the populations included into studies 1218.17 (Lina+Met) and 1218.18 (Lina+Met+SU). The proportions of patients who had been diagnosed with diabetes for more than 5 years were 54.9% (1218.17) and 73.3% (1218.18). Also, at the time of inclusion into the initial trial, 31.4% in the 1218.17 and 100% of patients

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in study 1218.18 took 2 or more OADs. Thus it appears that the diabetes was at a more advanced stage in patients included in the 1218.18 trial than in the 1218.17 trial. The smaller FPG change seen in study 1218.18 despite a similar HbA_{1c} reduction compared with the other 3 studies could indicate that in study 1218.18, linagliptin had a greater proportional effect on post-prandial glucose levels.

Table 4.5.3.3: 1 Mean changes from baseline in HbA_{1c} [%] and FPG [mg/dL] in EFF-C10 (1218.18 + 1218.40) over 102 weeks - FAS-OC

	HbA_{1c}				FPG			
	Met+SU/L5+Met+SU		L5+Met+SU		Met+SU/L5+Met+SU		L5+Met+SU	
	N	Mean HbA_{1c}	N	Mean HbA_{1c}	N	Mean FPG	N	Mean FPG
Baseline	182	8.13	544	8.13	181	159.9	535	158.7
Week 6	181	-0.19	537	−0.70	176	5.6	519	−12.2
Week 12	179	-0.22	533	−0.89	176	3.6	518	−9.3
Week 18	170	-0.19	527	−0.87	165	6.7	505	−6.7
Week 24	162	-0.14	514	−0.72	154	6.4	484	−6.1
Week 30	160	-0.65	498	−0.71	158	−15.3	484	−4.6
Week 42	150	-0.79	458	−0.67	148	−13.0	445	−5.4
Week 54	142	-0.79	420	−0.69	140	−16.8	411	−7.0
Week 66	134	-0.79	386	−0.68	121	−12.4	318	−4.4
Week 78	130	-0.66	355	−0.58	101	−10.3	265	0.8
Week 90	126	-0.56	334	−0.62	98	−10.6	247	−3.6
Week 102	109	-0.58	275	−0.64	86	−10.5	232	−0.7

Met = metformin; L5 = linagliptin 5mg qd; SU = sulphonylurea

Source: [U11-1151, Module 5.3.5.3, Tables 4.11.1.1 and 4.11.2.1]

Other indicators of persistence of efficacy: HbA_{1c} response (<6.5%, <7%), use of rescue medication, body weight (EFF-C10: 1218.18+1218.40)

Among the patients receiving triple therapy with Lina+Met+SU in both studies, the proportion of those achieving target HbA_{1c} levels remained largely constant over time. HbA_{1c} levels below 6.5% after 24 weeks were achieved by 14.0% of patients and after 102 weeks by 10.9% of patients. The target of HbA_{1c} below 7.0% was reached by 34.0% of patients after 24 weeks and by 34.2% of patients after 102 weeks of treatment with Lina+Met+SU.

The proportion of patients who received Lina+Met+SU in both studies and required rescue medication in the 102-week treatment period was 33.6%. As outlined above for the other combined study groupings including the 1218.40 trial, this proportion was substantially higher than the proportion of patients with rescue therapy at the end of the initial study 1218.18 (5.4% [U09-2458, Module 5.3.5.1, Table 15.2.3.1: 1]), also possibly caused by the stricter criteria for rescue therapy in the extension study 1218.40. Patients who had not taken rescue medication achieved an HbA_{1c} change from baseline of -0.58% at Week 102. Patients who took rescue medication either in the initial trial or in the follow-up trial had attained an HbA_{1c} change of -0.52% at Week 54, thereafter too few patients contributed data to allow conclusions.

Over 102 weeks of treatment, no relevant influence of linagliptin added to metformin+SU on the change in body weight was detected. The mean change from baseline in body weight in patients receiving triple combination therapy throughout was -0.55 kg after 102 weeks of treatment.

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5. OVERVIEW OF SAFETY

Generally, the analysis of safety is presented separately for linagliptin monotherapy, metformin monotherapy, the combination of linagliptin+metformin, and the triple combination of linagliptin+metformin+SU. Since metformin has been in clinical use for some 50 years in the European Union, no dedicated studies to assess the safety of metformin were conducted. However, information on metformin monotherapy is included in some of the safety analysis sets (SAFs) for the linagliptin mono and combination arms.

The linagliptin monotherapy submission comprised in total 37 studies (24 phase I, 4 phase II, and 9 phase III trials) involving 6198 patients with T2DM of whom 4061 took Lina 5 mg. The analysis of safety of linagliptin monotherapy presented here is based on data that were already presented as part of the linagliptin monotherapy submission, namely a pooled analysis of all placebo-controlled studies with Lina 5 mg conducted in patients with T2DM (SAF-2). It included 12 trials, among them 7 phase III trials, 3 phase II trials, and 2 phase I trials with a total of 2566 patients treated with Lina 5 mg qd.

In addition, new long-term safety data for linagliptin monotherapy are derived from a pooled analysis of studies 1218.16 (placebo-controlled 24 weeks) and 1218.17 (placebo-controlled 24 weeks, linagliptin as add-on to metformin), together with their 78-week open-label extensions as study 1218.40 (SAF-M1). The analysis was confined to patients who had received Lina 5 mg (n=336) or Lina 5 mg+Met (n=523) already in the initial trials. Hence, the treatment groups analysed in SAF-M1 were neither the result of a randomisation procedure nor were they placebo-controlled. However, otherwise they were parallel groups that were subjected to the same protocol in the extension phase. For the linagliptin monotherapy submission the cut-off date was 12 Jan 2010, for this submission, the cut-off date (second interim) was 15 Oct 2010; the final trial report will be available shortly.

The evaluation of safety of the Lina+Met combination, 14 clinical studies were analysed and arranged into 10 study groupings (SAF-C1 to SAF-C9 and SEA-2) including pooled analyses of several trials. An overview of the safety groupings is given in Table 5: 1 below.

The focus of the presentation is on SAF-C1, the safety data of the pivotal trial 1218.17 (n=700) which compared linagliptin with placebo in patients who took metformin as background medication over 24 weeks. This safety analysis is supported by data from a pooled analysis (SAF-C5) of all placebo-controlled trials that used metformin either as background medication or as study treatment (n=1971). Additionally, for the establishment of the side effect profile of Lina 5 mg+Met, an analysis set (SEA-2, n=1905) was generated that was based on SAF-C5 but considered solely patients who took linagliptin 5 mg resulting in an exclusion of 66 patients from study 1218.6 who took linagliptin 10 mg.

Data on long-term use of the Lina+Met combination is available from study grouping SAF-C6 (n=433) which combined data of the 24 week study 1218.46 with its double-blind extension trial 1218.52 to provide data over 78 weeks. This is supplemented by safety data from study 1218.20 that compared Lina+Met with Glim+Met over 104 weeks. The full analysis and description of all safety groupings is available in the SCS [Module 2.7.4, Section 1.1.6].

Safety of the triple combination therapy linagliptin+metformin+SU is evaluated based on SAF-C7 that contained the 1218.18 trial, a 24-week placebo-controlled trial of linagliptin added to background treatment of metformin + SU.

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Table 5: 1 Groupings of studies for the evaluation of safety of linagliptin and metformin combination

Safety grouping	Studies / Phase	Study characteristics and **objectives**	Treatment (weeks)	Patients, treated (N)
SAF-C1	1218.17 / III	Double-blind, placebo-controlled, metformin background, comparing linagliptin with placebo; **Safety data of the pivotal trial**	24	700
SAF-C2	1218.20 / III	Double-blind, active-controlled study, with metformin background comparing linagliptin with glimepiride; **Long-term safety of Lina+Met**	104	1551
SAF-C3	1218.62 /II	Double-blind, placebo-controlled, metformin background comparing linagliptin with placebo; **Safety of Lina+Met vs. Met**	12	491
SAF-C4	1218.46 [1]/ III	Double-blind, placebo-controlled study comparing different doses of linagliptin and metformin combination therapies with respective monotherapies; **Safety of Lina+Met combinations vs. mono components**	24	791
SAF-C5	1218.6 [2] / II 1218.17 / III 1218.46 [1] / III 1218.62 / II	Double-blind, placebo-controlled studies with metformin; comparing safety of Lina+Met with placebo+Met; **Safety of Lina+Met vs. Met in largest set**	≤24	1971
SEA-2	1218.6 [2] / II 1218.17 / III 1218.46 [1] / III 1218.62 / II	Double-blind, placebo-controlled studies with metformin; comparing safety of Lina+Met with PBO+Met limited to patients taking linagliptin 5 mg; **Establishment of side effect profile of Lina+Met**	≤24	1905
SAF-C6	1218.46 [1] / III 1218.52 [5] / III	Studies to assess long-term safety of linagliptin and metformin combination therapy; **Long-term safety of Lina+Met vs. Met**	≤78 [3]	433
SAF-C7	1218.18 / III	Double-blind, placebo-controlled study with metformin and SU background; comparing linagliptin with placebo; **Safety of Lina+Met+SU vs. Met+SU**	24	1055
SAF-C8	1218.6 [2/] II 1218.17 / III 1218.18 / III 1218.20 / III 1218.40 [4] /III 1218.46 / III 1218.52 / III 1218.62 / II	All studies with linagliptin plus metformin (metformin as background or given as free combination with linagliptin) to assess safety in a large patient population; **Safety of Lina+Met in the largest set (without comparison)**	≤104	3529 [6]
SAF-C9	1288.1 / I 1288.2 / I 1288.3 / I 1288.4 / I 1218.4 / I 1218.47 / I	Studies in healthy subjects[7]; **Safety in healthy subjects**	<2	354

[1] Without the open-label arm (linagliptin 2.5 mg+metformin 1000 mg, bid).
[2] Study 1218.6 had 5 treatment groups: placebo, Lina 1 mg, Lina 5 mg, Lina 10 mg, and open-label glimepiride; metformin was given as background medication; Lina 1 mg and open-label glimepiride were excluded from the analysis.
[3] The planned treatment duration of 1218.52 was 54 weeks, for this analysis the cut off date was 15 Oct 2010
[4] Including only patients from studies 1218.17 and 1218.18.
[5] Only patients who continued the respective treatment in 1218.52 were included in the analyses.
[6] Including the open-label arm (linagliptin 2.5 mg + metformin 1000 mg, bid) in study 1218.52.
[7] In studies 1288.1, 1288.2, 1288.3, and 1218.47 Lina+Met were given as free combination and as FDC. In study 1288.4, Lina+Met were administered as FDC, in study 1218.4 as free combination.

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The treated set (TS) was used for the analysis of safety and comprised all patients who received at least one dose of study medication. The screened set (SCR) consisted of all patients who were screened for the trial, had given informed consent, and had completed at least some screening procedures. Concomitant therapies were coded using the World Health Organization (WHO) Drug Dictionary, version Sept 2010. Concomitant diagnoses and adverse events were coded using the Medical Dictionary of Regulatory Affairs (MedDRA) version 13.1, with system organ class (SOC) and preferred term (PT).

For the analysis of adverse events, a treatment-emergent approach was chosen: All events with an onset date after the first dose of study medication up to a period of 7 days after the last dose of study medication were assigned to the randomised treatment period. Adverse events that occurred before first intake of randomised study medication were assigned to 'pre-treatment'; and adverse events that occurred after the last intake of study medication +7 days were assigned to 'post-treatment'. For patients participating in an extension trial, adverse events with an onset on or after the first day of treatment in the extension trial were assigned to the treatment of the extension trial.

Some events were of particular importance to the safety of linagliptin either because of scientific considerations or because of concerns that had arisen with other DPP-4 inhibitors. These 'adverse events of special interest' were hypoglycaemia, hypersensitivity reactions, renal events (including laboratory evaluations), hepatic events (including laboratory evaluations), severe cutaneous adverse reactions, and pancreatitis. For 'adverse events of special interest' other than hypoglycaemia, the analysis was based on standardised MedDRA queries (narrow Standard MedDRA Queries [SMQ]). Details on the definitions are available in the SCS [Module 2.7.4, Section 1.1.7].

A particular focus was also on the identification of hypoglycaemic events. For all studies (except 1218.6 which is therefore excluded from the analysis of hypoglycemic events), in addition to the standard reporting of adverse events, the investigators were asked to assess whether an adverse event was related to or a result of a hypoglycaemic episode. A detailed description of events contributing to the category 'investigator-defined hypoglycaemia' is provided in the SCS [Module 2.7.4, Section 1.1.7]. For the active-controlled trial 1218.20 (SAF-C2), a confirmatory analysis for hypoglycaemic events was performed as planned after 52 weeks and 104 weeks.

Since metformin is known to cause gastrointestinal side effects, additional analyses were performed. Drug-related gastrointestinal adverse events were analysed based on the SSC 'gastrointestinal adverse reactions'which included the MedDRA PTs 'vomiting', 'nausea', diarrhoea', and 'abdominal pain'. Furthermore, for the identification of side effects, the listed side effects of metformin [R11-0761, Module 2.7.5] were compared with the adverse events profile observed in SEA-2.

Note that the individual study results were re-calculated for this evaluation of safety, using the project database which takes into account information that was not availabe at the time of study database lock, rather than the individual study databases. Moreover, the analyses presented are partly based on pooled study data, modified statistical models, different analysis sets, and/or different imputation methods compared with the study reports. Therefore, small numerical differences between the study results presented in the individual study reports and here may occur.

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Analysis of cardiovascular risk

The evaluation of cardiovascular risk potentially associated with the use of linagliptin received particular attention. For the phase III studies, all reported treatment-emergent fatal events and events suggestive of stroke or myocardial ischaemia, including myocardial infarction, were reviewed and adjudicated in a blinded fashion by an independent Clinical Event Committee (CEC) which comprised 3 independent experts in cardiology (CEC Cardiology) and 3 independent experts of neurology (CEC Neurology). The CEC Cardiology reviewed all fatal events (without stroke events) and events suspected of myocardial ischaemia; the CEC Neurology adjudicated events suggestive of stroke (fatal and not-fatal strokes). The CEC evaluated whether pre-specified definitions for cardiovascular death, stroke, or myocardial ischaemia were fulfilled. Further details on the CEC, responsibilities, and clinical event definitions, are provided in the CEC Charter, which is included in the appendix of the CV meta-analysis report. Based on the blinded adjudications of the CEC a meta-analysis was conducted. The primary endpoint was a composite endpoint consisting of cardiovascular death (fatal stroke and fatal MI), non-fatal MI, non-fatal stroke, and unstable angina. In addition, the FDA-defined custom endpoint of major cardiovascular events (MACE) was analysed. Details regarding the statistical analyses and the full set of results are available in the CV meta-analysis report [U10-1736, Module 5.3.5.3]; however a summary of these results is provided in Section 5.6.

Laboratory evaluation

Samples for the determination of safety laboratory parameters (haematology, clinical chemistry, and urinalysis) were taken at almost all study visits in all trials. Laboratory values taken after the first dose of randomised treatment up to a period of 7 days after the last intake of treatment were assigned to the treatment phase. Laboratory analyses were performed on all safety groupings (SAF-C1 to SAF-C9, and SEA-2), however to a different extent per grouping: descriptive statistics and transitions relative to the reference range were performed for SAF-C2, SAF-C5, SAF-C6, and SAF-C7. Possible Clinically Significant Abnormalities (PCSA) and changes in renal function were performed on all SAFs except SAF-C9. Analyses to detect potential liver toxicity (including potential Hy's law cases) were performed on all SAFs. If applicable, reference to respective adverse event analyses were made for discussion of observations of laboratory analyses. Further details are available in the SCS [Module 2.7.4, Section 3].

Vital signs and physical examination

Evaluations of vital signs and physical examination were performed for SAF-C1 to SAF-C7. Changes from baseline in systolic and diastolic blood pressure and pulse rate were analysed descriptively. Clinically relevant abnormal findings in 12-lead electrocardiogram (ECG) measurements and in the physical examination were to be reported as adverse events. Further details are available in the SCS [Module 2.7.4, Section 4].

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5.1 SAFETY OF LINAGLIPTIN MONOTHERAPY

5.1.1 Exposure to linagliptin monotherapy

The planned study durations in the set of placebo-controlled trials (SAF-2) ranged from 12 days (1218.2) to 24 weeks (4 pivotal trials with different antidiabetic background medications). Trial 1218.23 had a planned duration of 52 weeks, however for SAF-2 only 12 week data of the placebo-controlled period were considered. Also for trial 1218.50 only the 18 week data of the placebo-controlled period were included. Median duration of treatment was almost identical in both treatment groups (168 and 169 days). Overall exposure in patient years was higher in the linagliptin group than in the placebo group due to the unbalanced allocation in most trials. For SAF-2 overall, 88.8% (placebo) and 92.9% (linagliptin) of patients completed the studies; 11.2% (placebo) and 7.1% (linagliptin) discontinued treatment prematurely. The proportions of patients who discontinued because of adverse events were slightly higher in the placebo group than in the linagliptin group (2.6% vs. 2.2%).

Table 5.1.1: 1 Exposure to study medication for SAF-2 and SAF-M1 – TS

	SAF-2 (placebo-controlled trials)		SAF-M1 (long-term linagliptin mono)	
	Placebo	Lina	Lina	Lina + Met
Number of patients, N (%)	1183 (100.0)	2566 (100.0)	336 (100.0)	523 (100.0)
Exposure categories, N (%)				
≥12 weeks	1007 (85.1)	2360 (92.0)	331 (98.5)	503 (96.2)
≥24 weeks	647 (54.7)	1679 (65.4)	321 (95.5)	485 (92.7)
≥52 weeks	0	0	275 (81.8)	442 (84.5)
≥78 weeks	0	0	270 (80.4)	424 (81.1)
Duration of treatment [days]				
Mean (±SD)	133.9 (51.5)	148.2 (42.4)	617 (211.0)	623 (212.4)
Median (min, max)	168 (1, 213)	169 (1, 214)	718 (1, 786)	719 (1, 794)
Overall patient years exposure	433.8	1041.4	567.9	891.5
Without rescue medication	n.a.	n.a.	522.3	831.5
With rescue medication	n.a.	n.a.	45.6	60.0

n.a. = not available

Source data: SAF-2 [U10-1607, Module 5.3.5.3, Table 4.2.1.1], SAF-M1 [U11-1152, Module 5.3.5.3, Table 8.2.1.1]

In regard to antidiabetic background medication in SAF-2, discontinuation rates in the placebo groups were generally higher than in the linagliptin groups where the incidences of premature discontinuations were very similar across all background medications (range for placebo 8.0% to 14.6%; range for linagliptin 5.8% to 8.5%). In patients who did not receive any antidiabetic background medication, the rates of premature discontinuations were 12.4% (placebo) and 6.5% (linagliptin); with metformin background, the discontinuation rates were similar (11.3% vs. 8.5%). Discontinuation rates were low with both SU background (8.3% vs. 6.2%) and metformin+SU background (8.0% vs. 7.2%).

The long-term treatment set of linagliptin monotherapy (SAF-M1) combines data of the 24-week placebo-controlled double-blind treatment periods in the initial trials (1218.16 and 1218.17) with data from the 78-week open-label uncontrolled extension (study 1218.40). It includes all patients who had completed the randomised treatment periods and continued into

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the extension phase regardless of the use of rescue medication. Essentially, this set comprises patients who were randomised to linagliptin in study 1218.16 and patients who were randomised to linagliptin and metformin (background therapy) in study 1218.17. Due to their origin in the different trials (1218.16 vs.1218.17), the groups differed already at baseline in a number of aspects such as race (Asian 46.4% vs. 23.5%), duration of diabetes more than 5 years (25.9% vs. 55.3%), and prior use of 2 or more antidiabetic drugs (0% vs. 31.4%). Thus, the treatment groups analysed are not the result of a randomisation procedure; however they had been subjected to the same protocol in the extension phase. The patient cohorts providing data until Week 78 represent a selection of those patients who had not had tolerability issues or had insufficient treatment response. This notwithstanding, premature discontinuation rates were similar and the predominant reasons were administrative issues and 'other' reasons accounting for 9.5% (Lina) and 11.7% (Lina+Met) of patients; 4.2% (Lina) and 3.6% (Lina+Met) of patients discontinued prematurely because of adverse events. Although this safety pooling includes patients who took rescue medication, the exposure with rescue medication contributed less than 10% to the total exposure. In SAF-M1 336 patients took linagliptin 5 mg and 523 patients linagliptin 5 mg and metformin as background medication. The planned study duration was a maximum of 102 weeks; the median treatment durations were 718 and 719 days and 85.1% (Lina) and 83.4% (Lina+Met) had completed the planned treatment period.

5.1.2 Adverse events with linagliptin monotherapy

The overall percentages of patients with adverse events in SAF-2 were comparable between treatments (placebo 53.8% vs. linagliptin 55.0%). (In the following, when percentages or numbers are given, the first [left-most] number always denotes the placebo group and the second [right-most] number the linagliptin group.) The proportions of patients with adverse events of mild (36.8% vs. 37.6%), moderate (15.6% vs. 15.6%), or severe (1.4% vs. 1.8%) intensity were similar. Drug-related adverse events were slightly more frequent in the linagliptin group than with placebo (8.5% vs. 10.4%). The most frequent drug-related adverse events were hypoglycaemia with a higher incidence in the linagliptin group (2.4% vs. 5.0%, occurring predominantly in a background of metformin+SU), and hyperglycaemia with a higher incidence in the placebo group (1.5% vs. 0.5%). The frequency of adverse events leading to premature discontinuations of study drug was higher in the placebo group than in the linagliptin group (3.6% vs. 2.3%). The difference was mainly due to higher incidences of hyperglycaemia (0.9% vs. 0.2%) and blood glucose increased (0.4% vs. 0.1%) in the placebo group. The incidence of serious adverse events was low in both treatment groups (2.5% vs. 2.7%). In SAF-2, there were 2 fatal serious adverse events in the linagliptin group and none in the placebo group (information on these cases is provided in the section below). A summary of adverse events by category is provided in the following table.

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Table 5.1.2: 1 Summary of adverse events for SAF-2 (all placebo-controlled trials) - TS

	Placebo N (%)	Linagliptin 5 mg N (%)
Number of patients	1183 (100.0)	2566 (100.0)
Patient years of exposure	433.8	1041.4
Patients with any adverse event	636 (53.8)	1412 (55.0)
Patients with adverse events of severe intensity	16 (1.4)	45 (1.8)
Patients with investigator-defined drug-related adverse events	101 (8.5)	268 (10.4)
Patients with adverse events of special interest [a]	10 (0.8)	16 (0.6)
Patients with adverse events leading to discontinuation	43 (3.6)	58 (2.3)
Patients with serious adverse events	29 (2.5)	69 (2.7)

[a] Hypersensitivity reactions, renal events, and hepatic events (based on investigator-reporting, excluding severe cutaneous adverse reactions and pancreatitis).

Source: [U10-1607, Module 5.3.5.3, Tables 5.1.2.1 and 4.2.1.1]

The analysis of adverse events revealed that on the level of system organ classes, there were only small differences between the treatment groups. However, on the level of preferred terms some differences were observed. As expected, hyperglycaemia occurred with a higher frequency in the placebo/background group than in the linagliptin group (10.6% vs. 5.0%). Conversely, hypoglycaemia was less frequent in the placebo group (4.1%) than in the linagliptin group (7.6%). Overall, infections and infestations were the most frequent adverse events. Musculoskeletal and connective tissue disorders; respiratory, thoracic and mediastinal disorders, and vascular disorders occurred with a slightly higher incidence in the linagliptin group, however the difference never exceeded 2%. In addition, slightly higher frequencies in the linagliptin group were also observed for skin and subcutaneous tissue disorders (2.6% vs. 4.0%); but no preferred term reached an overall incidence of 1% or was clearly increased compared with placebo. There were no indications of phototoxicity with linagliptin treatment. However, following a comprehensive search based on the High Level Term “photosensitivity conditions”, 4 patients with events were identified (including 2 cases without linagliptin treatment), none of the events was serious, 3 events were of mild and 1 of moderate intensity [phytophotodermatitis]. Following a medical evaluation of the events it was concluded that 3 cases had a clear alternative etiology and in 1 case an alternative etiology was likely (PatID 20465).

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Table 5.1.2: 2 Frequency of patients with adverse events occurring in more than 1% in either treatment group on the preferred term level, sorted by frequency in system organ class of the linagliptin group for the SAF-2 (all placebo controlled trials) - TS

System organ class Preferred term	Placebo N (%)	Linagliptin 5 mg N (%)
Number of patients	1183 (100.0)	2566 (100.0)
Patient years of exposure	433.8	1041.4
Patients with any adverse event	636 (53.8)	1412 (55.0)
Infections and infestations	244 (20.6)	491 (19.1)
Nasopharyngitis	65 (5.5)	150 (5.8)
Upper respiratory tract infection	53 (4.5)	84 (3.3)
Urinary tract infection	28 (2.4)	56 (2.2)
Influenza	16 (1.4)	37 (1.4)
Metabolism and nutrition disorders	208 (17.6)	408 (15.9)
Hypoglycaemia	49 (4.1)	195 (7.6)
Hyperglycaemia	125 (10.6)	128 (5.0)
Dyslipidaemia	13 (1.1)	31 (1.2)
Gastrointestinal disorders	127 (10.7)	269 (10.5)
Diarrhoea	27 (2.3)	53 (2.1)
Constipation	21 (1.8)	40 (1.6)
Nausea	14 (1.2)	28 (1.1)
Abdominal pain upper	15 (1.3)	18 (0.7)
Musculoskeletal and connective tissue disorders	102 (8.6)	264 (10.3)
Back pain	30 (2.5)	50 (1.9)
Arthralgia	21 (1.8)	47 (1.8)
Pain in extremity	11 (0.9)	34 (1.3)
Nervous system disorders	81 (6.8)	183 (7.1)
Dizziness	21 (1.8)	51 (2.0)
Headache	41 (3.5)	76 (3.0)
General disorders and administration site conditions	61 (5.2)	124 (4.8)
Asthenia	9 (0.8)	28 (1.1)
Fatigue	17 (1.4)	13 (0.5)
Investigations	49 (4.1)	102 (4.0)
Blood glucose increased	16 (1.4)	16 (0.6)
Respiratory, thoracic and mediastinal disorders	26 (2.2)	102 (4.0)
Cough	10 (0.8)	47 (1.8)
Vascular disorders	28 (2.4)	92 (3.6)
Hypertension	22 (1.9)	58 (2.3)

SAF-2 comprised trials that used the following background medications: none, metformin, metformin+SU, and 1 trial with an initial combination of pioglitazone.

Source data: [U10-1607, Module 5.3.5.3, Tables 5.2.2.1.1.4 and 4.2.1.1]

The most frequently reported drug-related adverse events were hypoglycaemia (PBO 2.4%, Lina 5.0%) and hyperglycaemia (PBO 1.5%, Lina 0.5%). This study grouping (SAF-2) included patients who received antidiabetic background medications such as metformin and sulfonylurea which are known to increase the risk of hypoglycaemia. And indeed, the higher for the overall incidence of hypoglycaemia originates from patients who received linagliptin added to a background of metformin and SU (Met+SU 14.8%, Lina+Met+SU 22.8%), but not as monotherapy or with other background medications.

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The proportions of patients who discontinued because of adverse events were small with 3.6% (PBO) and 2.3% (Lina). Adverse events belonging to the system organ class metabolism and nutrition disorders were the most frequent that led to discontinuation (1.4% vs. 0.5%). The most frequent adverse event which led to discontinuation was hyperglycaemia with 0.9% (PBO) and 0.2% (Lina). Hypoglycaemia caused 1 patient (0.1%) in the placebo group and 6 patients (0.2%) in the linagliptin group to discontinue treatment. No patient in the placebo group but 6 patients (0.2%) in the linagliptin group discontinued because of musculoskeletal and connective tissue disorders.
The numbers of patients with adverse events of special interest (based on narrow SMQs) were low and comparable between placebo and linagliptin: hepatic events (PBO 1.2%, Lina 1.0%), hypersensitivity reactions (PBO 0.5%, Lina 0.7%), and renal events (PBO 0.2%, Lina 0.1%). There was no case of severe cutaneous adverse reactions and no case of pancreatitis [U10-1607, Module 5.3.5.3, Tables 5.4.1.2.2, 5.4.2.2.2, 5.4.3.2.2, 5.4.4.2.2, and 5.4.5.2.2].

Deaths and other serious adverse events

There were 2 fatal events, both of them occurred in the linagliptin group: patient 50630, a 64-year old White female patient died due to cardio-respiratory arrest (study 1218.35); and patient 55591, a 68-year old Asian male patient, due to myocardial infarction (study 1218.50). None of the deaths was assessed as being related to the study medication.

In SAF-2, the frequency of serious adverse events was low and comparable between treatment groups (PBO 2.5% vs. Lina 2.7%). In the post-treatment period 2 patients who had been randomised to placebo and 5 patients who had received linagliptin had serious adverse events. In the system organ classes cardiac disorders (0.3% vs. 0.5%), skin and subcutaneous tissue disorders (0% vs. 0.1%), and vascular disorders (0.1% vs. 0.4%) slightly higher frequencies were observed in the linagliptin group than in the placebo group. Myocardial infarction (n=4), myocardial ischemia (n=1) and myocarditis (n=1) were only observed in the linagliptin group. For vascular disorders, the different incidences were mainly due to higher frequencies of hypertension (0.1%) and hypertensive crisis (0.1%) in the linagliptin group. The imbalance in serious adverse events is mainly brought about by the 1218.16 trial (0% vs. 0.4%). However, the apparent imbalance for vascular disorders and hypertension is not believed to be causally linked with linagliptin treatment because of differences at baseline.

Post treatment, in patients who had received placebo, diabetic retinopathy and bladder cancer were reported as serious adverse events. For patients who had received linagliptin, upper abdominal pain, pancreatitis, non-cardiac chest pain, post-operative wound infection, and decubitus ulcer were reported after stop of treatment.

Four patients (0.3%) in the placebo group and 14 patients (0.5%) in the linagliptin group had serious adverse events that caused discontinuation of study drug.

Adverse events with long-term treatment with linagliptin monotherapy SAF-M1

In the long-term set (SAF-M1) about 80% of patients had an adverse event and both treatment groups exhibited similar incidences over about 78 weeks. The incidence of adverse events of severe intensity was low (Lina 4.2%, Lina+Met 5.2%) as was the frequency of adverse events leading to discontinuation of the trial drug (Lina 3.6%, Lina+Met 3.4%).

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Table 5.1.2: 3 Adverse event overall summary for SAF-M1 (long-term) - TS

	Linagliptin N (%)	Lina+Met N (%)
Number of patients	336 (100.0)	523 (100.0)
Patient years of exposure	567.9	891.5
Patients with any adverse event	277 (82.4)	434 (83.0)
Patients with adverse events of severe intensity	14 (4.2)	27 (5.2)
Patients with investigator-defined drug-related adverse events	31 (9.2)	73 (14.0)
Patients with adverse events of special interest[1]	11 (3.3)	15 (2.9)
Patients with adverse events leading to discontinuation	12 (3.6)	18 (3.4)
Patients with serious adverse events	31 (9.2)	67 (12.8)

[1] Including hypersensitivity reactions, renal events, and hepatic events (based on investigator-reporting, excluding severe cutaneous adverse reactions and pancreatitis).

Source data: SAF-M1 [U11-1152, Module 5.3.5.3, Tables 8.2.1.1 and 8.2.2.1.1]

The most frequently reported adverse events were infections and infestations and metabolism and nutrition disorders. Differences between groups exceeding 5% were seen for infections and infestations (more in the Lina+Met group) mostly because of higher incidences of nasopharyngitis, urinary tract infections, and influenza. Metabolism and nutrition disorders were more frequent in the Lina group because of a higher incidence of hyperglycaemia. Hypoglycaemia, however, occurred with a lower frequency in the Lina group than in the Lina+Met group. Infections and infestations, musculoskeletal and connective tissue disorders, gastrointestinal disorders, and nervous system disorders occurred with a higher frequency in the Lina+Met group than in the Lina group. However, these apparent differences might be due to differences for concomitant diagnoses at baseline: infections and infestations (Lina 7.4% vs. Lina+Met 11.1%), musculoskeletal and connective tissue disorders (17.0% vs. 21.2%), gastrointestinal disorders (13.4% vs. 19.5%), and nervous system disorders (7.1% vs. 12.4%) (Data from study reports 1218.16 [U10-1103, Module 5.3.5.1] and 1218.17 [U09-2533, Module 5.3.5.1]) were less frequent in the Lina group than in the Lina+Met group. Also, respiratory, thoracic, and mediastinal disorders were more frequent in the Lina+Met group due to a higher frequency of cough. Hyperglycaemia was the most frequently reported preferred term in both groups.

Taken together, the analyis of SAF-M1 confirmed the safety analysis for the linagliptin monotherapy. However, on the level of preferred terms, higher incidences of nasopharyngitis, urinary tract infection, influenza, diarrhea, and cough were observed with Lina+Met combination therapy. A summary of adverse events for SAF-M1 is provided below.

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Table 5.1.2: 4 Frequency of patients with adverse events occurring in more than 5% in either treatment group on the preferred term level with their SOC, sorted by frequency in the Lina+Met group for SAF-M1 - TS

System organ class / Preferred term	Linagliptin N (%)	Lina+Met N (%)
Number of patients	336 (100.0)	523 (100.0)
Patient years of exposure	567.9	891.5
Patients with adverse events	277 (82.4)	434 (83.0)
Infections and infestations	109 (32.4)	232 (44.4)
Nasopharyngitis	23 (6.8)	57 (10.9)
Urinary tract infection	8 (2.4)	51 (9.8)
Upper respiratory tract infection	20 (6.0)	41 (7.8)
Influenza	8 (2.4)	33 (6.3)
Metabolism and nutrition disorders	160 (47.6)	223 (42.6)
Hyperglycaemia	123 (36.6)	153 (29.3)
Hypoglycaemia	8 (2.4)	33 (6.3)
Hypertriglyceridaemia	8 (2.4)	31 (5.9)
Musculoskeletal and connective tissue disorders	67 (19.9)	147 (28.1)
Back pain	15 (4.5)	41 (7.8)
Arthralgia	12 (3.6)	31 (5.9)
Gastrointestinal disorders	58 (17.3)	137 (26.2)
Diarrhoea	10 (3.0)	35 (6.7)
Nervous system disorders	38 (11.3)	97 (18.5)
Headache	15 (4.5)	34 (6.5)
Respiratory, thoracic and mediastinal disorders	23 (6.8)	57 (10.9)
Cough	9 (2.7)	32 (6.1)
Vascular disorders	30 (8.9)	54 (10.3)
Hypertension	20 (6.0)	41 (7.8)

Source data: [U11-1152, Module 5.3.5.3, Table 8.2.2.1.2 and Table 8.2.1.1]

Deaths and other serious adverse events during long-term treatment (SAF-M1)

The incidence of serious adverse events was lower in the Lina group than in the Lina+Met group (9.2% vs. 12.8%). Cardiac disorders (1.2% vs. 2.5%), renal and urinary disorders (0% vs. 1.7%), nervous system disorders (0.3% vs. 1.1%), and chest pain (0% vs. 1.0%) were more frequent in the Lina+Met group. Conversely vascular disorders were more frequent with Lina mono (1.5% vs. 0.2%). There were 2 fatal SAEs in the Lina group: patient 95701, a 63-year old White male patient died due to pulmonary embolism; and patient 93343, a 61-year old White male patient died due to cardiac tamponade. Both patients had continued from study 1218.16 into 1218.40; none of the deaths was assessed as being related to the study medication [U11-1152, Module 5.3.5.3, Listing 5.5.1]. An overview of serious adverse events is provided below.

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Table 5.1.2: 5 Frequency of patients with serious adverse events occurring in 1% or more of patients in either treatment group on the SOC level, sorted by frequency in the Lina+Met group for SAF-M1 - TS

	Lina N (%)	Lina+Met N (%)
Number of patients	336 (100.0)	523 (100.0)
Patient years of exposure	567.9	891.5
Patients with serious adverse events	31 (9.2)	67 (12.8)
Cardiac disorders	4 (1.2)	13 (2.5)
Infections and infestations	5 (1.5)	9 (1.7)
Renal and urinary disorders	0	9 (1.7)
Injury, poisoning and procedural complications	5 (1.5)	8 (1.5)
Musculoskeletal and connective tissue disorders	3 (0.9)	6 (1.1)
Neoplasms benign, malignant and unspecified (incl. cycsts and polyps)	3 (0.9)	6 (1.1)
Nervous system disorders	1 (0.3)	6 (1.1)
General disorders and administration site conditions	0	6 (1.1)
Chest pain	0	5 (1.0)
Vascular disorders	5 (1.5)	1 (0.2)

Source data: [U11-1152, Module 5.3.5.3, Table 8.2.2.1.9]

5.1.3 Clinical laboratory evaluation and vital signs for linagliptin monotherapy (SAF-2)

With regard to transitions, overall no clinically relevant findings or marked differences between placebo and linagliptin 5 mg were observed for any of the measured parameters (haematology, electrolytes, enzymes, substrates, and urine analysis). Further details are provided in [U10-1607, Module 5.3.5.3, Table 6.2.2.1].

PCSAs were reported for a several parameters (haematology and clinical chemistry), with low (<3.0%) and comparable incidences in both treatment groups. None of the observed increases or decreases was considered to be of clinical importance.

No patient fulfilled the criteria for Hy's law; the vast majority of patients (about 99%) in both treatment groups were categorised as having enzyme and total bilirubin levels of grade 0, indicating a very low risk of a potential liver injury. In both treatment groups, a few patients had laboratory values of grade 1 (placebo ≤1.0 %; linagliptin ≤0.8%). Very few patients in both treatment groups exhibited grade 2 and 3 laboratory findings. Further details are given in [U10-1607, Module 5.3.5.3, Table 6.3.2.1].

Based on MDRD staging, the majority of patients (placebo 71.2%, linagliptin 81.0%) had either normal renal function (eGFR ≥90 mL/min) or mild renal impairment (eGFR 60 to <90 mL/min) at baseline. The majority of patients maintaind their normal renal function or their mild renal impairment throughout the trials. In both treatment groups, the proportion of patients with moderate renal impairment (eGFR 30 to <60 mL/min) was low at baseline (PBO 3.8% , Lina 4.1%). The maximum frequency of patients with moderate renal impairment throughout the trials was 4.8% of patients in the placebo group and 6.1% of patients in the linagliptin group. Severe or end-stage renal impairment (eGFR <30 mL/min) was reported for 1 patient (0.1%) in the placebo group, both at baseline and throughout the

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trial. In the linagliptin group, 2 patients (0.1%) had severe or end-stage renal impairment at baseline and shifted to moderate renal impairment at the end of the trial (last value on treatment). During the observation period, 1 patient with normal renal function and 3 patients (0.1%) with moderate renal impairment at baseline shifted to severe or end-stage renal impairment in the linagliptin group. At the end of the trials, 2 patients (0.1%) were categorised as having severe or end-stage renal impairment in the linagliptin group (both having moderate renal impairment at baseline). The proportion of patients with missing values at baseline was relatively high (placebo 24.9%, linagliptin 14.8%). Further details are given in [U10-1607, Module 5.3.5.3, Table 6.6.2.1].

For SAF-M1 (long-term safety of linagliptin monotherapy) DILIs and changes in renal function were analysed. Overall, no liver toxicity was detected and the potential to induce liver toxicities was judged to be very low. The majority of patients in both treatment groups did not fulfil any of the defined criteria for evaluation of drug-induced liver injuries (96.7% Lina, 96.6% Lina+Met). No patient fulfilled the criteria for Hy's law. Further details are available in SCS [Module 2.7.4, Section 3.1]. Regarding renal function, the analysis of shifts from baseline to end of trial and to minimum or maximum value post-baseline did not indicate any difference between the Lina group and the Lina+Met group over 78 weeks of treatment. Details are available in the SCS [Module 2.7.4, Section 3.1]. Adverse events leading to discontinuation were checked manually with regard to laboratory abnormalities. Both treatment groups were comparable with regard to adverse events leading to discontinuation on the SOC investigations (Lina 0.9%, Lina+Met 1.0%). Further details are given in [U11-1152, Module 5.3.5.3, Table 8.2.2.1.6].

Vital signs

In the linagliptin monotherapy dossier, analyses of vital signs were not performed on SAF-2 (all placebo-controlled trials). For SAF-M1 vital signs were analysed by visit and for changes from baseline. In the Lina group, mean blood pressure and mean pulse rate remained largely unchanged (maximum changes in DBP and SBP <1.5 mmHg). In the Lina+Met group, a slight decrease in mean blood pressure and mean pulse rate was observed; with a maximum change from baseline for SBP of -1.8 mmHg, for DBP of -1.7 mmHg, and for pulse rate of -1.2 bpm. However, at Week 102, the mean changes were below 1 mmHg in both treatment groups in both systolic and diastalic blood pressure [U11-1152, Module 5.3.5.3, Table 8.4.1].

5.2 SAFETY OF METFORMIN MONOTHERAPY

As metformin is in clinical use for some 50 years in the EU, its safety profile is well established. Clinical trials have shown that metformin is generally well-tolerated and safe. The majority of side effects are gastrointestinal in nature and include nausea, vomiting, diarrhoea, abdominal pain, appetite loss, and taste disturbance. A large number of these gastrointestinal effects are mild and can be reduced by taking metformin with food, initiating therapy with the lowest possible dose, and slow dose titration. Nevertheless, approximately 5% of patients discontinue metformin therapy because they are intolerant to its gastrointestinal side effects [P97-1130, P10-09483, R10-4882, Module 2.7.5]. Furthermore, due to the gastrointestinal side effects of metformin, many patients are unable to take the recommended daily dose of at least 1500 mg metformin [R10-4375, Module 2.7.5].

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The most serious adverse event associated with metformin use is lactic acidosis. Lactic acidosis is rare with approximately 0.03 incidences per 1000 patient-years, but is associated with a high mortality. The risk of lactic acidosis is a concern in cases of metformin overdose or in special populations, such as patients with moderate renal impairment and those with congestive heart failure. Thus, while it is a serious adverse reaction, the risk of lactic acidosis can be minimised by adherence to prescribing recommendations and close monitoring of renal and cardiac function [R09-4255, R10-0861, R10-4375, Module 2.7.5].

In several studies, patients taking metformin experienced significantly fewer diabetes-related complications, including hypoglycaemia, than those taking other oral antidiabetic medications [R10-4882, Module 2.7.5]. In the United Kingdom Prospective Diabetes Study (UKPDS), metformin treatment was associated with 29.8 diabetes-related complications per 1000 patient-years, combined SU and insulin monotherapy was associated with 40.1 events per 1000 patient-years, and diet alone was associated with 43.3 events per 1000 patient years. Furthermore, UKPDS provided evidence that patients treated with metformin experience a significant reduction in the absolute risk of myocardial infarction [R09-4255, R10-0861, Module 2.7.5].

For this submission no dedicated studies with metformin monotherapy were conducted, however information on the safety of metformin monotherapy is included in the safety groupings in the respective treatment arms of some studies.

5.3 SAFETY OF COMBINATION THERAPY WITH LINAGLIPTIN + METFORMIN

5.3.1 Exposure to combination therapy with linagliptin + metformin

Overall, median exposures were similar between treatment groups in all study groupings. In SAF-C5 and SEA-2 (pooled metformin background studies) the planned study durations ranged from 12 weeks (1218.6 and 1218.62) to 24 weeks (1218.17 and 1218.46); however since the latter trials comprised more patients, the median exposures in both treatment groups were close to 24 weeks.

In SAF-C2 (active-controlled long-term study), 776 patients received Lina+Met and 775 patients received Lina+glimepiride (Glim). The planned study duration was 104 weeks. A large proportion of patients in both treatment groups (Lina+Met 80.5%, Glim+Met 81.3%) was exposed to randomised study medication for 78 weeks or more. The mean exposure was comparable: 627 days (Lina+Met) and 625 days (Glim+Met).

SAF-C6 (long-term safety, n=567 [switchers and non-switchers]), included patients who completed the 24 weeks of study 1218.46, did not use rescue medication, and continued in the double-blind extension (1218.52). The planned treatment duration of the extension trial was 54 weeks. At the time of the interim analysis about 50% of patients had been treated in the extension trial for 24 weeks or more.

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Table 5.3.1: 1 Exposure to study medication for SAF-C1, SAF-C5, and SAF-C6 - TS

	SAF-C1 (pivotal trial)		SAF-C5 (metformin background)		SAF-C6 (long term set)		
	Met	Lina+Met	Met	Lina+Met	Met1000	Lina2.5+ Met500	Lina2.5+ Met1000
Patients, N (%)	177 (100.0)	523 (100.0)	583 (100.0)	1388 (100.0)	147 (100.0)	143 (100.0)	143 (100.0)
Exposure, N (%)							
≥12 weeks	169 (95.5)	503 (96.2)	531 (91.1)	1260 (90.8)	134 (91.2)	132 (92.3)	135 (94.4)
≥24 weeks	161 (91.0)	455 (87.0)	394 (67.6)	690 (49.7)	128 (87.1)	126 (88.1)	131 (91.6)
≥52 weeks	0	0	0	0	74 (50.3)	74 (51.7)	76 (53.1)
≥78 weeks	0	0	0	0	3 (2.0)	3 (2.1)	4 (2.8)
Exposure [days]							
Mean (±SD)	165 (31.9)	167 (29.1)	146 (46.2)	130 (48.0)	327 (144.1)	333 (135.1)	335 (131.9)
Median (min, max)	171 (2, 196)	171 (1, 208)	169 (1, 196)	167 (1, 208)	365 (1, 558)	373 (3, 549)	369 (16, 561)
Overall patient years	80.1	238.6	232.5	495.3	131.8	130.4	131.1

SAF-C1: Linagliptin was administered as 5 mg qd, metformin was given as background medication.

SAF-C5: Linagliptin was administered as 2.5 mg bid, 5 mg qd, or 10 mg qd; metformin was given as background medication, monotherapy (500 mg or 1000 mg, bid), or free combination therapy (500 mg or 1000 mg, together with linagliptin 2.5 mg; all bid).

SAF-C6: Patients were treated with metformin 1000 mg bid, linagliptin 2.5 mg + metformin 500 mg (bid), or linagliptin 2.5 mg + metformin 1000 mg (bid). Only patients who did not switch treatments were included in the analysis.

Source data: SAF-C1 [U11-1152, Module 5.3.5.3, Table 4.1.1], SAF-C5 [U11-1152, Module 5.3.5.3, Table 4.5.1.1] and SAF-C6 [U11-1152, Module 5.3.5.3, Table 4.6.1]

In the pivotal trial (SAF-C1), similar proportions of patients in both treatment groups completed the study (Met 92.1%, Lina+Met 92.5%). Hence, the frequencies of premature discontinuations were also similar in both groups (Met 7.9%, Lina+Met 7.5%). The most frequent reason for premature discontinuation was refusal to continue medication (Met 2.3%, Lina+Met 2.5%), followed by the occurrence of adverse events (1.7% in both treatment groups).

In the pooled metformin background studies (SAF-C5), 88.5% (Met) and 92.2% (Lina+Met) of patients completed the planned treatment period. The proportion of premature discontinuation was higher in the Met group (11.5%) than in the Lina+Met group (7.8%). The most frequent reason for premature discontinuation in both treatment groups were administrative issues, i.e. non-compliance with the protocol, lost to follow-up, and refusal to continue with trial medication (Met 5.5%, Lina+Met 3.7%). As for exposure, the discontinuation rates for the SEA-2 were very similar as for SAF-C5.

In the active-controlled long-term set (SAF-C2) similar proportions of patients completed the trial period (Lina+Met 75.6% vs. Glim+Met 77.9%); the most frequent reasons for premature discontinuation were adverse events (Lina+Met 7.9% vs. Glim+Met 11.6%).

In the long-term safety set (SAF-C6), overall 80.4% completed the planned treatment period. The proportions of premature discontinuations were lowest in the Lina2.5+Met1000 group (16.1%) and highest in the Lina2.5+Met500 group (23.1%). The most frequent reasons for premature discontinuation were administrative issues (i.e. non-compliance with the protocol, lost to follow-up, and refusal to continue with trial medication) ranging from 5.6% in the

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Met1000 group to 10.5% in the Lina2.5+Met500 group and adverse events ranging from 4.9% in the Lina2.5+Met1000 group to 6.8% in the Met1000 group.

5.3.2 Adverse events with combination therapy with linagliptin + metformin

In the pivotal trial 1218.17 (SAF-C1) the incidences of adverse events were comparable in the metformin (57.1%) and the linagliptin + metformin group (53.9%). The incidence of adverse events of severe intensity was slightly higher in the Lina+Met group (1.1% vs. 2.1%). Conversely the adverse events considered related to study drug by the investigators were more frequent in the metformin group (11.3% vs. 7.3%). The proportions of patients who discontinued because of adverse events (2.3% vs. 1.7%) were similar in both treatment groups as was the incidence of serious adverse events (2.8% vs. 3.4%).

Table 5.3.2: 1 Adverse event overall summary for SAF-C1, SAF-C5, and SEA-2 - TS

	SAF-C1 (pivotal trial)		SAF-C5 (Met vs. Lina+Met)		SEA-2 (Met vs. Lina5+Met)	
	Met N (%)	Lina+Met N (%)	Met N (%)	Lina+Met N (%)	Met N (%)	Lina+Met N (%)
Patients (100.0%)	177	523	583	1388	583	1322
Patients with AEs	101 (57.1)	282 (53.9)	295 (50.6)	660 (47.6)	295 (50.6)	632 (47.8)
Patients with AEs of severe intensity	2 (1.1)	11 (2.1)	10 (1.7)	29 (2.1)	10 (1.7)	27 (2.0)
Patients with investigator-defined drug-related AEs	20 (11.3)	38 (7.3)	51 (8.7)	101 (7.3)	51 (8.7)	95 (7.2)
Patients with AEs of special interest[1]	3 (1.7)	2 (0.4)	10 (1.7)	11 (0.8)	n.a.	n.a.
Patients with AEs leading to discontinuation	4 (2.3)	9 (1.7)	15 (2.6)	33 (2.4)	15 (2.6)	31 (2.3)
Patients with SAEs	5 (2.8)	18 (3.4)	15 (2.6)	41 (3.0)	15 (2.6)	37 (2.8)

n.a. = not applicable, was not performed for SEA-2
AEs = adverse events; SAEs = serious adverse events
SAF-C5 and SEA-2 are identical except for 66 patients excluded from SEA-2 because they had taken 10 mg linagliptin.
SAF-C1: Linagliptin was administered as 5 mg qd, metformin was given as background medication.
SAF-C5: Linagliptin was administered as 2.5 mg bid, 5 mg qd, or 10 mg qd; metformin was given as background medication, monotherapy (500 mg or 1000 mg, bid), or free combination therapy (500 mg or 1000 mg, together with linagliptin 2.5 mg; all bid).
[1] Including hypersensitivity reactions, renal events, and hepatic events (based on investigator-reporting, excluding severe cutaneous adverse reactions and pancreatitis).

Source data: SAF-C1 [U11-1152, Module 5.3.5.3, Table 5.1.1.1], SAF-C5 [U11-1152, Module 5.3.5.3, Table 5.1.5.1], , SEA-2 [U11-1152, Module 5.3.5.3, Tables 4.1 to 4.6]

Generally, for the pivotal trial (SAF-C1) the overall frequency of adverse events and the frequencies in the various subcategories of adverse events were similar to those observed in the placebo and linagliptin groups in SAF-2 (i.e. for linagliptin monotherapy). The only exceptions were the frequencies of drug-related adverse events which were higher in the Met group of SAF-C1 than in the placebo arm of SAF-2 (11.3% vs. 8.5%) and somewhat lower in the Lina+Met group of SAF-C1 than in the linagliptin group of SAF-2 (7.3% vs. 10.4%) [U10-1607, Module 5.3.5.3, Table 5.1.2.1]. Furthermore, in regard to adverse event categories the observations made for the pivotal trial were corroborated by the analysis of the pool of placebo-controlled metformin background trials (SAF-C5 and SEA-2). Both overall and for individual adverse event categories similar incidences were observed.

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The analyses demonstrated that the incidences of the most frequent adverse events were similar in both treatment groups in SAF-C1. The most frequent adverse events on SOC level were infections and infestations (22.0% vs. 21.6%) and gastrointestinal disorders (11.9% vs. 11.1%) both with almost identical incidences in both treatment groups. Differences of 2% or more were observed for musculoskeletal and connective tissue disorders which were more frequent in the Lina+Met group (7.9% vs. 11.1%). Also injury, poisoning and procedural complications (2.3% vs. 5.0%) and respiratory, thoracic and mediastinal disorders (2.8% vs. 4.8%) were slightly more frequent in the Lina+Met group than in the Met group. Conversely, metabolism and nutrition disorders were substantially less frequent in the Lina+Met group than in the Met group (23.7% vs. 10.1%) mainly due to a lower incidence of hyperglycaemia (16.4% vs. 6.1%). Also hypoglycaemia was less frequent with Lina+Met combination therapy than with metformin alone (2.8% vs. 0.6%). Furthermore, investigations (8.5% vs. 4.0%) and psychiatric disorders (2.8% vs. 0.8%) were less frequent with the combination therapy than with metformin alone. Comparing the frequencies of adverse events in the Lina+Met group of the pivotal trial with those of the linagliptin group of SAF-2 (Lina mono submission) demonstrates similar incidences across system organ classes.

The comparison of the adverse event incidences in the pivotal trial (SAF-C1) with the incidences in the pooled analysis of all placebo-controlled trials with metformin background (SAF-C5, SEA-2) indicates a consistent safety profile of the Lina+Met combination. Differences in adverse events incidences found for the pivotal trial were also seen in the other analysis sets. Otherwise, the incidences of adverse events and SOCs were generally very similar in SAF-C1, SAF-C5, and SEA-2 (Table 5.3.2: 2).

Table 5.3.2: 2 Adverse events occurring in more than 2% of patients in either treatment group on the preferred term level with their SOCs in SAF-C1 and/or SAF-C5 and SOCs with frequencies above 2% in either treatment group in SAF-C5, sorted by frequency in the Lina+Met group of SAF-C5 - TS

	SAF-C1		SAF-C5		SEA-2	
	Met N (%)	Lina+Met N (%)	Met N (%)	Lina+Met N (%)	Met N (%)	Lina+Met N (%)
Patients (100.0%)	177	523	583	1388	583	1322
Patients with any adverse events	101 (57.1)	282 (53.9)	295 (50.6)	660 (47.6)	295 (50.6)	632 (47.8)
Infections and infestations	39 (22.0)	113 (21.6)	113 (19.4)	257 (18.5)	113 (19.4)	247 (18.7)
Nasopharyngitis	9 (5.1)	27 (5.2)	23 (3.9)	61 (4.4)	23 (3.9)	57 (4.3)
Urinary tract infection	9 (5.1)	16 (3.1)	18 (3.1)	39 (2.8)	18 (3.1)	37 (2.8)
Upper resp. tract infection*	4 (2.3)	15 (2.9)	13 (2.2)	32 (2.3)	13 (2.2)	32 (2.4)
Influenza	5 (2.8)	18 (3.4)	18 (3.1)	28 (2.0)	18 (3.1)	28 (2.1)
Gastrointestinal disorders	21 (11.9)	58 (11.1)	68 (11.7)	155 (11.2)	68 (11.7)	147 (11.1)
Diarrhoea	4 (2.3)	15 (2.9)	20 (3.4)	43 (3.1)	20 (3.4)	41 (3.1)
Nausea	3 (1.7)	6 (1.1)	12 (2.1)	24 (1.7)	12 (2.1)	21 (1.6)
Abdominal pain	4 (2.3)	2 (0.4)	5 (0.9)	4 (0.3)	5 (0.9)	4 (0.3)
Musculoskeletal and connective tissue disorders	14 (7.9)	58 (11.1)	45 (7.7)	127 (9.1)	45 (7.7)	124 (9.4)
Back pain	5 (2.8)	12 (2.3)	12 (2.1)	32 (2.3)	12 (2.1)	32 (2.4)
Arthralgia	3 (1.7)	11 (2.1)	11 (1.9)	24 (1.7)	11 (1.9)	24 (1.8)

(Table continued on next page)

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Table 5.3.2: 2 Adverse events occurring in more than 2% of patients in either treatment group on the preferred term level with their SOCs in SAF-C1 and/or SAF-C5 and SOCs with frequencies above 2% in either treatment group in SAF-C5, sorted by frequency in the Lina+Met group of SAF-C5 - TS - continued

	SAF-C1		SAF-C5		SEA-2	
	Met N (%)	Lina+Met N (%)	Met N (%)	Lina+Met N (%)	Met N (%)	Lina+Met N (%)
Patients (100.0%)	177	523	583	1388	583	1322
Patients with any adverse events	101 (57.1)	282 (53.9)	295 (50.6)	660 (47.6)	295 (50.6)	632 (47.8)
Metabolism and nutrition disorders	42 (23.7)	53 (10.1)	74 (12.7)	96 (6.9)	74 (12.7)	95 (7.2)
Hyperglycaemia	29 (16.4)	32 (6.1)	46 (7.9)	49 (3.5)	46 (7.9)	49 (3.7)
Hypoglycaemia	5 (2.8)	3 (0.6)	12 (2.1)	14 (1.0)	12 (2.1)	14 (1.1)
Hypertriglyceridaemia	5 (2.8)	2 (0.4)	9 (1.5)	4 (0.3)	9 (1.5)	4 (0.3)
Nervous system disorders	9 (5.1)	35 (6.7)	37 (6.3)	87 (6.3)	37 (6.3)	85 (6.4)
Headache	7 (4.0)	15 (2.9)	19 (3.3)	33 (2.4)	19 (3.3)	33 (2.5)
General disorders and administration site conditons	7 (4.0)	19 (3.6)	37 (6.3)	51 (3.7)	37 (6.3)	46 (3.5)
Investigations	15 (8.5)	21 (4.0)	37 (6.3)	51 (3.7)	37 (6.3)	48 (3.6)
Blood glucose increased	7 (4.0)	5 (1.0)	9 (1.5)	6 (0.4)	9 (1.5)	5 (0.4)
Injury, poisoning and procedural complications	4 (2.3)	26 (5.0)	13 (2.2)	50 (3.6)	13 (2.2)	49 (3.7)
Respiratory, thoracic and mediastinal disorders	5 (2.8)	25 (4.8)	14 (2.4)	49 (3.5)	14 (2.4)	47 (3.6)
Cough	3 (1.7)	11 (2.1)	5 (0.9)	24 (1.7)	5 (0.9)	23 (1.7)
Vascular disorders	7 (4.0)	23 (4.4)	22 (3.8)	39 (2.8)	22 (3.8)	38 (2.9)
Hypertension	6 (3.4)	17 (3.3)	17 (2.9)	29 (2.1)	17 (2.9)	28 (2.1)
Skin and subcutaneous tissue disorders	5 (2.8)	18 (3.4)	18 (3.1)	39 (2.8)	18 (3.1)	37 (2.8)
Cardiac disorders	2 (1.1)	12 (2.3)	8 (1.4)	29 (2.1)	8 (1.4)	26 (2.0)
Renal and urinary disorders	5 (2.8)	13 (2.5)	14 (2.4)	29 (2.1)	14 (2.4)	25 (1.9)
Psychiatric disorders	5 (2.8)	4 (0.8)	13 (2.2)	13 (0.9)	13 (2.2)	11 (0.8)

*Upper respiratory tract infection

SAF-C1: Linagliptin was administered as 5 mg qd; metformin was given as background medication.
SAF-C5 and SEA-2 are identical except for 66 patients excluded from SEA-2 because they had taken 10 mg linagliptin.

Source data: SAF-C1 [U11-1152, Module 5.3.5.3, Tables 5.2.1.1, 5.2.1.9, 5.2.5.1.1., 5.2.5.1.9; SEA-2 Table 4.1]

SAF-C5 comprised studies with different treatment durations (12 to 24 weeks) and different allocation ratios between the metformin and the Lina+Met group. Hence the overall exposure in the metformin group (232.5 PY) was lower than in the Lina+Met group (495.3 PY). This was also the case for SEA-2 (Met 232.5 PY vs. Lina+Met 480.3 PY). Therefore adverse events were additionally analysed using an exposure corrected approach, i.e. incidences per 100 PY (incidence densities). The overall adverse event rates were similar. However, some adverse events had higher incidence densities in the Lina+Met group. The biggest difference (6.58/100 PY) between the treatment groups was observed for musculoskeletal and connective tissue disorders mainly due to the absence of muscle spasms, bursitis, and osteoarthritis in the metformin group. Substantial differences were seen for injuries,

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poisoning and procedural complications (4.49/100PY) but without any particularly frequent event. Also respiratory, thoracic and mediastinal disorders had a higher incidence density (3.83/100 PY) in the Lina+Met group which was mainly accounted for by higher rates of cough. The higher rates for cardiac disorders in the Lina+Met group were due to slightly higher incidence densities of most preferred terms in this SOC without any particularly frequent event. The difference between treatmeent groups for reproductive system and breast disorders (1.87/100 PY) is mainly accounted for by the differences in benign prostatic hyperplasia (Met 0 vs. Lina+Met 0.98) and erectile dysfunction (Met 0 vs. Lina+Met 0.59).

Table 5.3.2: 3 Adverse events with a more than 2.0 (per 100 patient years) higher incidence density on SOC level in the Lina+Met group than in the Met group, sorted by the incidence densities in the Lina+Met group with PTs exhibiting a more than 1.0 (per 100 patient years) higher incidence density in the Lina+Met group than in the Met group - SAF-C5 - TS

	SAF-C5			
	Met		Lina+Met	
	N (%)	Rate / 100 patient years	N (%)	Rate / 100 patient years
Number of patients	583 (100.0)		1388 (100.0)	
Patients with any adverse event	295 (50.6)	187.85	660 (47.6)	191.99
Infections and infestations	113 (19.4)	53.45	257 (18.5)	56.59
Nasopharyngitis	23 (3.9)	9.88	61 (4.4)	12.23
Fungal infection	0	0	6 (0.4)	1.18
Respiratory tract infection	0	0	6 (0.4)	1.18
Musculoskeletal and connective tissue disorders	45 (7.7)	19.81	127 (9.1)	26.39
Back pain	12 (2.1)	5.11	32 (2.3)	6.36
Muscle spasms	0.2	0.42	11 (0.8)	2.17
Bursitis	0	0	8 (0.6)	1.58
Osteoarthritis	0	0	8 (0.6)	1.58
Injury, poisoning and procedural complications	13 (2.2)	5.55	50 (3.6)	10.04
Respiratory, thoracic and mediastinal disorders	14 (2.4)	6.00	49 (3.5)	9.83
Cough	5 (0.9)	2.12	24 (1.7)	4.76
Cardiac disorders	8 (1.4)	3.39	29 (2.1)	5.76

SAF-C5: Linagliptin was administered as 2.5 mg bid, 5 mg qd, or 10 mg qd; metformin was given as background medication, monotherapy (500 mg or 1000 mg, bid), or free combination therapy (500 mg or 1000 mg, together with linagliptin 2.5 mg; all bid).SAF

Source data: SAF-C5 [U11-1152, Module 5.3.5.3, Table 5.2.5.1.1]

5.3.2.1 Adverse event with the combination therapy with linagliptin and metformin in relevant subgroups

Since SAF-C5 was the largest set of trials with metformin background this set was used for the analysis of adverse events in relevant subgroups. The results for the subgroups are summarized below. Despite the fact that overall frequencies of drug-related adverse events, of adverse events leading to discontinuation and of serious adverse events were low, subgroup analysis were nevertheless performed. However, due to the low number of patients per subgroup category, clinically meaningful conclusions can not be based on these analyses. Therefore in the following, only the subgroups for common adverse events are described. The

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description is predominately based on exposure corrected incidences (incidence densities = events/100 PY). The subgroup analysis based on degrees of hepatic impairment could not be performed because this grouping comprised only 42 patients with hepatic impairment (defined as 2xULN of AST or 2xULN of ALT) out of which 27 patients had an adverse event.

Age: The overall incidence of adverse events was higher in the Met group than in the Lina+Met group in all age categories analysed with 50.6% vs. 47.8% in patients younger than 65 years of age, and with 49.2% vs. 45.8% in patients of 65 to 74 years of age. In patients 75 years of age or older, the overall frequency of adverse events appeared to be higher in both treatment groups (66.7% vs. 52.6%), however, this is based on a low number of patients in this age category (Met n=12; Lina+Met n=57) which prevents a comparison of treatments on SOC or PT level. Qualitatively the same differences as for the overall population were observed. [U11-1152, Module 5.3.5.3, Table 5.2.5.1.1]

Gender: For male patients, the overall incidences of adverse events were higher in the Met group than in the Lina+Met group (52.5% vs. 45.8%), while for female patients the incidences were similar (48.2% vs. 49.7%). The analysis by SOC and PT revealed some differences between genders and between treatment groups. Blood and lymphatic system disorders occurred in men with similar incidence densities in both treatment groups (3.79 vs. 3.66), in women however the incidence density in the Lina+Met group was higher (0.96 vs. 5.20). This difference was predominantly due to a higher rate of anaemia (0 vs. 3.02) in the Lina+Met group. Conversely, for cardiac disorders a difference between treatment groups was seen in men (3.01 vs. 6.97) mostly due to higher incidence densities of angina pectoris (0 vs. 1.45), and ventricular extrasystoles (0 vs. 1.45); however there was no difference in incidence densities between treatments in women (3.88 vs. 4.33). Musculoskeletal and connective tissue disorders were substantially more frequent in women (25.22 vs. 30.81) than in men (15.63 vs. 22.74); nevertheless the difference between the treatment groups was preserved. For women, the rate of gastrointestinal disorders was higher in the Lina+Met group (31.64 vs. 36.96) but this was not seen for men (30.98 vs. 29.32). For infections and infestations, in women the rate was higher in the Lina+Met group (53.19 vs. 66.37) while in was lower for men (53.64 vs. 48.55). For respiratory disorders and for injuries higher rates in the Lina+Met group were seen in both genders in a similar magnitude as for the overall population. [U11-1152, Module 5.3.5.3, Table 5.2.5.2.2.1]

Race: The analysis of the race subgroup was restricted to patients of White and of Asian race because only 17 patients in the SAF-C5 grouping were of Black race. Both for patients of White race and patients of Asian race, the frequencies of adverse events were higher in the Met group than in the Lina+Met group (White: 49.0% vs. 46.8%; Asian: 55.7% vs. 48.9%). For patients of White race, the incidence densities for cardiac disorders were lower in the Met group than in the Lina+Met group (2.94 vs. 7.12); this was not seen in Asian patients (4.68 vs. 2.25). While blood and lymphatic tissue disorders were of similar frequency in White patients in both treatment groups (1.18 vs. 2.71) the frequencies were higher in Asian patients (6.27 vs. 9.15). Likewise for infections and infestations, the treatment groups exhibited similar rates/100 PY for White patients (58.43 vs. 55.44) while for Asian patients a difference between the treatment groups was seen (38.98 vs. 57.21). [U11-1152, Module 5.3.5.3, Table 5.2.5.2.3.1]

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Ethnicity: Compared with the overall population the frequencies of adverse events were higher in patients of Hispanic/Latino ethnicity (63.9% vs. 56.3%), however this is based on a low number of Hispanic patients (Met n=72; Lina+Met n=158). The overall incidences of adverse events seen in non-Hispanic patients were similar to those observed for the entire population (49.0% vs. 46.5%). In regard to SOC and PT similar trends as for the overall population were observed for both Hispanic and non-Hispanic patients. Nevertheless, for infections and infestations the incidence densities were substantially higher in Hispanic patients (95.13 vs. 81.12) than in non-Hispanic patients (48.10 vs. 53.33); but the rates were comparable between treatment groups. For patients of Hispanic ethnicity, vascular disorders were more frequent in the Lina+Met group (3.18 vs. 15.27) predominantly because of higher incidence densities for hypertension (0 vs. 10.56). Similarly, for renal and urinary disorders an imbalance was seen (3.14 vs. 5.93). [U11-1152, Module 5.3.5.3, Table 5.2.5.2.4.1]

Geographic region: With regard to the incidences of adverse events analysed by geographical region, the observations were generally in line with those made for race and ethnicity. The number of patients from Africa was too small (total n = 23) to allow interpretation. The incidence of adverse events was higher in North America (Met 66.1% vs. Lina+Met 59.8%) and South America (62.7% vs. 58.0%) than in Europe (40.3% vs. 39.6%) and Asia (55.7% vs. 49.0%). Further details are available in the SCS [Module 2.7.4, Section 5.2.2.1 and [U11-1152, Module 5.3.5.3, Table 5.2.5.2.5.1].

Renal impairment: The analysis of renal impairment included 1077 patients without renal impairment (≥90 mL/min) and 894 patients with renal impairment (<90 mL/min). Overall, in both treatment groups, the incidences of adverse events were comparable between patients without (Met 52.3% vs. Lina+Met 48.5%) and patients with renal impairment (48.4% vs. 46.4%). No major difference in the adverse event pattern was noted between patients with normal renal function and patients with renal impairment. [U11-1152, Module 5.3.5.3, Table 5.2.5.2.6.1]

Metformin dose: Two subgroups were defined, patients receiving a total daily dose below 1500 mg (Met n=176, Lina+Met n=243) and patients receiving 1500 mg per day or more (Met n=407, Lina+Met n=1145). The incidences of adverse event were almost identical for both treatment groups in both dose categories (<1500 mg/day: Met 50.6% vs. Lina+Met 48.6%; ≥1500 mg/day Met 50.6% vs. Lina+Met 47.3%). The adverse event pattern did not appreciatively differ between these categories. [U11-1152, Module 5.3.5.3, Table 5.2.5.2.7.1]

P-gP/Cyp 3A4-inhibitors: Only few patients used P-gP/Cyp 3A4 inhibitors at baseline (Met n=17, Lina+Met n=43). Based on this low number of patients, the overall incidences of adverse event appeared higher than in the overall study population with 76.5% (Met) and 60.5% (Lina+Met). Particularly infections and infestations, gastrointestinal disorders, and skin and subcutaneous tissue disorder appeared to be occurring with higher frequencies than in the overall population. [U11-1152, Module 5.3.5.3, Table 5.2.5.3.3.1]

5.3.2.2 Drug-related adverse events

The incidences of adverse events that were assessed as related to study drug by the investigators was comparable between the Met (8.7%) and the Lina+Met (7.3%) group. The most frequent drug-related adverse events in the metformin group were metabolism and nutrition disorders (Met 10.30 vs. Lina+Met 5.37) and within this SOC hyperglycaemia (5.98

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vs. 1.97) and hypoglycaemia (2.96 vs. 2.37) were the most frequent events. In the Lina+Met group the most frequent drug-related adverse events were gastrointestinal disorders (Met 6.48 vs. Lina+Met 7.99). Within this SOC diarrhoea was the most frequent event with similar rates per 100 PY in both treatment groups (2.98 vs. 2.37). Overall incidences above 1% were observed for nervous system disorders (2.12 vs. 2.97) with dizziness (0.42 vs. 1.97) being more frequent in the Lina+Met group. All other system organ classes had incidences below 1% in both treatment groups. The patterns seen in the pooled set of metformin background trials (SAF-C5) are qualitatively and quantitatively similar to those observed in the pivotal trial 1218.17 (SAF-C1).

5.3.2.3 Adverse events leading to discontinuation

The percentage of patients who discontinued because of adverse events was low in both treatment groups (Met 2.6% vs. Lina+Met 2.4%). Gastrointestinal disorders were the most frequent adverse events that led to discontinuation, however with similar rates in both treatment groups (rate/100PY Met 2.11 vs. Lina+Met 1.96). Also for diarrhoea, the most frequent adverse events in this SOC, similar rates were seen in both treatment groups (0.84 vs. 0.98). Nervous system disorders appeared to be slightly more frequent in the Lina+Met group (0.42 vs. 0.98), without any adverse event of particularly elevated frequency. Again, as for drug-related adverse events the analysis based on SAF-C5 is both qualitatively and quantitatively similar to the observations for the pivotal trial 1218.17 (SAF-C1).

5.3.2.4 Serious adverse events

The incidences of serious adverse events were low in both treatment groups (Met 2.6% vs. Lina+Met 3.0%) and only adverse events in the system organ class injury, poisoning and procedural complications occurred with an incidence above 0.5% in any treatment group (Met: 0.2% vs. Lina+Met 0.6%). System organ classes with clearly higher rates per 100 patient years in the Lina+Met group were injury, poisoning and procedural complications and cardiac disorders. In some system organ classes, no serious adverse event was observed in the Met group but a few events occurred in the Lina+Met group, these were general disorders and administration site conditions, endocrine disorders, and respiratory, thoracic and mediastinal disorders. The data on serious adverse events in SAF-C5 is summarised in Table 5.3.2.4: 1 below.

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Table 5.3.2.4: 1 Serious adverse events occurring in at least 2 patients in either treatment group, sorted by frequency in the Lina+Met group for SAF-C5 - TS

	SAF-C5			
	Met		Lina+Met	
	N (%)	Rate / 100 patient years	N (%)	Rate / 100 patient years
Number of patients	583 (100.0)		1388 (100.0)	
Patients with serious AEs	15 (2.6)	6.38	41 (3.0)	8.16
Injury, poisoning and procedural complications	1 (0.2)	0.42	8 (0.6)	1.58
Cardiac disorders	1 (0.2)	0.42	7 (0.5)	1.38
Angina pectoris	0	0	4 (0.3)	0.79
Myocardial ischaemia	0	0	2 (0.1)	0.39
Infections and infestations	3 (0.5)	1.27	5 (0.4)	0.98
Renal and urinary disorders	2 (0.3)	0.85	5 (0.4)	0.98
Calculus ureteric	0	0	2 (0.1)	0.39
Nephrolithiasis	1 (0.2)	0.42	2 (0.1)	0.39
Renal mass	0	0	2 (0.1)	0.39
Musculoskeletal and connective tissue disorders	3 (0.5)	1.27	3 (0.2)	0.59
Intervertebral disc protrusion	1 (0.2)	0.42	2 (0.1)	0.39
Gastrointestinal disorders	1 (0.2)	0.42	3 (0.2)	0.59
General disorders and administration site conditions	0	0	3 (0.2)	0.59
Chest pain	0	0	2 (0.1)	0.39
Endocrine disorders	0	0	2 (0.1)	0.39
Respiratory, thoracic and mediastinal disorders	0	0	2 (0.1)	0.39
Nervous system disorders	1 (0.2)	0.42	2 (0.1)	0.39
Reproductive system and breast disorders	2 (0.3)	0.84	0	0

SAF-C5: Linagliptin was administered as 2.5 mg bid, 5 mg qd, or 10 mg qd; metformin was given as background medication, monotherapy (500 mg or 1000 mg, bid), or free combination therapy (500 mg or 1000 mg, together with linagliptin 2.5 mg; all bid).

Source data: SAF-C5 [U11-1152, Module 5.3.5.3, Table 5.2.5.1.6]

The analysis based on SEA-2, the safety set that was identical to SAF-C5 except for 66 patients who had taken 10 mg linagliptin, yielded very similar results, both qualitatively and quantitatively. When serious adverse events were analysed based on the largest set, i.e. the set of all studies that had used linagliptin and metformin (SAF-C8, n=3529), the findings were overall similar to those seen in SAF-C5. Cardiac disorders were the most frequently occurring serious adverse events (2.0%) followed by infections and infestations (1.7%), injury, poisoning and procedural complications (1.4%), and neoplasms (1.1%). For all other system organ classes, the frequencies were below 1%.

5.3.2.5 Adverse events with long-term treatment with combination therapy with linagliptin and metformin

Long-term safety was assessed in SAF-C2 (104 weeks) and SAF-C6 (78 weeks). Due to the longer treatment in SAF-C2, the overall frequencies of adverse events were about 40% higher

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than in the 24 week trials (SAF-C5). Over 2 years, the overall frequency of adverse events was lower in the Lina+Met group (85.4%) than in the Glim+Met group (91.1%). The percentages of patients with adverse events of severe intensity were comparable between treatment groups (Lina+Met 13.8% vs. Glim+Met 13.2%) as was the frequency of adverse events of special interest (Lina+Met 1.8%, Glim+Met 2.1%). However, the frequency of drug-related adverse events was substantially lower with Lina+Met (15.2%) than with Glim+Met (38.7%) mainly due to the difference in hypoglycaemia (Lina+Met 5.8%, Glim+Met 29.8%). Fewer patients in the Lina+Met group discontinued treatment because of adverse events (Lina+Met 7.7% vs. Glim+Met 11.1%). Discontinuation rates above 1% on the level of preferred terms were only seen in the SOC metabolism and nutrition disorders (Lina+Met 1.7%, Glim+Met 2.8%), with hypoglycaemia (0.4% Lina+Met, 2.5% Glim+Met) being the most frequently reported event leading to discontiuation. Furthermore, fewer patients in the Lina+Met group experienced serious adverse events (Lina+Met 17.4% vs. Glim+Met 20.9%).[U11-1152, Module 5.3.5.3, Table 5.2.2.1.5]

The adverse event profiles after 104 weeks (SAF-C2) and after 24 weeks (SAF-C5) were consistent, i.e. the 5 most frequent SOCs in SAF-C2 were also the most frequent in SAF-C5. The pattern of adverse events was slighlty different in SAF-C6 as investigations were the 5th most frequent SOC while they were less frequent in SAF-C2 and SAF-C5.

Within SAF-C2, the incidences of individual preferred terms and SOCs were largely comparable in both treatment groups with the exception of metabolism and nutrition disorders which were substantially less frequent with Lina+Met (18.6%) than with Glim+Met (44.1%) mainly caused by a much higher incidence of hypoglycaemia in the Glim+Met group (Lina+Met 7.1% vs. Glim+Met 34.8%). For most preferred terms, the Lina+Met group exhibited lower incidences than the Glim+Met group. Conversely, slightly higher incidences ≥2% in the Lina+Met group for individual preferred terms and SOCs were seen for arthralgia (Lina+Met 8.1% vs. Glim+Met 6.1%) and constipation (Lina+Met 4.3% vs. Glim+Met 2.1%), as well as for the SOCs injury, poisening and procedural complications (Lina+Met 16.4% vs. Glim+Met 13.8%) and for skin and subcutaneous tissue disorders (Lina+Met 15.3% vs. Glim+Met 12.3%).

For SAF-C6 some differences between treatment groups were apparent. However, these differences were consistent with the adverse event profile determined for Met versus Lina+Met in SAF-C5. The incidence of infections and infestations was lower in the Met1000 group than in the Lina+Met combination groups and the difference was predominately caused by a higher incidence of nasopharyngitis in the Lina+Met groups. The incidence of gastrointestinal disorders was lower in the Met1000 group (20.4%) than in the L2.5mg+M1000 group (25.9%) mainly because of a lower incidence of diarrhea (6.1% vs. 9.8%). The incidences of metabolism and nutrition disorders were higher in the L2.5+Met500 group than in both the Met1000 and the L2.5+Met1000 groups because of a higher incidence of hyperglycemia. Vascular disorders were more frequent in the L2.5+M1000 group than in both other groups, mainly because of a higher incidence of hypertension. Details are available in Table 5.3.2.5: 1 below.

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Table 5.3.2.5: 1 Frequency of patients with adverse events occurring in more than 5% in either treatment group on the SOC and preferred term level in SAF-C2 or SAF-C6, sorted by frequency in the Lina+Met group of SAF-C2 - TS

	SAF-C2		SAF-C6		
	Lina+Met N (%)	Glim+Met N (%)	Met1000 N (%)	L2.5+M500 N (%)	L2.5+M1000 N (%)
Number of patients	776 (100.0)	775 (100.0)	147 (100.0)	143 (100.0)	143 (100.0)
Patient years exposure	1332.5	1325.7	131.8	130.4	131.1
Patients with any adverse events	663 (85.4)	706 (91.1)	99 (67.3)	97 (67.8)	104 (72.7)
Infections and infestations	378 (48.7)	393 (50.7)	42 (28.6)	49 (34.3)	48 (33.6)
Nasopharyngitis	124 (16.0)	125 (16.1)	7 (4.8)	16 (11.2)	11 (7.7)
Upper respiratory tract infection	62 (8.0)	59 (7.6)	2 (1.4)	4 (2.8)	9 (6.3)
Bronchitis	52 (6.7)	61 (7.9)	6 (4.1)	5 (3.5)	1 (0.7)
Urinary tract infection	50 (6.4)	52 (6.7)	6 (4.1)	4 (2.8)	11 (7.7)
Musculoskeletal and connective tissue disorders	257 (33.1)	244 (31.5)	23 (15.6)	24 (16.8)	17 (11.9)
Back pain	71 (9.1)	65 (8.4)	6 (4.1)	6 (4.2)	9 (6.3)
Arthralgia	63 (8.1)	47 (6.1)	7 (4.8)	3 (2.1)	5 (3.5)
Pain in extremity	41 (5.3)	30 (3.9)	3 (2.0)	3 (2.1)	2 (1.4)
Gastrointestinal disorders	215 (27.7)	220 (28.4)	30 (20.4)	31 (21.7)	37 (25.9)
Diarrhoea	53 (6.8)	61 (7.9)	9 (6.1)	10 (7.0)	14 (9.8)
Nervous system disorders	149 (19.2)	181 (23.4)	12 (8.2)	18 (12.6)	15 (10.5)
Headache	50 (6.4)	40 (5.2)	7 (4.8)	6 (4.2)	5 (3.5)
Dizziness	26 (3.4)	48 (6.2)	1 (0.7)	2 (1.4)	2 (1.4)
Metabol.and nutrition disorders	144 (18.6)	342 (44.1)	27 (18.4)	38 (26.6)	23 (16.1)
Hypoglycaemia	55 (7.1)	270 (34.8)	6 (4.1)	6 (4.2)	3 (2.1)
Hyperglycaemia	40 (5.2)	41 (5.3)	10 (6.8)	16 (11.2)	6 (4.2)
Diabetes mellitus	15 (1.9)	13 (1.7)	8 (5.4)	8 (5.6)	5 (3.5)
Injury, poisoning and procedural complications	127 (16.4)	107 (13.8)	8 (5.4)	9 (6.3)	10 (7.0)
Skin and subcut. tissue disorders	119 (15.3)	95 (12.3)	9 (6.1)	7 (4.9)	10 (7.0)
General disorders and administration site conditions	114 (14.7)	120 (15.5)	13 (8.8)	12 (8.4)	15 (10.5)
Pyrexia	14 (1.8)	12 (1.5)	3 (2.0)	3 (2.1)	8 (5.6)
Respiratory, thoracic and mediastinal disorders	108 (13.9)	102 (13.2)	6 (4.1)	6 (4.2)	8 (5.6)
Cough	47 (6.1)	38 (4.9)	1 (0.7)	3 (2.1)	5 (3.5)
Investigations	100 (12.9)	104 (13.4)	22 (15.0)	19 (13.3)	20 (14.0)
Glom. filtration rate decreased	1 (0.1)	0	5 (3.4)	6 (4.2)	9 (6.3)
Vascular disorders	89 (11.5)	110 (14.2)	8 (5.4)	7 (4.9)	12 (8.4)
Hypertension	48 (6.2)	60 (7.7)	6 (4.1)	6 (4.2)	10 (7.0)
Psychiatric disorders	68 (8.8)	61 (7.9)	1 (0.7)	4 (2.8)	0
Eye disorders	65 (8.4)	71 (9.2)	4 (2.7)	1 (0.7)	3 (2.1)
Cardiac disorders	63 (8.1)	78 (10.1)	4 (2.7)	4 (2.8)	7 (4.9)
Renal and urinary disorders	57 (7.3)	58 (7.5)	7 (4.8)	4 (2.8)	6 (4.2)
Reproductive and breast disorders	44 (5.7)	42 (5.4)	2 (1.4)	7 (4.9)	2 (1.4)
Ear and labyrinth disorders	42 (5.4)	48 (6.2)	2 (1.4)	4 (2.8)	2 (1.4)
Blood and lymphatic system dis.	36 (4.6)	30 (3.9)	6 (4.1)	5 (3.5)	8 (5.6)

Source: SAF-C2 [U11-1152, Module 5.3.5.3, Tables 5.2.2.1.12, 4.2.1.1, 5.2.2.1.1], SAF-C6 [U11-1152, Module 5.3.5.3, Tables 5.2.6.12, 4.6.1, 5.2.6.1]

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For serious adverse events in SAF-C2, the frequency on the preferred term level was below 1% in either treatment group without clear trends. For most SOCs and preferred terms, the incidence of serious adverse events was lower in the Lina+Met group than in the Glim+Met group. Serious cardiac disorders (Lina+Met 3.2%, Glim+Met 4.3%) and serious nervous system disorders (Lina+Met 1.4%, Glim+Met 3.4%) were more frequent in the Glim+Met group than in the Lina+Met group; otherwise the frequencies were comparable between treatment groups.

In SAF-C6 (long-term safety), the frequency of serious adverse events was higher in the Met1000 group (6.1%) than in both combination treatment groups (L2.5+M500 4.2%, L2.5+M1000 4.9%). The frequency of serious adverse events on the preferred term level was below 1% in all treatment groups without any clear trends.

5.3.3 Clinical laboratory evaluation and vital signs

Clinical laboratory evaluation included analysis of haematology, clinical chemistry, and urinalysis. Throughout the clinical development emphasis was placed on the identification of drug-induced liver injuries (DILIs) and potential Hy's law cases and the changes in renal function (details are available in the SCS [Module 2.7.4, Section 3].)

5.3.3.1 Clinical laboratory evaluation and vital signs for linagliptin monotherapy

For the linagliptin monotherapy submission, analyses of laboratory data were performed on a number of safety groupings; for SAF-2 (all placebo-controlled trials, placebo vs. linagliptin), transitions relative to the reference range, possible clinically significant abnormalities (PCSAs), DILIS, and changes in renal function were analysed. In addition, adverse events leading to discontinuation were checked manually for laboratory abnormalities.

With regard to transitions, overall no clinically relevant findings or marked differences between placebo and linagliptin were observed for any of the measured parameters (haematology, electrolytes, enzymes, substrates, and urine analysis). Further details are provided in [U10-1607, Module 5.3.5.3, Table 6.2.2.1].

PCSAs were reported for a several parameters (haematology and clinical chemistry), with very low (<3.0% of patients) and comparable frequencies in both treatment groups. A possibly clinically significant increase in triglyceride levels was reported for 8.8% (PBO) and 7.8% (Lina). There were no patients with PCSAs reported in urine analysis. None of the observed increases or decreases was considered to be of clinical relevance. Further details are provided in [U10-1607, Module 5.3.5.3, Table 6.1.2.1].

No patient fulfilled the criteria for Hy's law; the vast majority of patients (about 99%) in both treatment groups were categorised as having enzyme and total bilirubin levels of grade 0, indicating a very low risk of a potential liver injury. In both treatment groups, some patients had liver enzyme values of grade 1 (placebo ≤1.0 %; linagliptin ≤0.8%). A few patients (<5) in both treatment groups had elevated liver enzyme levels but no pattern was discernable (details are available in [U10-1607, Module 5.3.5.3, Table 6.3.2.1]). There was no evidence for a treatment-induced difference in renal function; details are given in [U10-1607, Module 5.3.5.3, Table 6.6.2.1].

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For SAF-M1, the clinical laboratory evaluation focussed on parameters of liver function and of renal function. Liver enzyme levels were assessed in regard to transitions in relation to reference ranges, changes in mean values over time and from baseline, and the identification of DILIs. Overall the groups were comparable; differences between the Lina and Lina+Met groups exceeding 2% in the analysis of transistions from baseline to last value on treatment were seen for AST shifts from low or normal to high (Lina 2.9%, Lina+Met 5.3%) and for total bilirubin shifts from high and normal to low (Lina 3.7%, Lina+Met 8.5%). The treatment groups were very similar in regard to mean changes from baseline of liver enzymes. The evaluation for cases of DILI showed similar results in both groups; no patient qualified as Hy's law case, however 1 patient in the Lina group (PatID 62965) had ALT or AST increases above 3 ULN and concurrent elevation of bilirubin of more than 2 ULN. The frequency of patients with ALT or AST above 3 ULN was identical (1.5%) in both treament groups. [U11-1152, Module 5.3.5.3, Tables 8.3.1.1, 8.3.1.2, 8.3.1.3, 8.3.1.4]

The treatment groups were very similar in renal function parameters. Of the patients with normal renal function at baseline, 18.7% (Lina) and 23.2% (Lina+Met) had a shift to mild renal impairment at their last value on treatment and no patient (Lina) and 4 patients (1.3%) in the Lina+Met group had a shift to moderate renal impairment. Of the patients with mild renal impairment at baseline, the frequencies of shifts to moderate renal impairment were similar in both groups (Lina 6.4%, Lina+Met 6.7%). [U11-1152, Module 5.3.5.3, Table 8.3.2.1]

Vital signs

In the linagliptin monotherapy dossier, analyses of vital signs were not performed on SAF-2 (all placebo-controlled trials).

For SAF-M1 (long-term safety of linagliptin monotherapy), the mean blood pressure and mean pulse rate values were comparable between both patient groups at baseline. In both groups, minimal changes in mean blood pressure and mean pulse rate were observed over time. For the Lina group, with a maximum change from baseline in SBP of 1.1 mmHg at Week 12, for DBP of 1.3 mmHg at Week 30, and for pulse rate of 1.1 bpm at Week 30. In the Lina+Met group, with a maximum change from baseline for SBP of -1.8 mmHg at Week 6, for DBP of -1.7 mmHg at Weeks 30 and 90, and for pulse rate of -1.2 bpm at Week 66 [U11-1152, Module 5.3.5.3, Table 8.4.1] was observed.

5.3.3.2 Clinical laboratory evaluation and vital signs for metformin monotherapy

No separate studies with metformin monotherapy were conducted; however, information on clinical laboratory evaluation of metformin monotherapy is included in some of the SAFs in the respective metformin treatment arms. For details see SCS [Module 2.7.4, Section 3].

5.3.3.3 Clinical laboratory evaluation and vital signs for linagliptin and metformin combination therapy

Laboratory analyses were performed on all safety groupings (SAF-C1 to SAF-C9), however to a different extent per grouping: descriptive statistics and transitions relative to the reference ranges were performed for SAF-C2, SAF-C5, SAF-C6, and SAF-C7. PCSAs and changes in renal function were performed on all SAFs except SAF-C9. Analyses to detect

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potential liver toxicity were performed on all SAFs. Laboratory data at baseline, last value on treatment, and changes from baseline were analysed descriptively for SAF-C2, SAF-C5, and SAF-C6, and SAF-C7. The description will focus on SAF-C5. Details on other safety sets are available in the SCS [Module 2.7.4, Section 3.3].

The analysis of PCSAs did not reveal meaningful differences between the Met and Lina+Met groups and for most parameters less than 2% of patients had PCSAs. Laboratory parameters for which ≥2% had PCSAs were: decrease in haematocrit (Met 1.6%, Lina+Met 2.1%), increase in eosinophils (Met 2.1%, Lina+Met 1.7%), increase in amylase (Met 1.8%, Lina+Met 2.1%), increase in triglycerides (Met 7.3%, Lina+Met 7.8%), and increase in uric acid (Met 2.0%, Lina+Met 1.9%). [U11-1152, Module 5.3.5.3, Table 6.1.5.1]

In regard to transitions relative to reference ranges, the treatment groups exhibited very similar patterns and differences between treatment grous in shifts from baseline to last value on treatment were generally below 2%. Exceptions were phosphate for which 1.2% (Met) and 3.4% (Lina+Met) of patients had shifts from low or normal to high values. Also for ALT (Met 8.4%, Lina+Met 6.4%) and glucose (Met 50.0%, Lina+Met 42.6%) differences in shifts from low or normal to high values were observed. For the urine albumin/creatinine ratio, the frequency of shifts from low or normal to high values was lower in the Met group (12.2%) than in the Lina+Met group (15.0%). For creatine kinase (CK-MB), no patient in the Met group but 9 patients (2.1%) in the Lina+Met group had shifts from low or normal to high values for their last observation on treatment. [U11-1152, Module 5.3.5.3, Table 6.2.5.1]

The treatment groups exhibited similar frequencies for transistions between MDRD stages; differences were generally below 5% for any category. For the mean changes from baseline in serum creatinine the treatment groups showed very similar patterns. Haematological parameters (including blood differential) remained essentially unchanged in both treatments. Also, mean levels of electrolytes and liver enzymes were stable. Both treatment groups had a small increase (from baseline to last value on treatment) in mean amylase levels (Met 3 U/L, Lina+Met 6 U/L). As expected, in both groups, mean glucose levels decreased (Met -15.0 mg/dL, Lina+Met -28.0 mg/dL). Other substrates such as total cholesterol, HDL, LDL, urea, creatinine, and bilirubin remained stable. For triglycerides, small reductions in mean values were seen in both groups (Met -6 mg/dL, Lina+Met -18 mg/dL).. The mean urine albumine/creatinine ratio increased in the Met group (+13.9 µg/mg) and minimally decreased in the Lina+Met group (-1.4 µg/mg); this was supported by an analysis of the mean differences from baseline (Met +14.2 µg/mg, Lina+Met -14.2 µg/mg). [U11-1152, Module 5.3.5.3, Table 6.4.5.1] The evaluation of DILIs could not identfy any case of Hy's law in either treatment group and very few patients had elevations of AST or ALT of more than 3 ULN (Met 0.7%, Lina+Met 0.4%). [U11-1152, Module 5.3.5.3, Table 6.5.5.1]

Vital signs

Blood pressure and pulse rates at baseline were comparable between treatment groups. Over the 24-week treatment period only minimal changes in mean values were observed both treatment groups (<1 mmHg for blood pressure and <1 bpm for pulse rate). The frequencies of patients with marked increases in SBP (≥30 mmHg) were similar in both treatment groups (Met 4.9%, Lina+Met 4.5% for maximum value post baseline) as well as for DBP (≥20 mmHg) with 4.4% (Met) and 3.8% (Lina+Met) for the maximum value post baseline.

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5.3.3.4 Clinical laboratory evaluation and vital signs for long-term treatment

For SAF-C2, the analysis of clinical laboratory parameters was uneventful and confirmed the results of the analysis based on SAF-C5. Both treatment groups were very similar for most clinical laboratory parameters. In regard to PCSAs the treatment groups showed very similar percentages (<2% difference); a larger difference was only seen for possible significant increases in uric acid (Lina+Met 8.4%, Glim+Met 11.2%). [U11-1152, Module 5.3.5.3, Table 6.1.2.1] For transitions in regard to reference range, differences between treatment groups were seen in the shifts (baseline to the last value on treatment) from low or normal to high values for GGT (Lina+Met 5.7% vs. Glim+Met 9.1%), CK MB enzymatic (Lina+Met 8.3% vs. Glim+Met 11.5%), glucose (Lina+Met 53.7% vs. Glim+Met 42.4%), triglycerides (Lina+Met 19.9% vs. Glim+Met 25.1%), and for uric acid (Lina+Met 8.6% vs. Glim+Met 13.6%). For red blood cells a small difference was seen for shifts from high or normal to low values (Lina+Met 14.5% vs Glim+Met 10.5%) and for LDH (Lina+Met 11.2% vs. Glim+Met 9.2%). [U11-1152, Module 5.3.5.3, Table 6.2.2.1] There was no difference between the treatment groups in regard to renal function (MDRD stages). The treatment groups were also very similar for differences from baseline for most parameters. For creatine kinase a mean decrease of -8 U/L was seen in the Lina+Met group whereas the Glim+Met group showed a mean increase of +22 U/L. [U11-1152, Module 5.3.5.3, Table 6.4.2.2]

In regard to liver function analysis, no patient qualified as potential Hy's law case and the percentages of patients with ALT or AST >3 ULN were similar in both treatment groups (Lina+Met 1.2% vs. Glim+Met 1.5%). [U11-1152, Module 5.3.5.3, Table 6.5.2.1]

Also the analysis of clinical laboratory parameters for SAF-C6 confirmed the assessment based on SAF-C5, no safety concerns in regard to clinical laboratory evaluations were identified for up to 78 weeks of treatment. In regard to PCSAs, all 3 treatment groups were similar. Differences between the treatment groups exceeding 2% between Met1000 and the combination treatment groups were only seen for decreses in haematocrit (M1000 3.0%, L2.5M500 0.7%, L2.5M1000 0%), haemoglobin (M1000 3.7%, L2.5M500 0%, L2.5M1000 0.7%), increases in amylase M1000 0.7%, L2.5M500 3.7%, L2.5M1000 5.0%), which were however not associated with any clinical signs and symptoms of pancreatitis; and increases in uric acid (M1000 5.1%, L2.5M500 0.7%, L2.5M1000 2.9%). [U11-1152, Module 5.3.5.3, Table 6.1.6.1]. In regard to transitions relative to reference ranges from baseline to the last value on treatment all 3 treatment groups showed a comparable profile. Differences of more than 5% (1 patient represents 0.7%) between M1000 and the combination therapy groups were only seen for haemoglobin (M1000 7.1%, L2.5M500 1.6%, L2.5M1000 5.3%) and LDL (M1000 16.7%, L2.5M500 9.4%, L2.5M1000 11.5%) for shifts from high or normal to low values. For white blood cells (M1000 1.5%, L2.5M500 3.1%, L2.5M1000 7.4%) and for triglycerides (M1000 14.4%, L2.5M500 7.7%, L2.5M1000 11.6%) differences were seen for shifts from low or normal to high values. [U11-1152, Module 5.3.5.3, Table 6.2.6.1] There were only small differences between the treatment groups in regard to renal function (MDRD stages). The differences from baseline in clinical laboratory parameters observed were similar in all 3 treatment groups. The greatest differences in mean changes were seen for glucose (M1000 -38 mg/dL, L2.5M500 -48 mg/dL, L2.5M1000 -55 mg/dL), alkaline phosphatase (M1000 -19 U/L, L2.5M500 -19 U/L, L2.5M1000 -26 U/L), and creatine kinase (M1000 -21 U/L, L2.5M500 -25 U/L, L2.5M1000 -49 U/L). The analysis of liver function parameters did not show any difference between the treatment groups; no patient qualified as

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Hy's law case and the percentages of patients with AST or ALT >3 ULN were similar (M1000 2.03%, L2.5M500 2.1%, L2.5M1000 1.4%). [U11-1152, Module 5.3.5.3, Table 6.5.6.1]

Vital signs

For SAF-2, Pulse rate was essentially unchanged after 104 weeks of treatment (change from baseline to end of treatment <1 bpm) in both treatment groups. Also mean SBP values did not appreciatively change (change from baseline <1 mmHg) in both treatment groups. For DBP a slight reduction was seen in the Lina+Met group (-1.2 mmHg) and to a lesser extent in the Glim+Met group (-0.6 mmHg). Furthermore the proportion of patients with marked increases in SBP (≥30 mmHg) for the maximum value post baseline was similar in both treatment groups (Lina+Met 14.7% vs. Glim+Met 11.6%). Similar proportions in both treatment groups had marked increases in DBP (≥20 mmHg) for their maximum post baseline observation (Lina+Met 10.5% vs. Glim+Met 9.6%). [U11-1152, Module 5.3.5.3, Tables 7.2.1, 7.2.5]

In SAF-C6 for Week 54, the changes in vital signs were small and comparable in all treatment groups. For pulse rate, changes from baseline were below 2 bpm in any treatment group. For SBP in the Met1000 group the mean change from baseline was +1.2 mmHg while the Lina+Met groups showed mean reductions of -1.7 mmHg (L2.5M500) and -3.4 mmHg (L2.5M1000). The maximum change in DBP was 1 mmHg in any treatment group. The proportions of patients with marked increases in SBP (≥30 mmHg) for their maximum value post baseline was comparable between the Met1000 group (6.3%) and the Lina+Met combination groups (8.7% and 7.8%). Similarly the percentages of patients with marked increases in DBP (≥20 mmHg) for the maximum value post baseline were largely comparable (M1000 8.5%, L2.5+M500 7.2%, L2.5+M1000 4.3%). [U11-1152, Module 5.3.5.3, Table 7.6.1]

5.4 SAFETY OF COMBINATION THERAPY WITH LINAGLIPTIN + METFORMIN + SU

5.4.1 Exposure to combination therapy with linagliptin + metformin + SU

The planned study duration in study 1218.18 (SAF-C7) was 24 weeks and the median exposure of 170 days in both treatment groups was in accordance with this. Overall 92.0% (Met+SU) and 92.7% (Lina+Met+SU) completed the planned treatment period and hence the proportions of premature discontinuations were similar in both treatment groups (8.0% vs. 7.3%). The most frequent reasons for premature discontinuation were adverse events (Met+SU 1.9%, Lina+Met+SU 2.9%), non-compliance with study protocol (Met+SU 1.5%, Lina+Met+SU 2.4%), refusal to continue trial medication (Met+SU 3.0%, Lina+Met+SU 1.8%), and lack of efficacy (Met+SU 1.5%, Lina+Met+SU 0.3%). [U11-1152, Module 5.3.5.3, Table 2.7.1]

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Table 5.4.1: 1 Exposure to study medication for SAF-C7 (Lina+Met+SU) - TS

	Met+SU	Lina+Met+SU
Number of patients, N (%)	263 (100.0)	792 (100.0)
Exposure categories, N (%)		
≥12 weeks	257 (97.7)	763 (96.3)
≥24 weeks	231 (87.8)	700 (88.4)
Exposure [days]		
Mean (±SD)	166 (25.4)	166 (27.7)
Median (min, max)	170 (6, 213)	170 (3, 214)
Overall patient years	119.5	358.9

SAF-C7: Linagliptin was administered as 5 mg qd, metformin and SU were given as background medication.
Source data: SAF-C7 [U11-1152, Module 5.3.5.3, Table 4.7.1]

5.4.2 Adverse events with combination therapy with linagliptin + metformin + SU

Overall, 61.2% (Met+SU) and 67.3% (Lina+Met+SU) of patients were reported with an adverse event. The frequency of patients with adverse events of severe intensity (1.5% vs. 2.5%), of adverse events leading to treatment discontinuation (1.9% vs. 3.2%), and of adverse events of special interest (0.4% vs. 1.3%) were generally low but slightly lower in the Met+SU than in the Lina+Met+SU groups. The incidence of serious adverse events was comparable between treatment groups (Met+SU 4.2% vs. Lina+Met+SU 3.2%). The percentage of patients with drug-related adverse events was higher in the Lina+Met+SU group (18.3%) than in the Met+SU group (12.2%). This was mainly due to a higher incidence of hypoglycemia (Met+SU 7.6% vs. Lina+Met+SU 14.5%). [U11-1152, Module 5.3.5.3, Tables 5.1.7.1, 5.2.7.3] Due to the mode of action of insulin secretagogues such as SUs, patients on combination therapy are known to be at risk of hypoglycaemic events. Similar observations have also been reported for other DPP-4 inhibitors when administered as add-on therapy on the combination of metformin and SU [R09-5501, Module 2.7.5].

The most frequent adverse events were metabolism and nutrition disorders, followed by infections and infestations, and gastrointestinal disorders. This profile is consistent with that observed for SAF-C5. The higher overall incidence of adverse events in the Lina+Met+SU group are largely accounted for by higher incidences of adverse events in the SOCs metabolism and nutrition disorders (including hypoglycaemia Met+SU 14.8% vs. Lina+Met+SU 22.7%), musculoskeletal and connective tissue disorders, investigations, vascular disorders, and respiratory and thoracic disorders. For the SOC investigations some small differences were seen for increased AST (0% vs. 0.5%), increased creatine phosphokinase (0.8% vs. 1.3%), and increased glycosylated haemoglobin (0% vs. 0.8%) with higher incidences in the Lina+Met+SU group.

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Table: 5.4.2: 1 Frequency of adverse events occurring in more than 2% of patients in either treatment group on the SOC and preferred term level, sorted by frequency in the Lina+Met+SU group for SAF-C7 - TS

	SAF-C7	
	Met+SU N (%)	Lina+Met+SU N (%)
Number of patients	263 (100.0)	792 (100.0)
Patients with any adverse event	161 (61.2)	533 (67.3)
Metabolism and nutrition disorders	72 (27.4)	250 (31.6)
Hypoglycaemia	39 (14.8)	180 (22.7)
Hyperglycaemia	28 (10.6)	49 (6.2)
Infections and infestations	77 (29.3)	172 (21.7)
Upper respiratory tract infection	25 (9.5)	47 (5.9)
Nasopharyngitis	12 (4.6)	41 (5.2)
Urinary tract infection	15 (5.7)	26 (3.3)
Gastrointestinal disorders	47 (17.9)	102 (12.9)
Diarrhoea	9 (3.4)	20 (2.5)
Abdominal pain upper	6 (2.3)	9 (1.1)
Musculoskeletal and connective tissue disorders	24 (9.1)	98 (12.4)
Arthralgia	4 (1.5)	21 (2.7)
Back pain	8 (3.0)	13 (1.6)
Nervous system disorders	30 (11.4)	78 (9.8)
Headache	12 (4.6)	33 (4.2)
Dizziness	12 (4.6)	29 (3.7)
General disorders and administration site conditions	18 (6.8)	61 (7.7)
Asthenia	5 (1.9)	19 (2.4)
Injury, poisoning and procedural complications	15 (5.7)	41 (5.2)
Investigations	10 (3.8)	40 (5.1)
Skin and subcutaneous tissue disorders	13 (4.9)	39 (4.9)
Vascular disorders	6 (2.3)	35 (4.4)
Hypertension	5 (1.9)	20 (2.5)
Renal and urinary disorders	11 (4.2)	34 (4.3)
Respiratory, thoracic and mediastinal disorders	7 (2.7)	33 (4.2)
Cough	3 (1.1)	19 (2.4)
Psychiatric disorders	8 (3.0)	18 (2.3)

SAF-C7: Linagliptin was administered as 5 mg qd, metformin and SU were given as background medication.

Source data: SAF-C7 [U11-1152, Module 5.3.5.3, Table and 5.2.7.9]

Overall, the number of patients with serious adverse events was low and comparable between both treatment groups. The frequencies of individual preferred terms were below 1% in each treatment group without clear trends. The most frequently reported serious adverse events occurred in the SOCs injury, poisoning and procedural complications (Met+SU 1.5%, Lina+Met+SU 0.9%); renal and urinary disorders (Met+SU 0.8%, Lina+Met+SU 0.1%); infections and infestations (Met+SU 1.1%, Lina+Met+SU 0.5%); and cardiac disorders (Met+SU 0.8%, Lina+Met+SU 0.4%). There were no fatal events in this safety grouping. Further details are available in the SCS [Module 2.7.4, Section 2.1.4.3].

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5.4.3 Clinical laboratory evaluation and vital signs for combination therapy with linagliptin + metformin + SU

Overall, no clinically relevant findings or significant differences between both treatment groups were observed for any of the measured parameters. In general, the mean values at baseline and last value on treatment for all haematology parameters, differential counts, electrolytes, enzymes, and substrates were comparable between both treatment groups. In regard to PCSA the treatment groups exhibited a very similar profile. PCSAs were generally below 3% in both treatment groups except for increases in amylase (Met+SU 3.8% vs. Lina+Met+SU 4.8%), increases in triglycerides (Met+SU 8.8% vs. Lina+Met+SU 9.0%), and increases in uric acid (Met+SU 1.9% vs. Lina+Met+SU 3.2%). Also, for renal function stages, no meaningful difference between treatment groups was observed. No patient met the criteria for Hy's law and equal proportions of patients had ALT or AST >3 ULN (0.4% in both groups). [U11-1152, Module 5.3.5.3, Tables 6.1.7.1, 6.5.7.1]

Vital signs

Mean changes in pulse rate from baseline to end of treatment were below 2 bpm in both treatment groups. Changes in systolic or diastolic blood pressure were below 1 mmHg in both treatment groups. [U11-1152, Module 5.3.5.3, Tables 7.7.1]

5.5 ADVERSE EVENTS OF SPECIAL INTEREST

The analysis of the adverse events of special interest was based on narrow SMQs and not on investigator-reported significant adverse events to cover the medical concepts for hypersensitivity reactions, renal events, hepatic events, severe cutaneous adverse reactions, and pancreatitis in a more complete way. For hypersensitivity reactions, skin and subcutaneous tissue disorders, and renal and urinary disorders the system organ classes were checked manually in addition. For complete definitions see SCS [Module 2.7.4, Section 1.1.7].

Overall, the numbers of patients with adverse events of special interest were low in each SAF, with no clear trend in any treatment group. No patients were reported with severe cutaneous adverse reactions in any of the safety groupings. Few patients were reported with hypersensitivity reactions (≤1.2%) and renal events (≤1.3%) in each SAF. The number of patients with hepatic events differed across SAFs, ranging from no patients in the Met group of SAF-C3 to 8.2% in the Met group of SAF-C6. The number of patients with pancreatitis was low (≤0.2%).

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Table 5.5: 1 Overview of adverse events of special interest based on SMQs - TS

	Hypersensitivity reactions[1] N (%)	Renal events[2] N (%)	Hepatic events[3] N (%)	Severe cutaneous adverse reactions[4] N (%)	Pancreatitis[5] N (%)
SAF-C1					
Met (N=177)	0	1 (0.6)	2 (1.1)	0	0
Lina+Met (N=523)	1 (0.2)	0	4 (0.8)	0	0
SAF-C2					
Lina+Met (N=776)	8 (1.0)	4 (0.5)	34 (4.4)	0	1 (0.1)
Glim+Met (N=775)	9 (1.2)	10 (1.3)	35 (4.5)	0	0
SAF-C3					
Met (N=44)	0	0	0	0	0
Lina+Met (N=447)	0	0	4 (0.9)	0	0
SAF-C4					
Placebo (N=72)	0	0	0	0	0
Lina (N=142)	1 (0.7)	0	5 (3.5)	0	0
Met (N=291)	0	0	10 (3.4)	0	0
Lina+Met (N=286)	0	0	4 (1.4)	0	0
SAF-C5					
Met (N=583)	0	1 (0.2)	12 (2.1)	0	0
Lina+Met (N=1388)	1 (0.1)	0	14 (1.0)	0	1 (0.1)
SEA-2					
Met (N=583)	0	1 (0.2)	12 (2.1)	0	0
Lina+Met (N=1322)	1 (0.1)	0	13 (1.0)	0	1 (0.1)
SAF-C6					
Met1000 (N=147)	0	1 (0.7)	12 (8.2)	0	0
Lina2.5+Met500 (N=143)	0	0	9 (6.3)	0	0
Lina2.5+Met1000 (N=143)	1 (0.7)	0	6 (4.2)	0	0
SAF-C8					
Lina+Met (N=3529)	26 (0.7)	15 (0.4)	129 (3.7)	0	6 (0.2)

[1] Based on SMQs 'anaphylactic reaction' and 'angioedema'.
[2] Based on SMQ 'acute renal failure'.
[3] Based on SMQs 'liver-related investigations, signs and symptoms', 'cholestasis and jaundice of hepatic origin', 'hepatitis, non-infectious', and 'hepatic failure, fibrosis, cirrhosis and other liver damage-related conditions'.
[4] Based on SMQ 'severe cutaneous adverse reaction'.
[5] Based on SMQ 'acute pancreatitis'.

Source data: SAF-C1 [U11-1152, Module 5.3.5.3, Tables 5.3.2.1.1, 5.3.3.1.1, 5.3.4.1.1, 5.3.5.1.1, and 5.3.6.1.1], SAF-C2 [U11-1152, Module 5.3.5.3, Tables 5.3.2.2.1, 5.3.3.2.1, 5.3.4.2.1, 5.3.5.2.1, and 5.3.6.2.1], SAF-C3 [U11-1152, Module 5.3.5.3, Tables 5.3.2.3.1, 5.3.3.3.1, 5.3.4.3.1, 5.3.5.3.1, and 5.3.6.3.1], SAF-C4 [U11-1152, Module 5.3.5.3, Tables 5.3.2.4.2.1, 5.3.3.4.2.1, 5.3.4.4.2.1, 5.3.5.4.2.1, and 5.3.6.4.2.1], SAF-C5 [U11-1152, Module 5.3.5.3, Tables 5.3.2.5.1, 5.3.3.5.1, 5.3.4.5.1, 5.3.5.5.1, and 5.3.6.5.1], SAF-C6 [U11-1152, Module 5.3.5.3, Tables 5.3.2.6.1, 5.3.3.6.1, 5.3.4.6.1, 5.3.5.6.1, and 5.3.6.6.1], SAF-C8 [U11-1152, Module 5.3.5.3, Tables 5.3.2.8.1, 5.3.3.8.1, 5.3.4.8.1, 5.3.5.8.1, and 5.3.6.8.1],SEA-2 [U11-1152, Module 5.3.5.3, Section 4.6]

Hypoglycaemic events

The analysis of hypoglycaemic events was based on investigator-reporting. For the pivotal trial 1218.17 (SAF-C1), the incidence of hypoglycaemia was higher in the Met group (2.8%) than in the Lina+Met group (0.6%) and this was corroborated in the set of metformin-controlled studies SAF-C5 (Met 2.5% vs. Lina+Met 1.4%). Severe hypoglycaemic events, i.e. events requiring assistance, were limited to the metformin group in SAF-C5. As shown by the analysis of trial 1218.18 (SAF-C7), the addition of an SU to metformin led to a substantially

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higher frequency of hypoglycaemic events (16.0%) which was even higher in the triple combination group of Lina+Met+SU (23.7%). Over long-term treatment of about one year (SAF-C6) the frequency of hypoglycaemic events appear to be similar with metformin and Lina+Met. However the incidences of hypoglycaemia in the Lina+Met groups were higher than after 24 weeks and yet higher after about 2 years of treatment (SAF-C2). In SAF-C2, the difference between both treatment groups in hypoglycaemic events (Lina+Met 7.5% vs. Glim+Met 36.1%) was significant and in favour of linagliptin ($p<0.0001$). The majority of patients in all SAFs did not require assistance in case of a hypoglycaemic episode, and large proportions of patients in all SAFs had an onset of a first episode after 28 days. Few patients were reported with a hypoglycaemic episode while rescue medication had been given. With combination therapy of Lina+Met only 1 patient had a severe hypoglycaemic event (requiring assistance) during long-term treatment. However, in the triple combination of Lina+Met+SU the incidence also of severe hypoglycaemic events appears to be increased.

Table 5.5: 2 Summary of investigator-reported hypoglycaemic events by SAFs - TS

	Number of patients N (100.0%)	Median exposure [days]	Investigator-defined hypoglycaemia N (%)	Severe hypoglycemia[1] N (% of patients with hypoglycaemic events)
SAF-C1 (1218.17)				
Met	177	171	5 (2.8%)	0
Lina+Met	523	171	3 (0.6%)	0
SAF-C5 (metformin-controlled studies)				
Met	512	169	13 (2.5%)	1 (7.7%)
Lina+Met	1256	167	17 (1.4%)	0
SAF-C7 (1218.18) (background Met+SU)				
Met+SU	263	170	42 (16.0%)	2 (4.8%)
Lina+Met+SU	792	170	188 (23.7%)	5 (2.7%)
SAF-C6 (1218.46/52) (long-term)				
Met 1000 bid	147	365	6 (4.1%)	1 (16.7%)
Lina2.5+Met500 bid	143	373	7 (4.9%)	0
Lina2.5+Met1000 bid	143	369	4 (2.8%)	0
SAF-C2 (1218.20) (active-controlled)				
Glim+Met	775	729	280 (36.1%)	12 (4.3%)
Lina+Met	776	729	58 (7.5%)	1 (1.7%)

[1] Requiring the assistance of another person to actively administer carbohydrate, glucagon or other resuscitative actions.

Source: [U11-1152, Module 5.3.5.3, SAF-C1 Tables 5.3.1.1.2 and 4.1.1, SAF-C2 Tables 5.3.1.2.2 and 4.2.1.1, SAF-C5 Tables 5.3.1.5.3 and 4.5.1.1, SAF-C6 Tables 5.3.1.6.2 and 4.6.1, SAF-C7 5.3.1.7.2 and 4.7.1]

Pancreas disorders

In total, 10 cases of pancreas disorders were reported: 8 cases occurring during treatment with Lina+Met, and 2 cases during treatment with Glim+Met. In the post-treatment period, 2 cases of pancreas disorders were reported. None of the cases has been necrotising, haemorrhagic, or fatal. All of these cases but one (patient No. 20904, trial 1218.20, pancreatic carcinoma) were included in the linagliptin monotherapy SCS.

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Table 5.5: 3 Overview on pancreatic disorders during treatment and during post-treatment with linagliptin + metformin -TS

Preferred term	Patient No./BI trial/SAF	Age [years] / gender	Race	Intensity	Drug-related	Serious
During treatment						
Pancreatic carcinoma	20159/1218.20/C2, C8	71 / Female	White	Severe	No	Yes
Pancreatic cyst	80524/1218.18/C8	59 / Female	White	Mild	No	No
Pancreatitis	9106/1218.6/C5, C8	42 / Male	White	Mild	No	No
Acute pancreatitis	94364/1218.40/C8	34 / Female	Asian	Moderate	No	Yes
Chronic pancreatitis	85360/1218.18/C8	68 / Female	White	Mild	No	No
Chronic pancreatitis	28310/1218.20/C2, C8	70 / Male	White	Moderate	No	No
Chronic pancreatitis	96346/1218.40/C8	55 / Female	White	Moderate	No	No
Chronic pancreatitis	96363/1218.40/C8	56 / Female	White	Moderate	No	Yes
Post-treatment						
Pancreatic carcinoma, non-resectable	96822/1218.40/C8	70 / Male	White	Severe	No	Yes
Chronic pancreatitis	5026/1218.6/C5, C8	56 / Female	White	Mild	No	No

Source data: CTR 1218.6 [U08-1056, Module 5.3.5.1, Appendix 16.2, Listings 4.1.1 and 7.2.2], CTR 1218.18 [U09-2458, Module 5.3.5.1, Appendix 16.2, Listings 4.1 and 7.2.2], CTR 1218.20 [U10-1465, Module 5.3.5.1, Appendix 16.2, Listings 4.1 and 7.2.2], CTR 1218.40 [U10-1468, Module 5.3.5.2, Appendix 16.2.4, Listing 1 and Appendix 16.2.7, Listing 2.2]

5.6 RESULTS OF CARDIOVASCULAR SAFETY ANALYSIS

Eight trials with a total of 5239 patients with T2DM were included in a cardiovascular meta-analysis. The primary endpoint was based on adjudicated events and was a composite endpoint consisting of cardiovascular death (including fatal stroke and fatal MI), non-fatal MI, non-fatal stroke, and hospitalisation due to unstable angina. This analysis indicated that treatment with linagliptin was not associated with an increased cardiovascular risk compared with a pooled comparator group (placebo, glimepiride, and voglibose). For further details, including results on subgroup analyses, refer to the report [U10-1736, Module 5.3.5.3].

5.7 DEATHS

In total, there were 18 fatal events (14 during treatment, 4 during post-treatment) reported in the studies and groupings used for the evaluation of safety (SAF-C2, SAF-C4, SAF-C5, SAF-C6, and SAF-C8; there were no deaths in SAF-C1, SAF-C3, SAF-C7, and SAF-C9). In addition, there were 2 fatal events which occurred during washout, i.e. at a time when the patients did not receive any study medication. Out of the 14 deaths occurring during randomised treatment, 9 patients were receiving Lina+Met, 1 patient received Met monotherapy, and 4 patients received Gim+Met.
Of the 4 events occurring post-treatment, 1 patient had received Lina+Met, 1 patient had received Glim+Met, and 2 patients had received Lina+Met+SU. None of the deaths was considered as related to the study medication. The patients who died were predominately men (13 male, 5 female) and of White race (14 White, 3 Asian, 1 Black); the age ranged from 53 to 72 years. For 10 patients, death was due to cardiac complications. An analysis based on exposure-corrected incidences demonstrates that death rates are similar across treatments and studies.

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Table 5.7: 1 Exposure corrected estimates of death incidence rates by study – all SAFs

		Met	Lina+Met
1218.20	Patients at risk / total exposure [patient years]	775 / 1326 PY	776 / 1333 PY
	Deaths on treatment	4	4
	Incidence / Rate per 100 patient years	0.52% / 0.30	0.52% / 0.30
	Deaths post-treatment	1	1
	Incidence	0.13%	0.13%
1218.46/	Patients at risk / total exposure [patient years]	147 / 132 PY	286 / 262 PY
1218.52	Deaths on treatment	1	2
	Incidence / rate per 100 patient years	0.68% / 0.76	0.70% / 0.76
1218.40	Patients at risk / total exposure [patient years]	n.a.	2121 / 1887
	Deaths on treatment	n.a.	3
	Incidence / rate per 100 patient years	n.a.	0.14% / 0.16
	Deaths post-treatment	n.a.	2
	Incidence	n.a.	0.09%

n.a. = not applicable as all patients received linagliptin 5 mg in the extension trial (+backgound medication)

Source data: [U11-1152, Module 5.3.5.3, Tables 4.2.1.1 and 4.6.1, Listing 5.5.1], [U11-1261, Module 5.3.5.2, Table 15.3.1: 1]

5.8 RESULTS OF THE ANALYSIS OF SIDE EFFECTS

The analysis of side effects was based on SEA-2 which comprised all trials comparing placebo+metformin with linagliptin 5 mg+metformin. SEA-2 is very similar to SAF-C5 but excluded all patients that had taken linagliptin doses other than linagliptin 5 mg. Based on this safety set, a special analysis was undertaken to identify any adverse events that are possibly associated with linagliptin+metformin treatment. The analysis compared results for 1322 patients treated with linagliptin 5 mg+metformin and 583 patients treated with placebo+metformin. Since SEA-2 set does not provide a true placebo versus Lina+Met comparison, potential side effects were identified (according to Criteria 1 to 3, see below) by comparing frequencies of adverse events with linagliptin+metformin treatment to the frequencies of adverse events with placebo+metformin treatment, for adverse events which are not known to be related to metformin treatment (for these, placebo+metformin is set as "placebo").The criteria employed were:

(1) Adverse events that occurred with a frequency of 2% or more with linagliptin+metformin treatment and had a 2-fold higher frequency in the linagliptin+metformin group than in the comparator group, or when the frequency in the comparator group was zero, and/or

(2) Adverse events that were likely related based on medical plausibility, and/or

(3) Adverse events that had a consistent pattern across antidiabetic background treatments; i.e., their frequencies in the linagliptin groups were consistently higher than in the placebo groups with every antidiabetic background medication

Undesirable effects of metformin (Glucophage® SPC), are lactic acidosis, decrease of vitamin B12 absorption, taste disturbance, abdominal pain, nausea, vomiting, diarrhea, loss of appetite, liver-function test abnormalities, hepatitis, erythema, urticaria, and pruritus [R10-5537]. Based on these, potential side effects were identified by Criterion 4:

(4) Adverse events with linagliptin+metformin with a frequency comparable to the frequency of the respective adverse event with placebo+metformin

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The analysis used the overall comparison of placebo+Met versus Lina+Met as well as the comparison of placebo versus Lina+Met. No analysis was performed for demographic subgroups.

The summary of identified side effects was based on the available evidence from placebo-controlled studies and considered also the side effects known for the monotherapy treatment with the components of this fixed-dose combination. The entries in the table were calculated on MedDRA preferred terms or, where appropriate, included additional terms in order to identify and display a particular medical concept.

Table 5.8: 1 Side effects identified for the fixed-dose combination of linagliptin and metformin – SEA-2

SOC	Linagliptin (monotherapy)	Linagliptin+Metformin
Infections and infestations	Nasopharyngitis[1]	Nasopharyngitis[1,2]
Immune system disordrers	Hypersensitivity[1]	Hypersensitivity[1]
Respiratory, thoracic & mediastinal disorders	Cough[1]	Cough[1,2]
Gastrointestinal disorders		Decreased appetite[3]
		Diarrhea[2,3]
		Nausea[2,3]
	Pancreatitis[1]	Pancreatitis[1]
		Vomiting [2,3]
Skin and subcutaneous tissue disorders		Pruritus[2,3]

[1] Identified in the linagliptin monotherapy application
[2] Identified based on analyses on SEA-2
[3] Labeled metformin side effects Glucophage® EU-SPC [R10-5537, Module 2.7.5])

When linagliptin and metformin treatment was combined with a sulfonylurea, hypoglycaemia was observed as an additional side effect.

5.9 PREGNANCY AND LACTATION

There are no adequate and well-controlled studies with linagliptin in pregnant women. Because animal reproduction studies are not always predictive of human response, linagliptin like other antidiabetic medications should be avoided during pregnancy.

Pregnant and nursing mothers were excluded from the studies in the linagliptin clinical development programme both for the monotherapy as well as for the combination therapy with metformin. In all studies, women of childbearing potential participating in the trial were asked to use appropriate contraceptive precautions to prevent pregnancies. All women with child-bearing potential had to undergo pregnancy testing at screening, and all were tested negative. However, in total 9 women became pregnant while exposed to study medication (6 pregnancies were reported in the studies analysed for the linagliptin + metformin combination therapy, in addition 3 more pregnancies were reported in studies not analysed for this FDC dossier; data status 02 March 2011). Of these pregnancies one child is still to be born, 4 children are healthy (n=3) or the health status is unknown (n=1); the pregnancies were aborted (n=2), the child was born dead (n=1), or the child had congenital abnormalities (n=1). Full details including narratives are available in the SCS [Module 2.7.4, Section 5.4.1].

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There are limited data from the use of metformin in pregnant women. Metformin was not teratogenic in rats at a dose of 200 mg/kg/day associated with 4 times human exposure. At higher doses (500 and 1000 mg/kg/day, associated with 11 and 23 times human exposure), teratogenicity of metformin was observed in the rat [U10-2386].

No studies on lactating animals have been performed with the combination of metformin and linagliptin. Non-clinical studies with the individual active substances have shown excretion of both metformin and linagliptin into milk in lactating rats [U08-1929; R10-5537, Module 2.7.5]. Metformin is excreted with milk in humans [R10-4318, Module 2.7.5]. It is not known whether linagliptin is excreted into human milk. Linagliptin should not be used during breast feeding.

5.10 OVERDOSAGE, DRUG ABUSE AND DEPENDENCE, WITHDRAWAL

Overdose

Based on its mode of action and on its pharmacokinetic characteristics, it is unlikely that acute overdosing of linagliptin will affect vital functions or cause life threatening hypoglycaemia. In the event of an overdose, the usual supportive measures should be employed, e.g. removal of unabsorbed material from the gastrointestinal tract, clinical monitoring, and institution of supportive therapy if required.

In a randomised, double-blind, placebo-controlled trial in healthy male subjects, single doses of linagliptin of up to 600 mg (equivalent to 120 tablets) were administered orally and were overall well tolerated. Generally, the occurrence of adverse events was not dose dependent. Neither clinically relevant changes of laboratory parameters nor of ECG parameters were observed (BI trial 1218.1) [U05-2072, Module 5.3.3.1].

For metformin, information on overdose is given in [R10-5537, Module 2.7.5]. Briefly, hypoglycaemia has not been seen with doses up to 85 g. However, metformin overdose may lead to lactic acidosis which requires immediate hospitalisation. The most effective method to remove lactate and metformin is haemodialysis.

Drug abuse

Since the pharmacological properties, as well as non-clinical and clinical data do not indicate an impact on the central nervous system suggestive for stimulant, depressant, hallucinogenic or mood-elevating effects the potential for abuse is believed to be minimal. For metformin, information is available in [R10-5537, Module 2.7.5].

Neither linagliptin nor metformin is a controlled substance. Linagliptin has not been systematically studied in animals or humans for its potential for abuse, tolerance, or physical dependence. In clinical trials to date no potential for abuse was observed with the linagliptin am metformin combination.

Withdrawal

In the trials performed with the combination of linagliptin and metformin no withdrawal phenomena were observed. Information on metformin is given in [R10-5537, Module 2.7.5].

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5.11 EFFECTS ON ABILITY TO DRIVE OR OPERATE MACHINERY OR IMPAIRMENT OF MENTAL ABILITY

No studies on the effects on the ability to drive and use machines have been performed with linagliptin. Information on metformin is given in [R10-5537, Module 2.7.5].

5.12 POSTMARKETING EXPERIENCE

The application for a marketing authorisation for linagliptin monotherapy is currently under evaluation and hence no post-marketing data are available. As this is a new drug application for the linagliptin and metformin fixed-dose combination, no post-marketing data are available. Conversely, metformin has been used in Europe for over 50 years and extensive post-marketing data is available for. Refer to the current SPC for further information [R10-5537, Module 2.7.5].

6. BENEFITS AND RISKS CONCLUSIONS

The clinical development programme for the combination therapy comprised 14 clinical trials, among them, 2 phase II-trials, 4 phase III-trials and 2 long-term follow-up trials of the phase III-studies. In total 3529 patients with type 2 diabetes received treatment with linagliptin and metformin (Lina+Met). Overall, 2694 patients were treated for at least 24 weeks, 2081 patients for at least 52 weeks, and 1756 patients for more than 78 weeks. As evidenced by the demographic and baseline characteristics, the clinical programme included a representative spectrum of patients with type 2 diabetes mellitus.

Efficacy

Overall, 7 trials contributed to the demonstration of efficacy of the combination therapy with Lina+Met. The pivotal placebo-controlled trial 1218.17 investigated the efficacy of linagliptin as add-on therapy to metformin over 24 weeks. Additional key trials for the establishment of efficacy were studies 1218.62, 1218.46, and 1218.20. Trial 1218.62, a placebo-controlled study over 12 weeks, showed comparable efficacy of the linagliptin 2.5 mg twice daily (bid) regimen and the linagliptin 5 mg qd regimen when added to metformin. Study 1218.46 tested 2 Lina+Met combinations versus Met in a partial first-line setting. Trial 1218.20 was a double-blind active-controlled trial that tested Lina+Met against glimepiride+metformin (Glim+Met) over 104 weeks of treatment.

Long-term efficacy of the Lina+Met combination is shown by data from the patients who completed study 1218.17 and continued into the open-label 78-week extension study 1218.40. Furthermore longterm data (54 weeks) are available from study 1218.52, which is a double-blind metformin monotherapy-controlled extension trial in patients who completed the 24-week treatment period of study 1218.46 without requiring rescue medication.

Efficacy of Lina+Met in combination with SU over 24 weeks was investigated in study 1218.18, where linagliptin 5 mg qd or placebo were added to ongoing therapy with metformin and an SU. Data on long-term efficacy is available from patients who completed study 1218.18 and continued into trial 1218.40.

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Changes in HbA_{1c}

The pivotal trial 1218.17 showed that linagliptin added to ongoing metformin background therapy was superior to metformin alone in regard to the HbA_{1c} change from baseline. After 24 weeks of treatment a statistically significant and clinically meaningful adjusted mean treatment difference in HbA_{1c} of -0.64% in favour of Lina+Met combination was observed. Study 1218.62 demonstrated equivalence of Lina 2.5 mg bid to Lina 5 mg qd. The adjusted mean difference in HbA_{1c} changes from baseline to Week 12 was -0.74% (Lina 2.5 bid+Met vs. Met). Study 1218.46 was conducted in a partial first-line setting (47.5% treatment-naïve patients) and yielded similar results to the pivotal trial although the treatment effect in the Lina+Met combination therapy groups versus metformin was slightly smaller than in the pivotal trial with -0.58% (L2.5+M500 vs. M500) and -0.51% (L2.5+M1000 vs. M1000), $p<0.001$ for both comparisons. The Lina+Met combination therapy groups also achieved a substantially greater reduction in HbA_{1c} than linagliptin 5 mg monotherapy with differences of -0.77% (L2.5+M500 vs. Lina 5) and -1.14% (L2.5+M1000 vs. Lina 5); p-value was below 0.0001 for both comparisons.

Study 1218.20 demonstrated non-inferiority in the reduction of HbA_{1c} of Lina+Met to Glim+Met. After 104 weeks of treatment, a mean HbA_{1c} decrease from baseline was observed in both treatment groups (Lina+Met -0.16% vs. Glim+Met -0.36%; adjusted mean difference 0.20%). The 2-sided 97.5% CI (0.09%, 0.30%) for the treatment difference excluded the pre-specified non-inferiority margin of 0.35% and the p-value for the non-inferiority test ($p = 0.0004$) was below $\alpha=0.0125$. It was therefore concluded that Lina+Met was non-inferior to Glim+Met.

Across all studies, including the pivotal trial 1218.17, the additional reductions in HbA_{1c} achieved with Lina+Met combination therapies over Met monotherapy (as background or as study treatment) were in the range of 0.51% to 0.64%. Hence all trials demonstrated a clinically meaningful reduction in HbA_{1c} with combination therapy of linagliptin and metformin.

FPG and other measures of efficacy

Also for FPG substantial and clinically meaningful reductions were observed in all trials. The reductions in FPG were in the range of -13.7 mg/dL after 12 weeks to -21.1 mg/dL after 24 weeks of treatment; the p-values for the comparisons Of Lina+Met versus Met were always below 0.005. Hence the results seen with FPG support the findings of the primary endpoint analysis for HbA_{1c}.

Also the analyses of other measures of efficacy such as percentage of patients achieving glycaemic control defined as HbA_{1c} below 7% or below 6.5% and post-prandial glucose (2hPPG) measurements corroborated the superior efficacy of Lina+Met over Met. In addition, substantially fewer patients in the Lina+Met groups than in the Met groups needed rescue medication. For study 1218.17, only 7.8% of patients in the Lina+Met group but 18.9% of patients in the Met group used rescue medication. This was confirmed by results from the other trials.

In addition, the combination therapy of Lina+Met proved to be weight neutral in comparison with metformin (1218.17: -0.4 kg in both treatment groups). Weight neutrality could be confirmed over 104 weeks of treatment (1218.20) in which patients receiving Lina+Met lost

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on average 1.4 kg while patients taking Glim+Met gained 1.3 kg, resulting in a treatment difference of -2.7 kg.

Superior efficacy of Lina+Met over Met could also be shown in relevant subgroups of patients. Thus factors such as age, gender, race (White vs. Asian), ethnicity, geographical region, and baseline BMI did not have an influence on the treatment effect of the Lina+Met combination therapy. Furthermore, the presence of washout of prior OADs and the time since diagnosis of diabetes did not influence the treatment effect of Lina+Met combination therapy.

The reductions of HbA_{1c} were largely independent of metformin dose (500 mg, 850 mg or 1000 mg). The magnitude of the Lina+Met treatment effect was largely consistent across the different metformin doses with adjusted mean changes from baseline of -0.51% to -0.70% after 24 weeks of treatment. A post-hoc analysis of study 1218.46 showed that the combination of linagliptin 2.5 mg bid + metformin 500 mg bid was at least as effective in reducing HbA_{1c} and FPG levels as monotherapy with metformin 1000 mg bid. Therefore, linagliptin 2.5 mg/metformin 500 mg FDC could be considered a useful alternative to metformin monotherapy for patients intolerant to the recommended daily metformin dose. In sumamry, the data support all 3 intended FDC posologies.

Analysis of the long-term trials demonstrated that the Lina+Met treatment effect in regard to the reduction of HbA_{1c} and FPG persists over at least 104 weeks. This could also be demonstrated across trials for other efficacy outcomes such as percentage of patients achieving glycaemic control, need for rescue medication, and for weight neutrality.

Trial 1218.18 demonstrated the superior efficacy of a triple combination of Lina+Met+SU over Met+SU with an adjusted mean difference in HbA_{1c} of -0.62, $p<0.0001$. The superiority of the triple combination was also seen for all other measures of efficacy, i.e. FPG, percentage of patients achieving glycaemic control, use of rescue therapy, and body weight. The persistence of the treatment effects could be demonstrated for up to 102 weeks of treatmentin trial 1218.40.

Safety

The evaluation of safety of Lina+Met combination therapy for the treatment of patients with T2DM was based on 6 phase I studies, 2 phase II studies, and 6 phase III studies. Data were analysed in 10 study groupings: SAF-C1 to SAF-C9, including the analysis of safety for the triple combination of linagliptin+metformin+SU (SAF-C7). A subset of SAF-C5 (only including patients treated with the linagliptin 5 mg daily as this is the proposed dose) was used for identification of potential side effects (SEA-2). Results of the linagliptin mono SCS have been summarised based on SAF-2. In addition, new long-term safety data of linagliptin monotherapy over 78 weeks of treatment was analysed (SAF-M1).

Linagliptin monotherapy

In SAF-M1, infections and infestations, musculoskeletal and connective tissue disorders, gastrointestinal disorders, and nervous system disorders occurred with a higher frequency in the Lina+Met group than in the Lina group. However, these apparent differences in adverse events might be due to differences for concomitant diagnoses at baseline: infections and infestations (Lina 7.4% vs. Lina+Met 11.1%), musculoskeletal and connective tissue disorders (17.0% vs. 21.2%), gastrointestinal disorders (13.4% vs. 19.5%), and nervous

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system disorders (7.1% vs. 12.4%) (Data from study reports 1218.16 [U10-1103; Module 5.3.5.1] and 1218.17 [U09-2533, Module 5.3.5.1]) were less frequent in the Lina group than in the Lina+Met group. In addition, respiratory, thoracic, and mediastinal disorders (6.8% vs. 10.9%) were more frequent in the Lina+Met group due to a higher frequency of cough (2.7% vs. 6.1%) in the Lina+Met group. Hyperglycaemia was the most frequently reported preferred term in both groups (Lina: 36.6%, Lina+Met 29.3%).

Taken together, the analyis of SAF-M1 confirmed the safety analysis for the linagliptin monotherapy. However, on the level of preferred terms, higher incidences of nasopharyngitis, urinary tract infection, influenza, diarrhea, and cough were observed with Lina+Met combination therapy. Analysis of adverse events by time interval did neither show clinically meaningful increases over time nor notable differences between treatments.

Linagliptin + metformin combination therapy

Analysis of safety was initially based on the pivotal trial (1218.17, SAF-C1) and this analysis was compared with data obtained from SAF-C5 (and SEA-2), a pooled analysis of all metformin-controlled trials.

In SAF-C1, the most frequent adverse events were infections and infestations (22.0% vs. 21.6%). Gastrointestinal disorders also occurred with almost identical incidences in both treatment groups (11.9% vs. 11.1%). Differences of 2% or more were observed for musculoskeletal and connective tissue disorders which were more frequent in the Lina+Met group (7.9% vs. 11.1%). Also injury, poisoning and procedural complications (2.3% vs. 5.0%) and respiratory, thoracic and mediastinal disorders (2.8% vs. 4.8%) were slightly more frequent in the Lina+Met group. Conversely, metabolism and nutrition disorders were substantially less frequent in the Lina+Met group than in the Met group (23.7% vs. 10.7%) mainly due to a lower incidence of hyperglycaemia (16.4% vs. 6.1%). Also hypoglycaemia was less frequently with Lina+Met combination therapy than with metformin alone (2.8% vs. 0.6%). Furthermore, investigations (8.5% vs. 4.0%) and psychiatric disorders (2.8% vs. 0.8%) were less frequent with the combination therapy than in the Met group. Comparing the frequencies of adverse events in the Lina+Met group of the pivotal trial with those of the linagliptin group in SAF-2 demonstrates very similar or even lower incidences across system organ classes.

The comparison of the adverse event incidences in the pivotal trial (SAF-C1) with the incidences in the pooled analysis of all placebo-controlled trials with metformin background (SAF-C5, SEA-2) indicated a consistent safety profile of the Lina+Met combination. Overall incidences of adverse events and SOCs were generally very similar in SAF-C1, SAF-C5, and SEA-2. Differences in adverse events incidences found in the pivotal trial were also seen in the other analysis sets. These observations were confirmed in the analysis of the other safety sets. The analysis of the safety by relevant subgroups indicated that the safety profile of the Lina+Met combination was not influenced by age, gender, race (White and Asian), ethnicity, geographic region, renal impairment, or metformin dose.

In total, 10 cases of pancreas disorders were reported: 8 cases occurring during treatment with Lina+Met, and 2 cases during treatment with Glim+Met. In the post-treatment period, 2 cases of pancreas disorders were reported. None of the cases has been necrotising, haemorrhagic, or fatal. Photosensitivity reactions were reported by 4 patients: 3 cases

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occurring during treatment with Lina+Met and 1 case during treatment with Met. None of the events was serious, 1 event was considered drug-related. Angioedema was reported by 2 patients during treatment with linagliptin + metformin; one event was serious, both events were not assessed as drug-related.

In total, 18 deaths were reported: 14 deaths during treatment and 4 deaths occurring in the post-treatment period. In addition, 2 more deaths occurred during washout. Of the 14 deaths occurring during treatment, 9 deaths were reported during treatment with Lina+Met and 5 deaths during treatment with Met (monotherapy or in combination with glimepiride); however exposure-corrected incidences of death were very similar across treatment groups within studies. No death was considered related to the study medication.

Taken together, in the clinical studies included in this submission, the overall incidences of adverse events, drug-related adverse events, adverse events of severe intensity, adverse events leading to discontinuation, and serious adverse events were largely comparable for combination therapy with linagliptin+metformin versus monotherapies. The data reveal no indication of an increased safety burden for the combination over that seen with the components.

The observed adverse events profile is in accordance with the known profile of individual components such as gastrointestinal disorders (e.g., decreased appetite, nausea, vomiting, and diarrhea) and pruritus associated with metformin; and nasopharyngitis, hypersensitivity, cough, and pancreatitis for linagliptin. Hypoglycemia is identified as a side effect only when linagliptin and metformin are combined with sulfonylureas.

In summary, the benefits of the combination therapy with linagliptin and metformin are greater reductions in HbA_{1c} and FPG over metformin and linagliptin monotherapies, low need for rescue therapy, weight neutrality, lower incidence of hypoglycemic events (vs glimepiride +metformin), good tolerability also among special populations, and the convenience and enhanced compliance expected with a single-tablet formulation.

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7. LITERATURE REFERENCES

P97-1130 Davidson MB, Peters AL. An overview of metformin in the treatment of type 2 diabetes mellitus. Am J Med 1997; 102 (1), 99 - 110

P08-07949 Elrishi MA, Khunti K, Jarvis J, Davies MJ The dipeptidyl-peptidase-4 (DPP-4) inhibitors: a new class of oral therapy for patients with type 2 diabetes mellitus. Pract Diabetes Int 2007; 24 (9), 474 - 482

P10-09483 Dunn CJ, Peters DH. Metformin: a review of its pharmacological properties and therapeutic use in non-insulin-dependent diabetes mellitus. Drugs 1995; 49 (5), 721 – 749

R04-2186 The Diabetes Control and Complications Trial Research Group. The effect of intensive treatment of diabetes on the development and progression of long-term complications in insulin-dependent diabetes mellitus. N Engl J Med 1993; 329 (14), 977 - 986

R07-0040 King H, Aubert RE, Herman WH. Global burden of diabetes, 1995 - 2025: prevalence, numerical estimates, and projections. Diabetes Care 1998; 21 (9), 1414 - 1431

R09-4255 Glucophage 500 mg and 850 mg film coated tablets (Merck Serono) (summary of product characteristics, date of revision of the text: 01/10/2007, last updated on the eMC: 01/05/2009)

R09-5501 Januvia 100 mg film-coated tablets: sitagliptin (Merck Sharp & Dohme) (package leaflet, last approved in August 2009). 2009

R10-0861 Glucophage, metformin hydrochloride, manufacturer's standard 500 mg and 850 mg tablets: oral antihyperglycemic agent (sanofi-aventis Canada) (product monograph, date of revision: October 28, 2009, s-a version 4.0 dated October 28, 2009): health professional information; scientific information; consumer information

R10-1386 Melikian C, White J, Vanderplas A, Dezii CM, Chang E. Adherence to oral antidiabetic therapy in a managed care organization: a comparison of monotherapy, combination therapy, and fixed-dose combination therapy. Clin Ther 2002;24(3):460-467.

R10-1387 Guillausseau PJ. Influence of oral antidiabetic drugs compliance on metabolic control in type 2 diabetes.A survey in general practice. Diabetes Metab 2003;29:79-81.

R10-2970 Scheen AJ. Clinical pharmacokinetics of metformin. Clin Pharmacokinet 1996; 30 (5), 359 - 371

R10-4318 Hale TW, Kristensen JH, Hackett LP, Kohan R, Ilett KF. Transfer of metformin into human milk. Diabetologia 2002;45:1509-1514.

R10-4375 Glucophage (metformin hydrochloride tablets), Glucophage XR (metformin hydrochloride extended-release tablets) (Bristol-Myers Squibb Company) (NDA 20-357/S-031, NDA 21-202/S-016, version: 08/27/2008)

R10-4454 Actoplus Met (pioglitazone hydrochloride and metformin hydrochloride) tablets (Takeda Pharmaceutical), Rx only (revised proposed patient information sheet, August 8, 2005)

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R10-4455 Avandamet: brand of rosiglitazone maleate and metformin hydrochloride tablets (GlaxoSmithKline) (prescribing information, 2002)

R10-4456 Eucreas 50 mg/850 mg film-coated tablets, Eucreas 50 mg/1000 mg film-coated tablets (Novartis Pharma) (summary of product characteristics, date of first authorisation/renewal of the authorisation 14.11.2007)

R10-4457 Metaglip (glipizide and metformin HCl) tablets 2.5 mg/250 mg, 2.5 mg/500 mg, 5 mg/500 mg (Bristol-Myers Squibb), Rx only (issued October 2002)

R10-4882 Scarpello JHB, Howlett HCS. Metformin therapy and clinical uses. Diabetes Vasc Dis Res 2008; 5 (3), 157 - 167

R10-4894 Schernthaner G, Barnett AH, Betteridge DJ, Carmena R, Ceriello A, Charbonnel B, Hanefeld M, Lehmann R, Melecki MT, Nesto R, Pirags V, Scheen A, Seufert J, Sjoholm A, Tsatsoulis A, DeFronzo R. Is the ADA/EASD algorithm for the management of type 2 diabetes (January 2009) based on evidence or opinion? A critical analysis. Diabetologia 2010; 53, 1258 - 1269

R10-4896 Bosi E. Metformin - the gold standard in type 2 diabetes: what does the evidence tell us? Diabetes Obes Metab 2009; 11 (Suppl 2), 3 - 8

R10-5341 Glucophage (metformin hydrochloride) tablets, Glucophage XR (metformin hydrochloride) extended-release tablets (Bristol-Myers Squibb), Rx only (product information, revised January 2009)

R10-5537 Glucophage 500 mg and 850 mg film coated tablets (Merck Serono) (summary of product characteristics, date of revision of the text: 04/2010, last updated on the eMC: 17/08/10). website: medicines.org.uk (2010)

R11-0761 European Agency for the Evaluation of Medicinal Products (EMEA) Committee for Proprietary Medicinal Products (CPMP): summary information on a referral opinion following an arbitration pursuant to Article 11 of Council Directive 75/319/EEC as amended, for Glucophage/Glucophage Forte/Risidon/Dianben, International Nonproprietary Name (INN): metformin (CPMP/4082/00). website: ema.europa.eu/docs/en_GB/document_library/Referrals_document/Glucophage_30/WC500011044.pdf (access date: 9 March 2011) ; London: EMEA (2001)